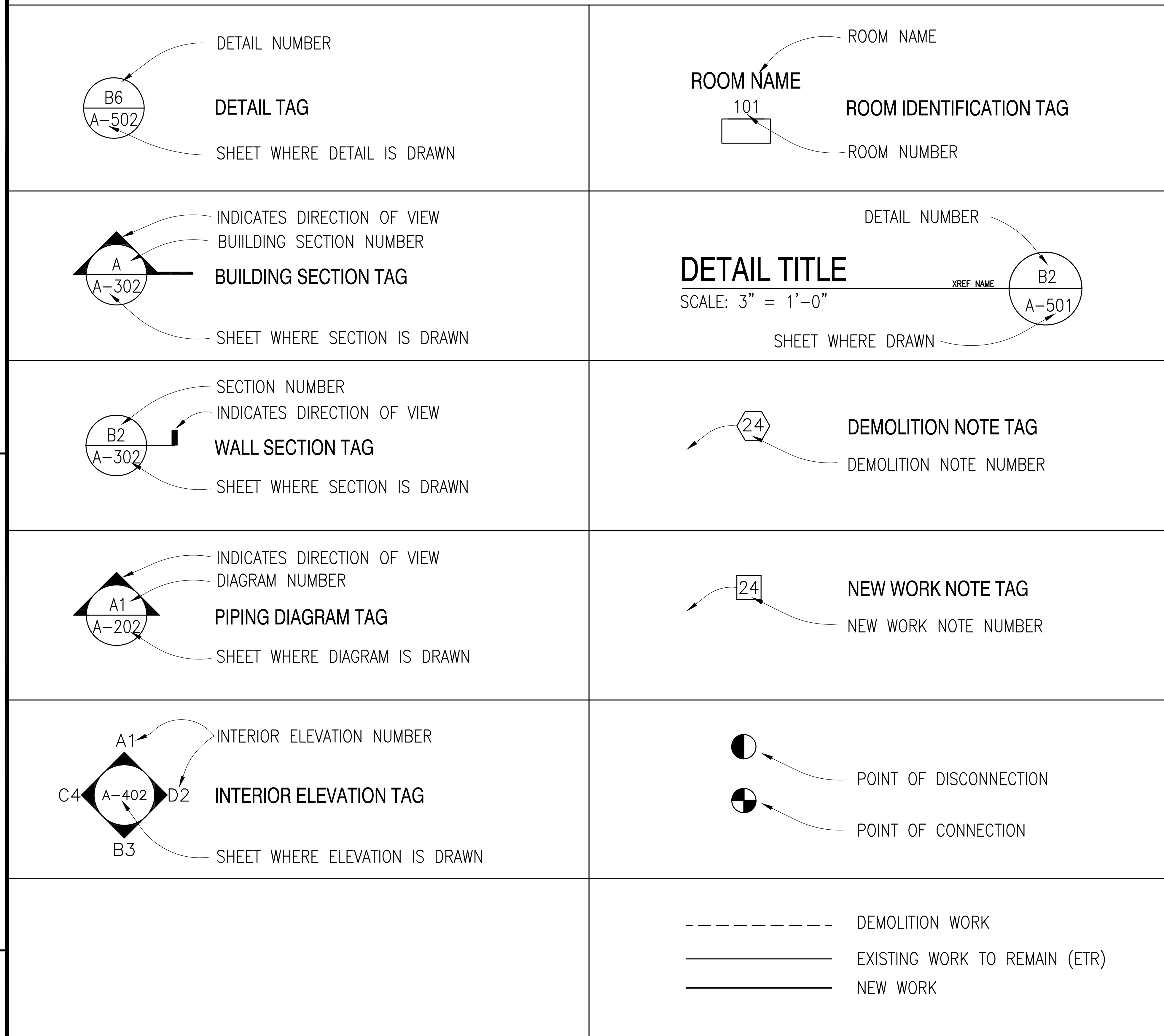
DRAWING SYMBOLS



LEGEND

	DOMESTIC COLD WATER PIPING
	HOT WATER PIPING
	HOT WATER RECIRCULATION PIPING
V	VENT PIPING (SANITARY)
W	WASTE PIPING (SANITARY)
D	CONDENSATE DRAIN PIPING (A/C)
R	REFRIGERANT PIPING
HWS	HOT WATER SUPPLY PIPING
HWR	HOT WATER RETURN PIPING
CWS	CHILLED WATER SUPPLY PIPING
CWR	CHILLED WATER RETURN PIPING
HPS	HIGH PRESSURE STEAM PIPING
HPC	HIGH PRESSURE CONDENSATE PIPING
LPS	LOW PRESSURE STEAM PIPING
LPC	LOW PRESSURE CONDENSTATE PIPING
PCR	PUMPED CONDENSTATE RETURN (STEAM)
	GATE VALVE
	BALL VALVE
	FLOW CONTROL BALANCING VALVE
	2-WAY CONTROL VALVE
	3-WAY CONTROL VALVE
	PIPE CAP
	DIRECTION OF FLOW (PUMPED)
	DIRECTION OF PITCH (GRAVITY FLOW)
(A) 200	DIFFUSER/GRILLE MARK (CFM)
(X) 300	EXISTING DIFFUSER/GRILLE MARK (CFM)
(T)	THERMOSTAT
(S)	HVAC SYSTEM EMERGENCY SHUTDOWN SWITCH
1	PHASE NUMBER MARK
VRF-#	VARIABLE REFRIGERANT FLOW INDOOR UNIT
ODU #	NEW EQUIPMENT TAG
	MANUAL BALANCING DAMPER
M	MOTORIZED BALANCING DAMPER
	RISE IN DUCT (POINT IS HIGH)
	FIRE DAMPER
	NEW FLEXIBLE DUCTWORK (INSULATED)
	SUPPLY AIR DUCT SECTION (POSITIVE)
	RETURN/EXHAUST AIR DUCT SECTION (NEGATIVE)
	DUCT UP THRU FLOOR/ROOF
	FLAT OVAL DUCTWORK (IN WIDTH X IN HEIGHT)
ø	DIAMETER (ROUND DUCTWORK)
12X10	DUCTWORK SIZE (IN WIDTH X IN HEIGHT)
▲	RA OPENING IN WALL, ABOVE CEILING, 16X12 (HOLD TIGHT TO STRUCTURE)
	EXHAUST FAN
	SUPPLY AIR DIFFUSER
	RETURN OR EXHAUST AIR GRILLE
	VRF FLOOR FAN COIL UNIT
	VRF CEILING CASSETTE FAN COIL UNIT
	VRF WALL MOUNTED FAN COIL UNIT
	HRP (HEAT RECOVERY PORT)

ABBREVIATIONS

A/C	AIR CONDITIONING
AC	ABOVE CEILING
ACU	AIR CONDITIONING UNIT
AD	ACCESS DOOR
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
APD	AIR PRESSURE DROP
AS	AIR SEPARATOR
BAS	BUILDING AUTOMATION SYSTEM
BDD	BACKDRAFT DAMPER
CER	CEILING EXHAUST REGISTER
CFT	CHEMICAL FEED TANK
CFM	CUBIC FEET PER MINUTE
CONV	CONVECTOR
CPS	CONDENSATE PUMP SET
CR	CONDENSATE RETURN
CRG	CEILING RETURN GRILLE
CRU	COMPUTER ROOM UNIT
CSD	CEILING SUPPLY DIFFUSER
CTG	CEILING TRANSFER GRILLE
CTR	COOLING TOWER RETURN
CTS	COOLING TOWER SUPPLY
COND	CONDENSING UNIT
CUH	CABINET UNIT HEATER
CW	COLD WATER
DB	DRY BULB
DC	DRY COOLER
DDC	DIRECT DIGITAL CONTROL
DEG., °	DEGREES
DIA., ø	DIAMETER
DOAS	DEDICATED OUTDOOR AIR SYSTEM
DOM HWH	DOMESTIC HOT WATER HEATER
DSS	DUCTLESS SPLIT SYSTEM
DS	DOWN IN SHAFT
DSA	DUCT SOUND ATTENUATOR
DW	DOUBLE WALL (DUCTWORK)
EA	EXHAUST AIR
EAT	ENTERING AIR TEMPERATURE
ECS	ELECTRONIC COOLING SYSTEM
EF	EXHAUST FAN
EMD	END OF MAIN DRIP (STEAM)
ER	EXHAUST REGISTER
ERU	ENERGY RECOVERY UNIT
ESP	EXTERNAL STATIC PRESSURE
ET	EXPANSION TANK
ETR	EXISTING TO REMAIN
EV	EVAPORATOR
FA	FRESH AIR (ERU)
FCU	FAN COIL UNIT
FD	FIRE DAMPER
FT	FLASH TANK (STEAM)
FPB	FAN POWERED BOX
FTR	FIN TUBE RADIATION
GIH	GRAVITY INTAKE HOOD
GWR	GLYCOL WATER RETURN
GWS	GLYCOL WATER SUPPLY
HWH	HOT WATER HEATER
HX	HEAT EXCHANGER
LD	LINEAR DIFFUSER
MBH	1000 BTU'S PER HOUR
MCA	MINIMUM CIRCUIT AMPS
MOD	MOTOR OPERATED DAMPER
OA	OUTSIDE AIR
OBD	OPPOSED BLADE DAMPER
OF	ON FLOOR
OG	ON GRADE
OR	ON ROOF
OW	ON WALL
P	PUMP
RA	RETURN AIR
RAD	RISE AND DRIP
RAF	RETURN AIR FAN
PRV	POWER ROOF VENTILATOR
PRS	PRESSURE REDUCE STATION
RH	RELATIVE HUMIDITY
RL	REFRIGERANT LIQUID
RR	RETURN REGISTER
RS	REFRIGERANT SUCTION
RTU	ROOFTOP UNIT
SA	SUPPLY AIR
SAT	SUPPLY AIR TEMPERATURE
SB	SECURITY BAR ASSEMBLY
SH	STEAM HUMIDIFIER
SR	SUPPLY REGISTER
STM	STEAM
TA	TRANSFER AIR
TC	THRU CEILING
TF	THRU FLOOR
TR	THRU ROOF
TYP.	TYPICAL (AT ALL SIMILAR CONDITIONS)
UH	UNIT HEATER
UON	UNLESS OTHERWISE NOTED
VAV	VARIABLE AIR VOLUME
VF	VENTILATION FAN (OA SUPPLY)
VFD	VARIABLE FREQUENCY DRIVE
VRF	VARIABLE REFRIGERANT FLOW
WB	WET BULB
WC	WATER COLUMN
WEF	WALL EXHAUST FAN
WH	WALL HOOD
WL	WALL LOUVER
WPD	WATER PRESSURE DROP
WSR	WALL SUPPLY REGISTER

GENERAL NOTES

1. INSTALL ALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER'S WRITTEN INSTRUCTIONS BY A MANUFACTURER'S CERTIFIED CONTRACTOR.
2. PATCH ALL HOLES FROM REMOVAL OF PIPING, DUCTWORK, AND/OR EQUIPMENT WITH MATERIALS MATCHING EXISTING ADJACENT CONSTRUCTION.
3. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, SPECIFIED AND AS REQUIRED BY CODE.
4. UNLESS INDICATED OTHERWISE, PIPING AND DUCTWORK TO BE REMOVED SHALL INCLUDE ASSOCIATED HANGERS, INSULATION, AND OTHER APPURTENANCES.
5. ALL PIPING AND DUCTWORK INSTALLATIONS SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFFSET PIPING AND DUCTWORK AROUND OBSTRUCTIONS AT NO ADDITIONAL COST TO THE GOVERNMENT.
6. COORDINATE WITH ARCHITECTURAL AND ELECTRICAL DRAWINGS PRIOR TO INSTALLING MECHANICAL EQUIPMENT.
7. IT IS THE INTENTION OF THE DRAWINGS AND SPECIFICATIONS TO COVER ALL WORK AND MATERIAL FOR A COMPLETE INSTALLATION. ANY EQUIPMENT, TRIM HARDWARE AND/OR DEVICES USUALLY UTILIZED IN CLASS OF WORK, THOUGH NOT SPECIFICALLY MENTIONED OR SHOWN ON THE DRAWINGS, BUT WHICH MAY BE NECESSARY FOR THE SATISFACTORY COMPLETION OF WORK (AS DETERMINED BY THE CONTRACTING OFFICER) SHALL BE FURNISHED AND INSTALLED BY THE CONTRACTOR AS PART OF HIS TOTAL WORK, AT NO ADDITIONAL COST TO THE GOVERNMENT.
8. GENERAL NOTES ON THIS DRAWING ARE APPLICABLE TO EACH MECHANICAL DRAWING OF THIS SET. GENERAL NOTES SPECIFIC TO INDIVIDUAL DRAWINGS WILL BE SHOWN ON THE RESPECTIVE DRAWING.

GENERAL NOTES

9. ENTIRE INSTALLATION, INCLUDING MATERIALS, EQUIPMENT, AND WORKMANSHIP, SHALL CONFORM WITH ALL APPLICABLE LAWS, CODES, AND REGULATIONS OF MUNICIPAL, STATE, AND FEDERAL AUTHORITIES, ALSO THE LATEST EDITION OF THE STATE BUILDING CODES, APPLICABLE ARI STANDARDS AND OTHER REGULATORY BODIES HAVING JURISDICTION OVER THE CLASS OF WORK. WHERE APPLICABLE, MATERIALS AND EQUIPMENT SHALL HAVE STAMPS OR SEALS OF ARI, ASME, UL, OR ASTM. THE CONTRACTOR SHALL MAKE TESTS FOR ACCEPTANCE AND APPROVAL BY CODE AND THE REQUIREMENTS OF APPLICABLE REGULATORY AGENCIES. REQUIRED TESTS SHALL BE PERFORMED IN THE PRESENCE OF THE GOVERNMENT UNLESS OTHERWISE WAIVED IN WRITING.
10. THE CONTRACTOR SHALL OBTAIN AND PAY FOR ALL PERMITS, LICENSES, DOCUMENTS, AND SERVICES RELATED TO INSTALLATION OF THE WORK.
11. THE CONTRACTOR SHALL COORDINATE HIS WORK WITH OTHER TRADES IN ORDER TO RESOLVE ANY CONFLICTS THAT MIGHT ARISE DUE TO THE LOCATION OF EQUIPMENT OR THE USE OF THE AVAILABLE SPACE.
12. RUN ALL PIPING AND DUCTWORK CONCEALED ABOVE CEILINGS OR IN WALLS OR CHASES UNLESS OTHERWISE NOTED.
13. EXISTING DUCTWORK, PIPE, AND EQUIPMENT SIZES NOTED ARE FOR THE CONVENIENCE OF THE CONTRACTOR ONLY AND ARE NOT WARRANTED TO BE CORRECT. FIELD VERIFY ALL DIMENSIONS PRIOR TO CONSTRUCTION. THE CONTRACTOR SHALL FIELD VERIFY ALL EXISTING DUCT, PIPE, AND EQUIPMENT SIZES AND THEIR RESPECTIVE LOCATIONS BEFORE PROCEEDING WITH ANY WORK.
14. CUT OPENINGS, AS REQUIRED, IN THE EXISTING CONSTRUCTION FOR THE INSTALLATION OF NEW DUCTWORK, PIPING, AND EQUIPMENT. PATCH AND REPAIR TO MATCH THE EXISTING ADJACENT CONSTRUCTION. NEW OPENINGS IN EXISTING CONCRETE MASONRY CONSTRUCTION SHALL BE SAW CUT OR DRILLED.
15. LAYOUT OF PIPING IS DIAGRAMMATIC. RUN ALL EXPOSED PIPING AS HIGH AS POSSIBLE UNLESS OTHERWISE NOTED. ALLOW FOR RISES, DROPS, AND OFFSETS AS REQUIRED.
16. TAB WORK: (SEE SPECIFICATION SECTION 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC).
17. OPENINGS CREATED WHERE DEMOLISHED PIPING/HVAC DUCTWORK PASS THRU WALLS AND/OR ROOFS (AND NOT TO BE REUSED FOR NEW CONSTRUCTION) SHALL BE PATCHED TO MATCH EXISTING SURFACES.
18. DUCTWORK THAT IS ASSOCIATED WITH THIS BUILDING SECTION TO BE DEMOLISHED.
19. EACH EXHAUST AND RETURN DIFFUSER SHALL HAVE AN INDIVIDUAL BALANCE DAMPER LOCATED IN THE DUCT FOR THAT INDIVIDUAL DIFFUSER.

SYM | DESCRIPTION | DATE | APPR

NAVFAC

SEAL

A/E INFO

APPROVED David W. Grunfeld
FOR COMMANDER NAVFAC
ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON
SATISFACTORY TO | DATE 06/18/2012
DES JLS | DRW JLS | CHK JM
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION SLW

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND – MID-ATLANTIC
NORTHEAST IPT
NAVAL STATION – NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON
GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
MECHANICAL LEGEND, NOTES & ABBREVIATIONS

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624697
SHEET 124 OF 206
M-001

DRAWFORM REVISION: 10 MARCH 2009

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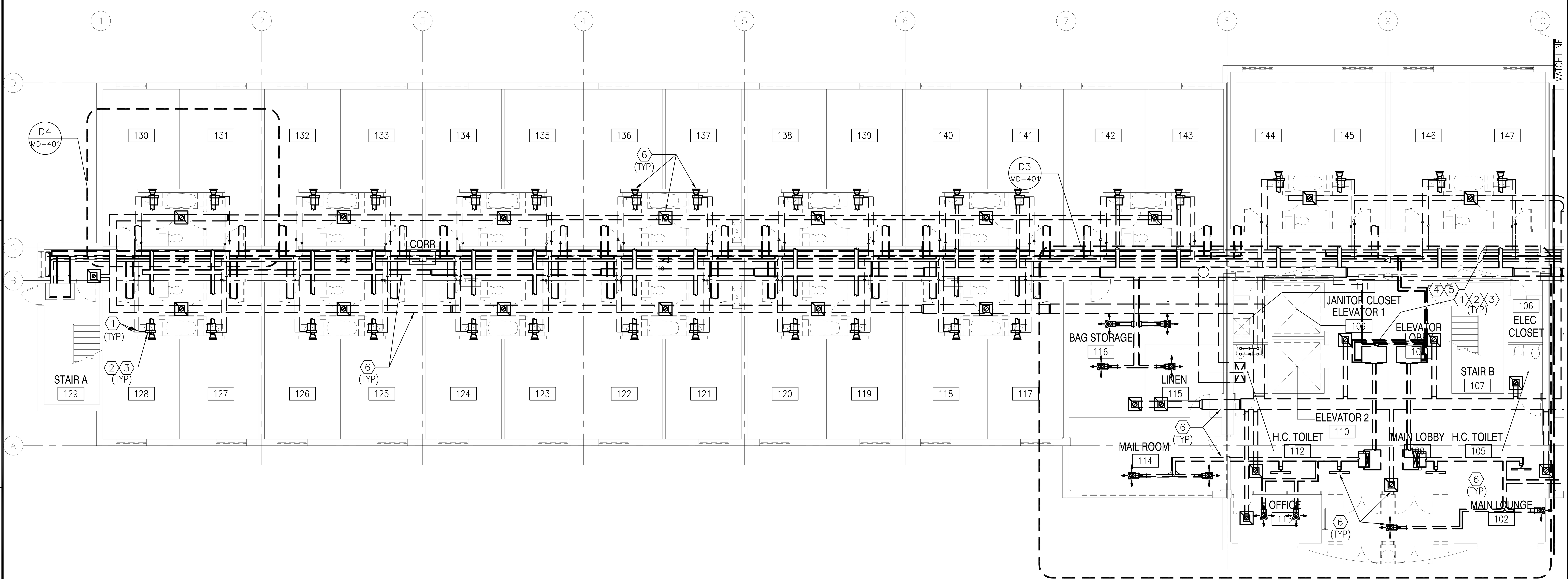


DEMOLITION FIRST FLOOR DUCTWORK PLAN - NORTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES, AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVE. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

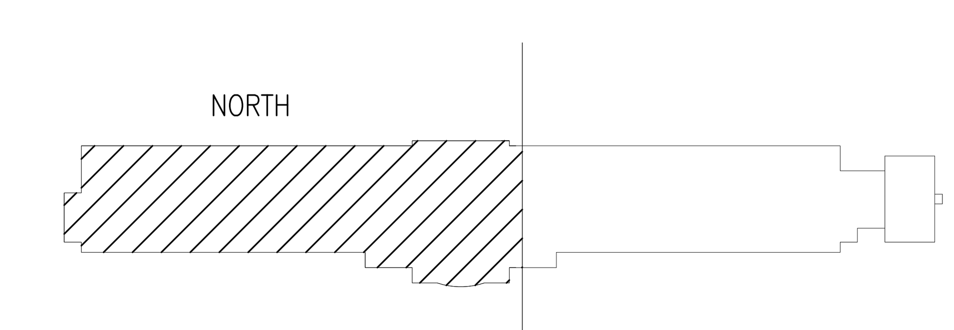
DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
4. SEE DRAWING MD-102 FOR CONTINUATION OF PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
5. SEE DRAWING MD-102 FOR PIPING CONTINUATION.
6. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN DUCTWORK, DIFFUSERS, DAMPERS, AND GRILLS TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.

GRAPHIC SCALE

1/8" = 1'-0" 0 4' 8' 16'

KEY PLAN



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ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON
SATISFACTORY TO DATE 06/18/2012
DES JLS DRW MDS CHK JM
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION S.W

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NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL DEMOLITION PLAN - FIRST FLOOR - NORTH

SCALE:	AS NOTED
EPROJECT NO.:	1127117
CONSTR. CONTR. NO.	N40085-12-R-1715
NAVFAC DRAWING NO.	12624698
SHEET	125 OF 206

MD101

DRAWFORM REVISION: 10 MARCH 2009

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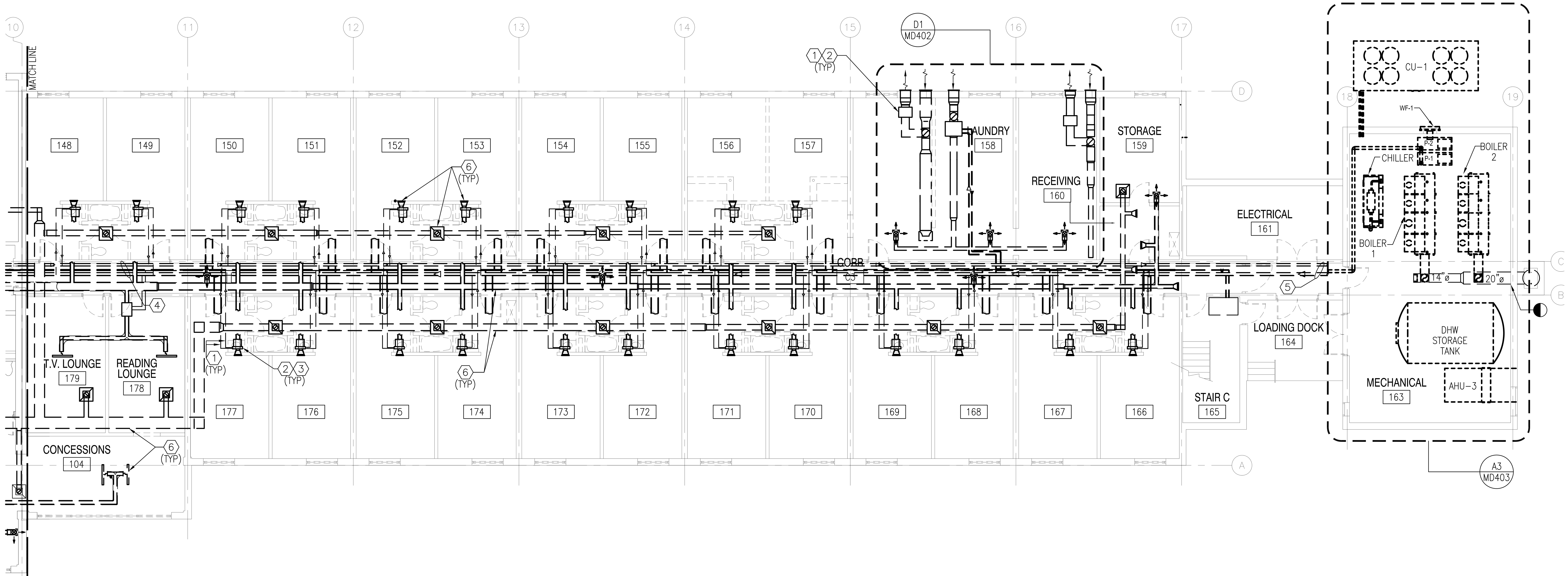


MECHANICAL DEMOLITION PLAN - FIRST FLOOR - SOUTH

SCALE: 1/8" = 1'-0"

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

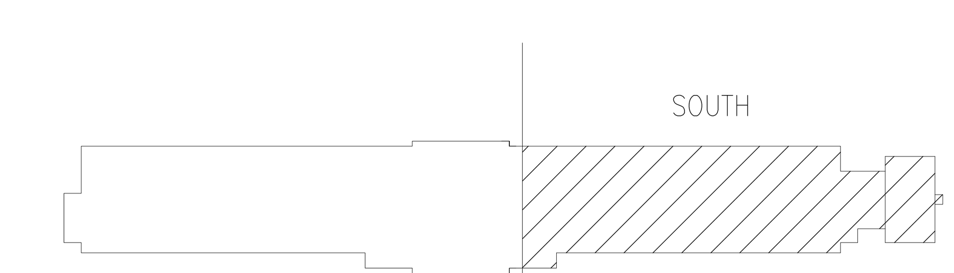
DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
4. SEE DRAWING MD-101 FOR CONTINUATION OF PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
5. SEE DRAWING MD-403 FOR PIPING ORIGIN.
6. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN DUCTWORK AND DIFFUSERS, DAMPER, AND GRILLS TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.

GRAPHIC SCALE

1/8" = 1'-0" 0 4' 8' 16'

KEY PLAN



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SATISFACTORY TO DATE 06/18/2012
DES JLS DRW MDS CHK JM
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION S|W

DEPARTMENT OF THE NAVY
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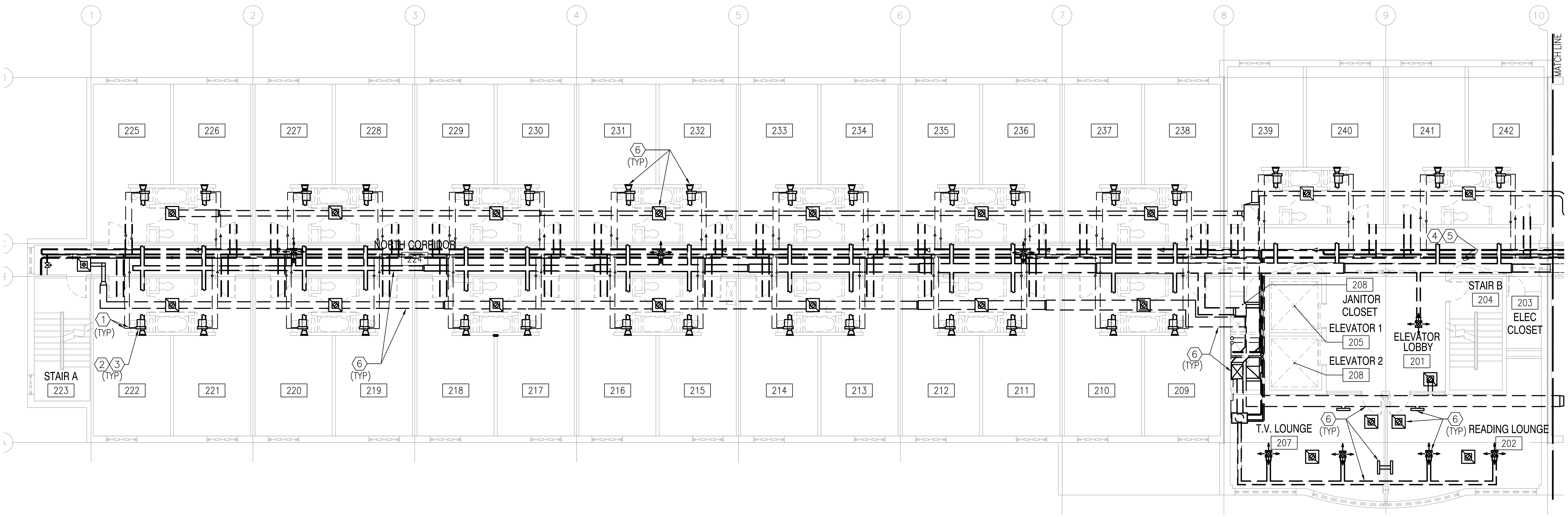
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
MECHANICAL DEMOLITION PLAN - FIRST FLOOR - SOUTH

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624699
SHEET 126 OF 206
MD102

DRAWFORM REVISION: 10 MARCH 2009



MECHANICAL DEMOLITION PLAN - SECOND FLOOR - NORTH

SCALE: 1/8" = 1'-0"

TRUE NORTH

PLAN NORTH

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

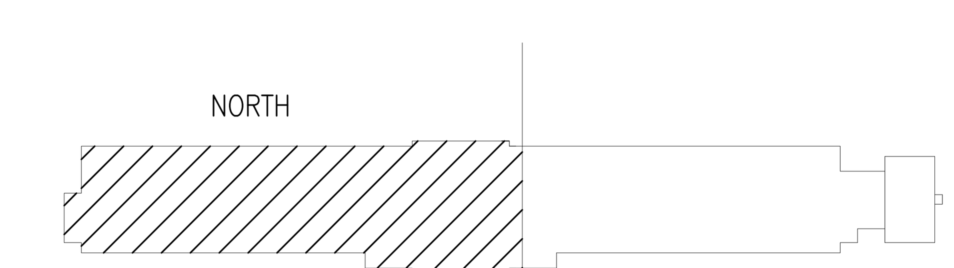
DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
4. SEE DRAWING MD-104 FOR CONTINUATION OF PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
5. SEE DRAWING MD-104 FOR PIPING CONTINUATION.
6. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN DUCTWORK AND DIFFUSERS, DAMPER, AND GRILLS TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.

GRAPHIC SCALE

1/8" = 1'-0" 0 4' 8' 16'

KEY PLAN



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ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON

SATISFACTORY TO DATE 06/18/2012

DES JLS DRW MDS CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND – MID-ATLANTIC
NAVAL STATION – NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON GROTON, CT

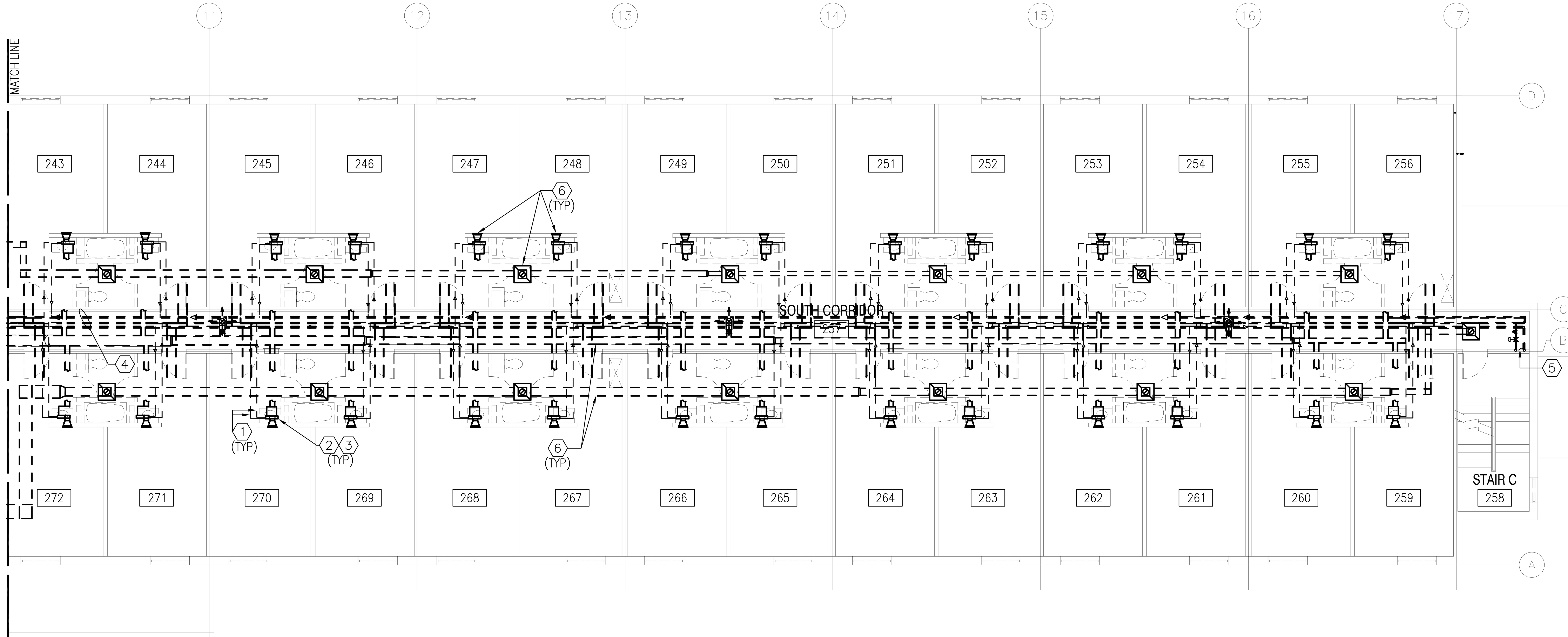
REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL DEMOLITION PLAN – SECOND FLOOR – NORTH

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624700
SHEET 127 OF 206

MD103

DRAWFORM REVISION: 10 MARCH 2009



MECHANICAL DEMOLITION PLAN - SECOND FLOOR - SOUTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

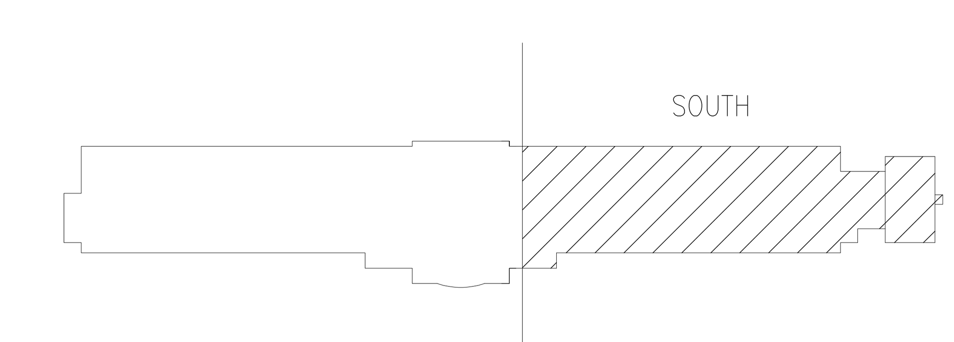
DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
4. SEE DRAWING MD-103 FOR CONTINUATION OF PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
5. SEE DRAWING MD-403 FOR PIPING ORIGIN.
6. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN DUCTWORK AND DIFFUSERS, DAMPER, AND GRILLS TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.

GRAPHIC SCALE

1/8" = 1'-0"   0   4'   8'   16'

KEY PLAN



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SATISFACTORY TO   DATE 06/18/2012
DES JLS   DRW MDS   CHK JM
PM/DM RS/NW
BRANCH MANAGER
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FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
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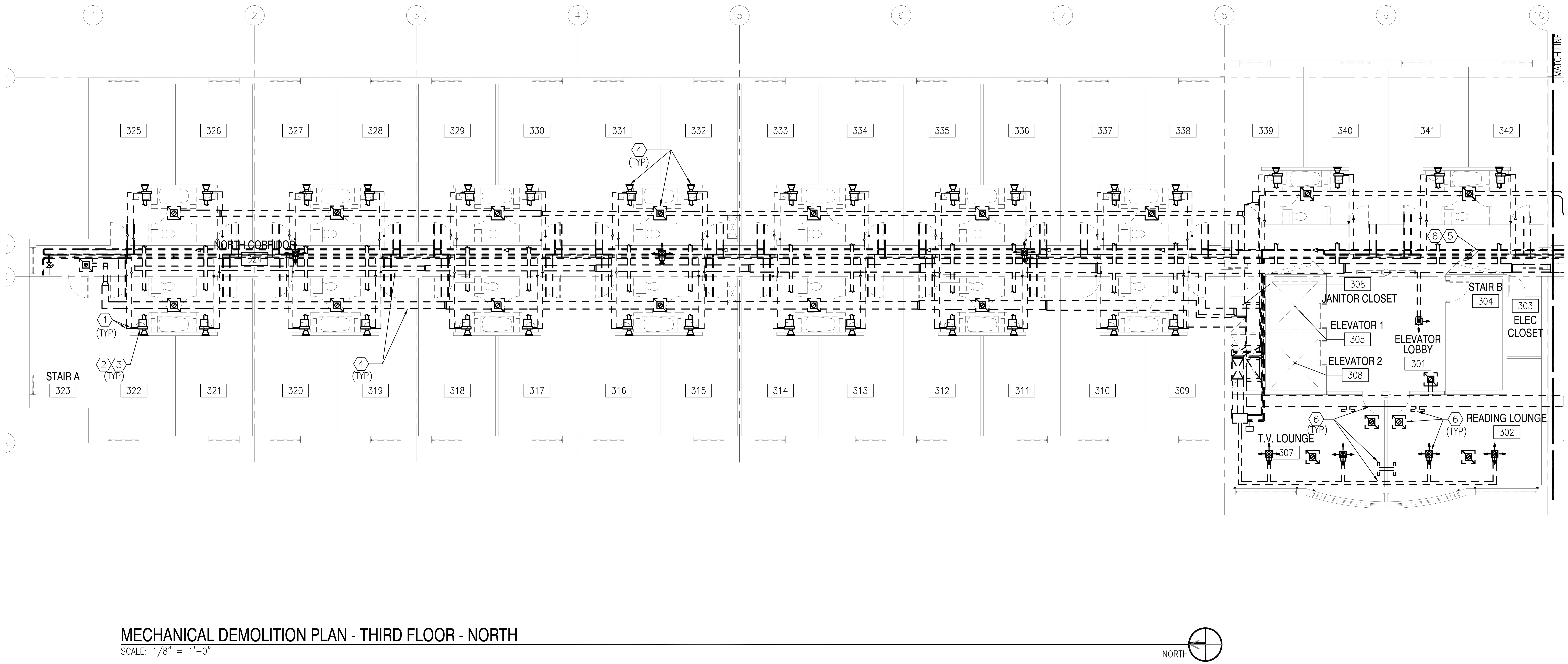
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON
GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
MECHANICAL DEMOLITION PLAN - SECOND FLOOR - SOUTH

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624701
SHEET 128 OF 206
MD104

DRAWFORM REVISION: 10 MARCH 2009

FILE NAME: P:\Connecticut\New_London\Buildings\B-534\2013_1127117_RM10-7303_Repair Sub A School BQ 534\B_Design\Drawings\Working\Mechanical\BLD0534-1127117-MD-104.dwg LAYOUT NAME: MD-104 PLOTTED: Thursday, July 12, 2012 - 10:39am USER: joseph.t.smith4



MECHANICAL DEMOLITION PLAN - THIRD FLOOR - NORTH

SCALE: 1/8" = 1'-0"

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.

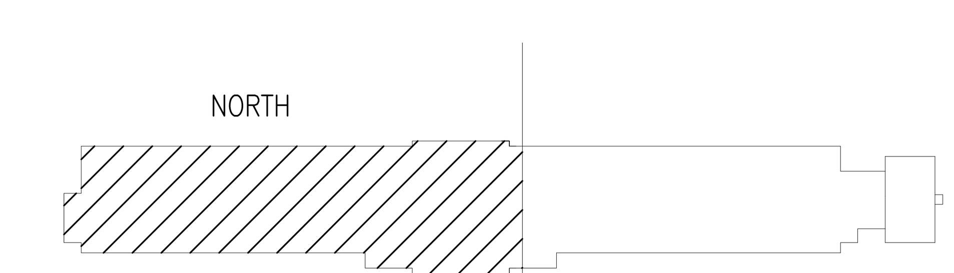
DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN DUCTWORK AND DIFFUSERS, DAMPER, AND GRILLS TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
5. SEE DRAWING MD-106 FOR CONTINUATION OF PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
6. SEE DRAWING MD-106 FOR PIPING CONTINUATION.

GRAPHIC SCALE

1/8" = 1'-0"   0 4' 8' 16'

KEY PLAN



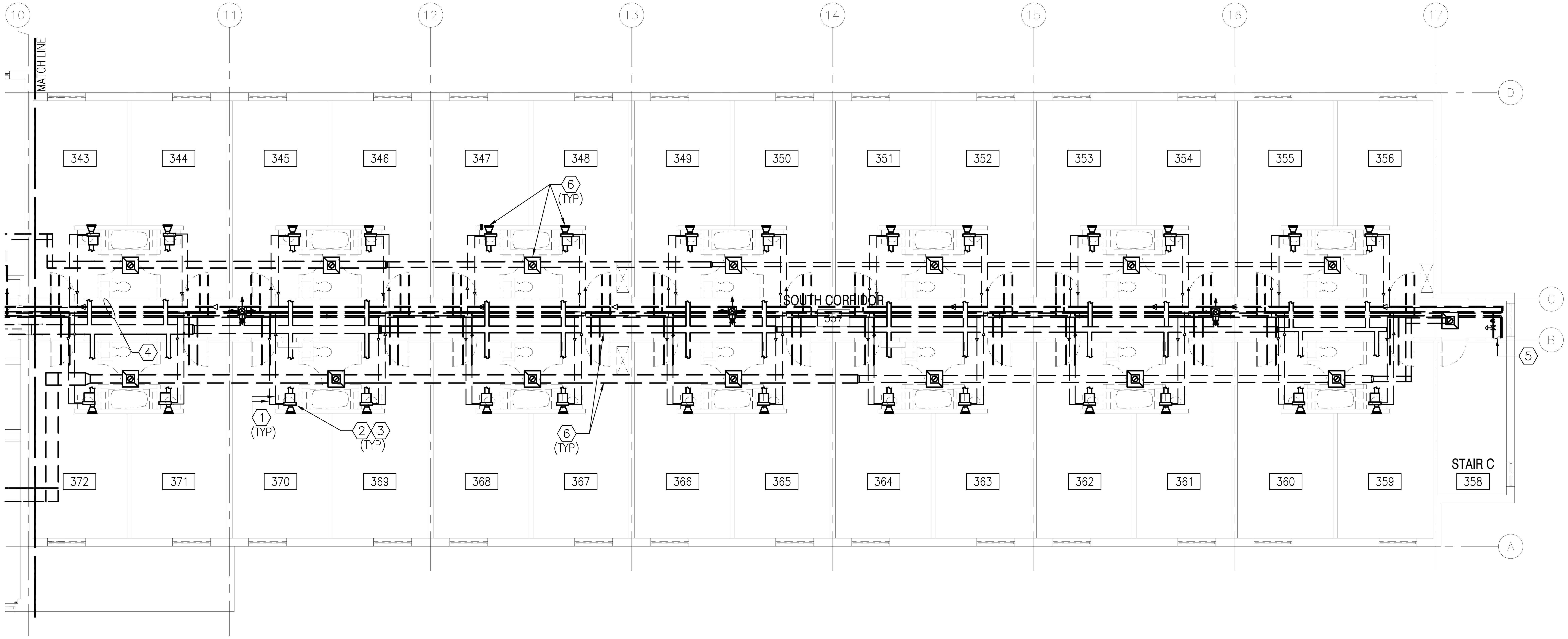
APPROVED David W. Greenfield
FOR COMMANDER NAVFAC
ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON
SATISFACTORY TO DATE 06/18/2012
DES JLS DRW MDS CHK JM
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION SLW

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
MECHANICAL DEMOLITION PLAN - THIRD FLOOR - NORTH

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624702
SHEET 129 OF 206
MD105

DRAWFORM REVISION: 10 MARCH 2009

FILE NAME: P:\Connecticut\New_London\Buildings\B-534\2013_1127117_RM10-7303_Repair Sub A School BQ 534\B_Design\Drawings\Working\Mechanical\BLD0534-1127117-MD-105.dwg LAYOUT NAME: MD-105 PLOTTED: Thursday, July 12, 2012 - 10:40am USER: joseph.smith4



MECHANICAL DEMOLITION PLAN - THIRD FLOOR - SOUTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

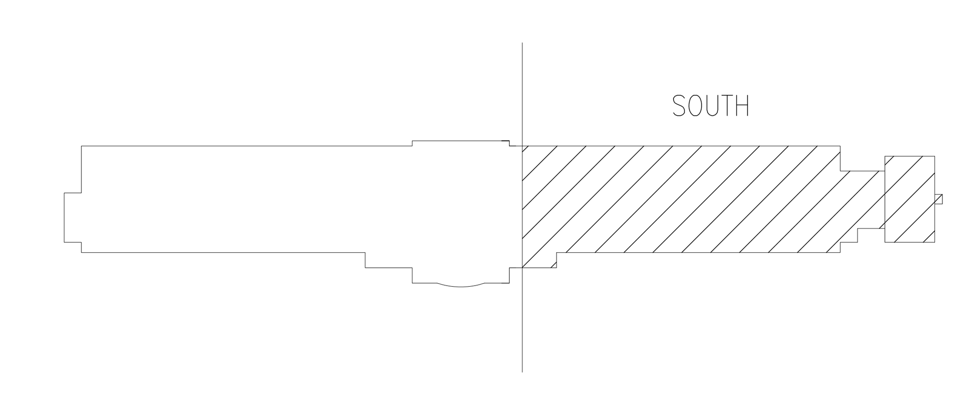
DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
4. SEE DRAWING MD-105 FOR CONTINUATION OF PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
5. SEE DRAWING MD-403 FOR PIPING ORIGIN.
6. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN DUCTWORK AND DIFFUSERS, DAMPER, AND GRILLS TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.

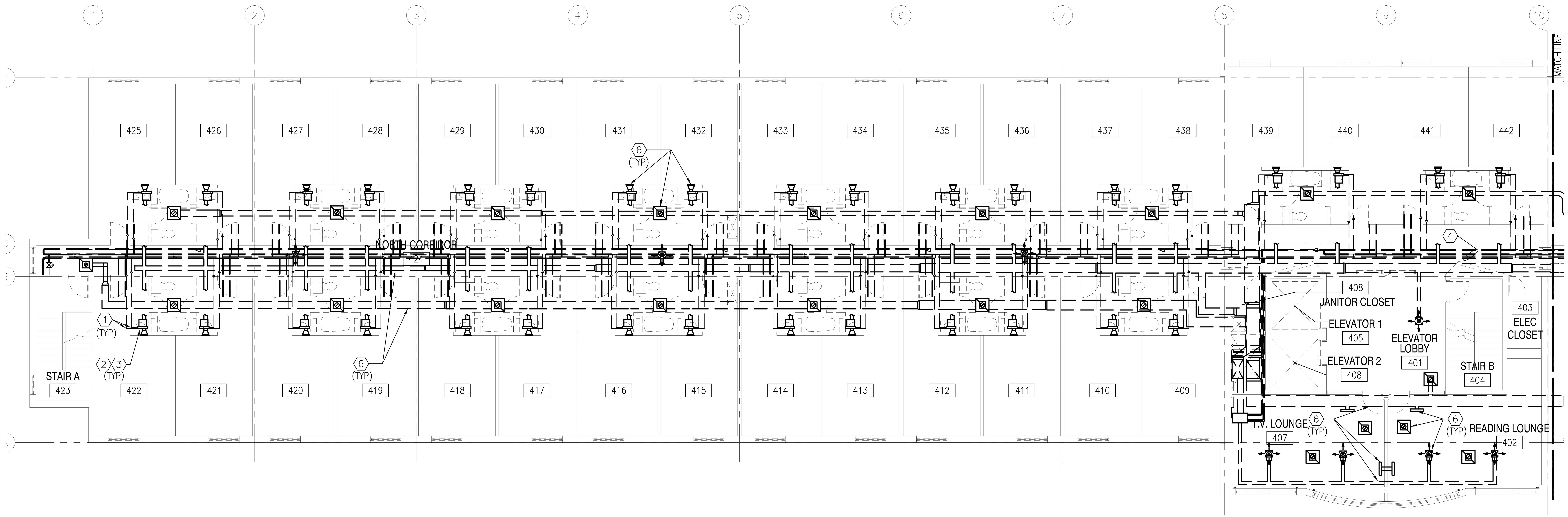
GRAPHIC SCALE

1/8" = 1'-0"   0   4'   8'   16'

KEY PLAN



APPROVED
FOR COMMANDER NAVFAC
ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON
SATISFACTORY TO   DATE 06/18/2012
DES JLS   DRW MDS   CHK JM
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION S/W
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON
GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
MECHANICAL DEMOLITION PLAN - THIRD FLOOR - SOUTH
SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624703
SHEET 130 OF 206
MD106
DRAWFORM REVISION: 10 MARCH 2009



MECHANICAL DEMOLITION PLAN - FOURTH FLOOR - NORTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

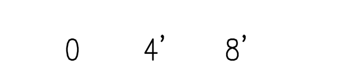
1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

DEMOLITION KEYNOTES

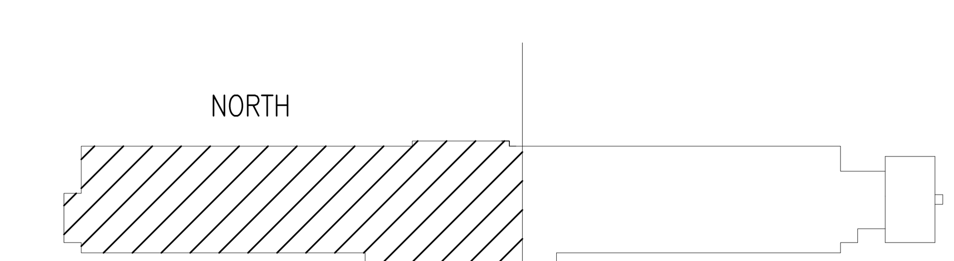
1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
4. SEE DRAWING MD-108 FOR CONTINUATION OF PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
5. SEE DRAWING MD-108 FOR PIPING CONTINUATION.
6. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN DUCTWORK AND DIFFUSERS, DAMPER, AND GRILLS TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.

GRAPHIC SCALE

1/8" = 1'-0"



KEY PLAN



SYM | DESCRIPTION | DATE | APPR

A/E INFO

SEAL

APPROVED David Greenfield

David W Greenfield

FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON

SATISFACTORY TO DATE 06/18/2012

DES JLS | DRW MDS | CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION SLW

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NORTHEAST IPT
NAVAL STATION – NORFOLK, VIRGINIA

NAVSUBASE NEW LONDON
GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL DEMOLITION PLAN – FOURTH FLOOR – NORTH

SCALE: AS NOTED

EPROJECT NO.: 1127117

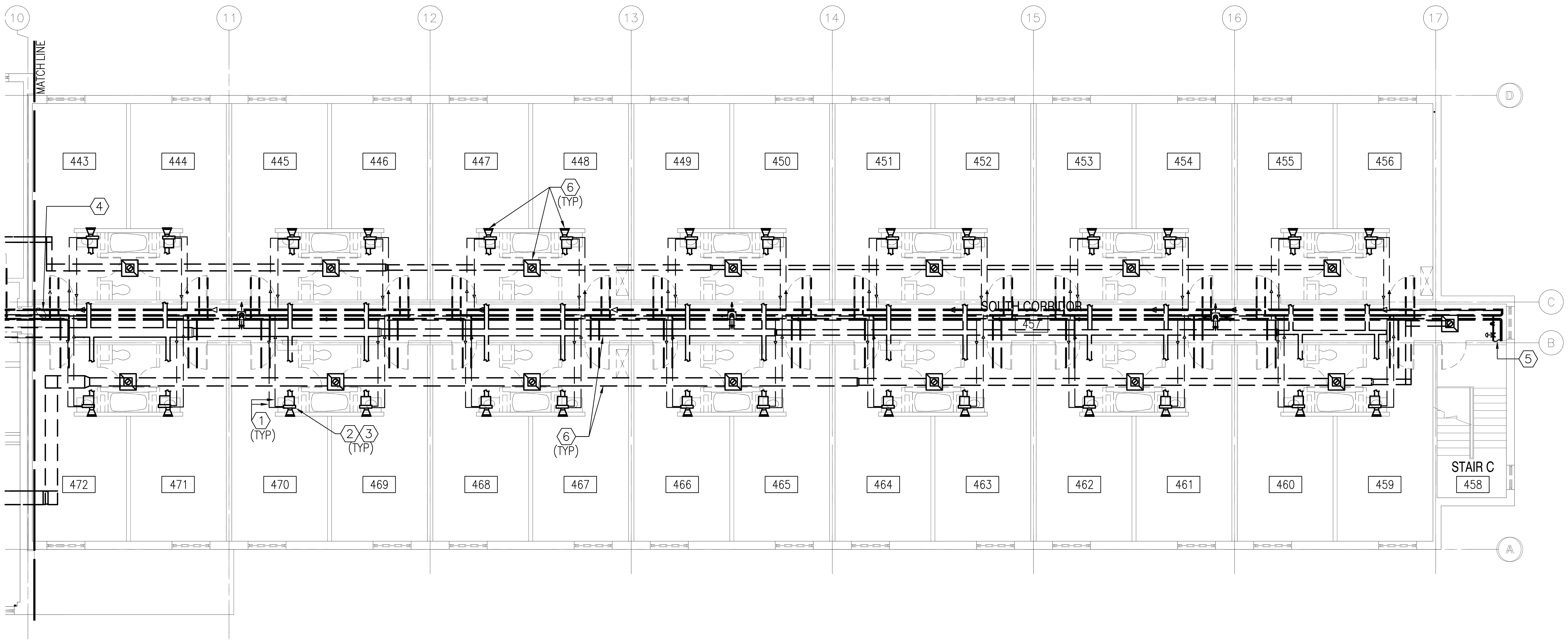
CONSTR. CONTR. NO. N40085-12-R-1715

NAVFAC DRAWING NO. 12624704

SHEET 131 OF 206

MD107

DRAWFORM REVISION: 10 MARCH 2009



MECHANICAL DEMOLITION PLAN - FOURTH FLOOR - SOUTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

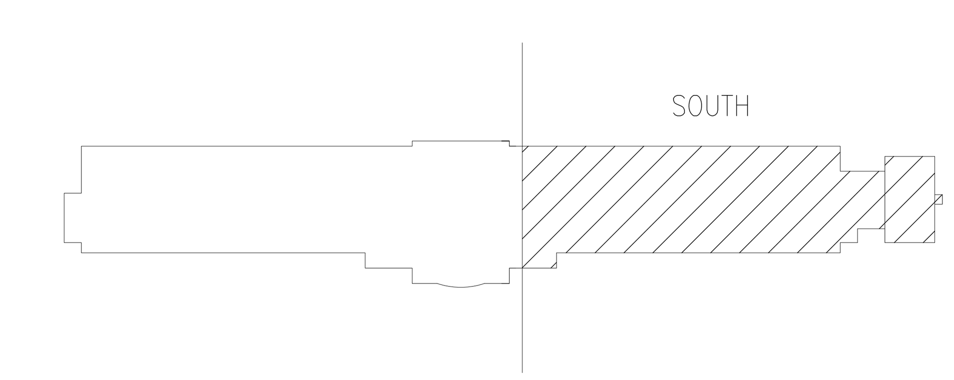
DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.
4. SEE DRAWING MD-107 FOR CONTINUATION OF PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW.
5. SEE DRAWING MD-403 FOR PIPING ORIGIN.
6. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN DUCTWORK AND DIFFUSERS, DAMPER, AND GRILLS TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH NEW FINISHES.

GRAPHIC SCALE

1/8" = 1'-0"   0 4' 8' 16'

KEY PLAN



SYM | DESCRIPTION | DATE | APPR

NAVFAC

SEAL

A/E INFO

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO   DATE 06/18/2012

DES JLS   DRW MDS   CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NORTHEAST IPT
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON   GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL DEMOLITION PLAN - FORTH FLOOR - SOUTH

SCALE: AS NOTED

EPROJECT NO.: 1127117

CONSTR. CONTR. NO. N40085-12-R-1715

NAVFAC DRAWING NO. 12624705

SHEET 132 OF 206

MD108

DRAWFORM REVISION: 10 MARCH 2009

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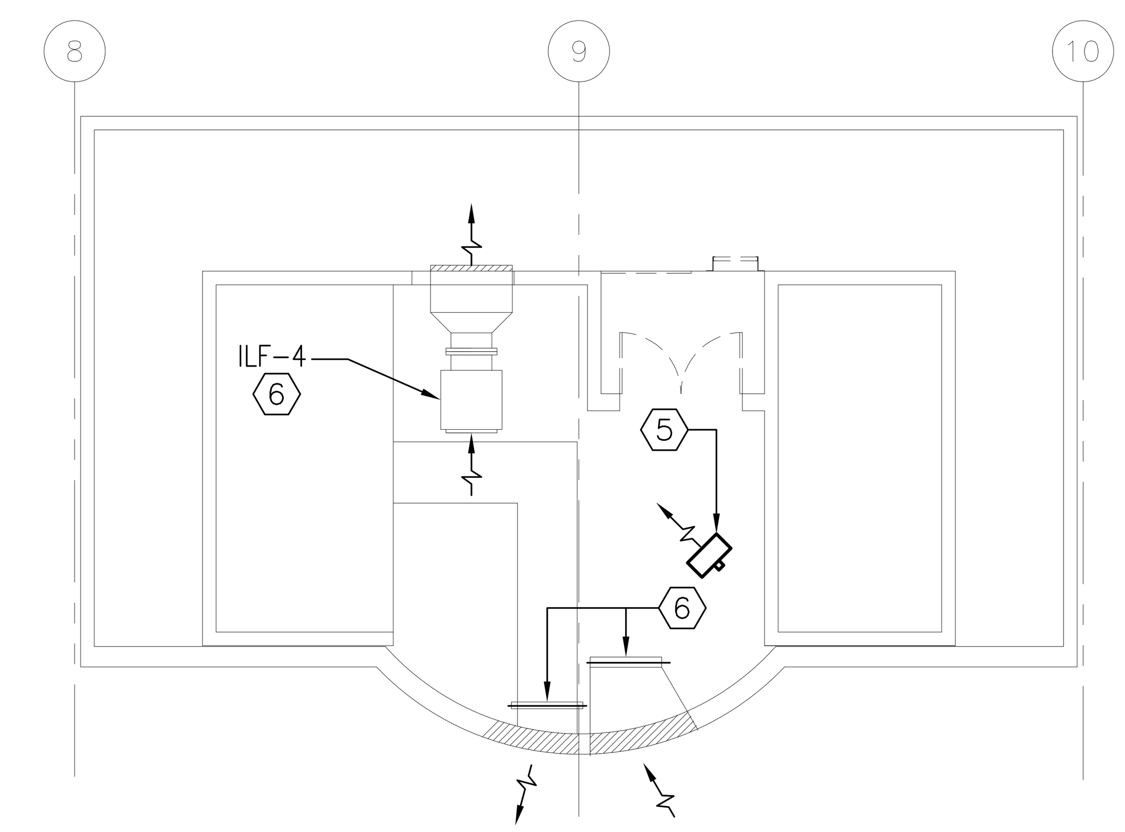
GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO AHU UNIT.
2. AHU UNIT AND INLINE EXHAUST FAN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING AND DUCTWORK. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
4. SEE DRAWING MD-403 FOR PIPING ORIGIN.
5. REMOVE EXISTING UNIT HEATER
6. CHECK WORKING CONDITION OF INLINE FAN AND DAMPERS. RESET CONTROLS FOR ALL EQUIPMENT AND MOTORIZED DAMPERS AT NEW SET POINT. SEE SHEET M-109.

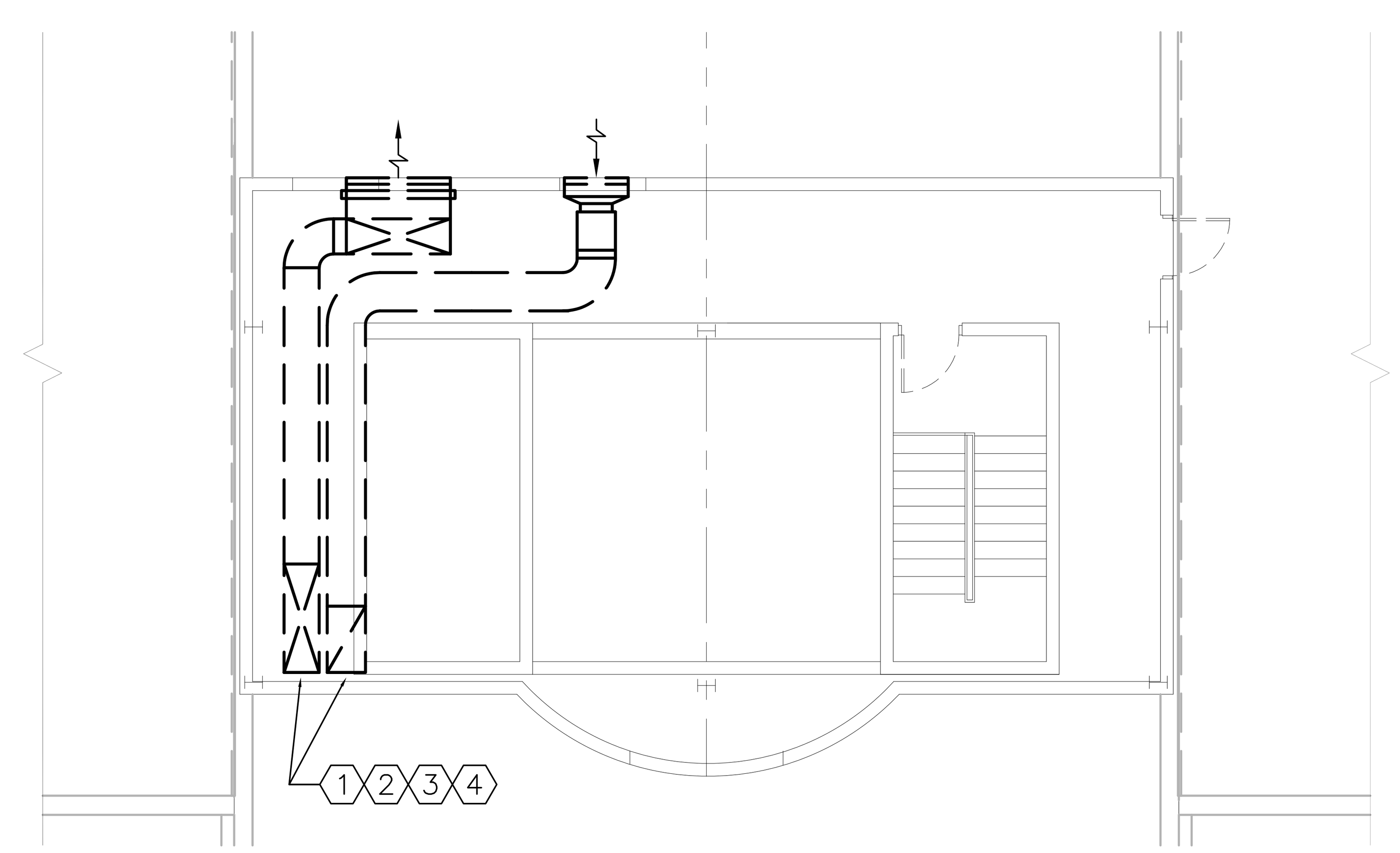


MACHINE ROOM FLOOR PLAN

SCALE: 1/8" = 1'-0"

TRUE NORTH

PLAN NORTH



PENTHOUSE FLOOR PLAN

SCALE: 1/8" = 1'-0"

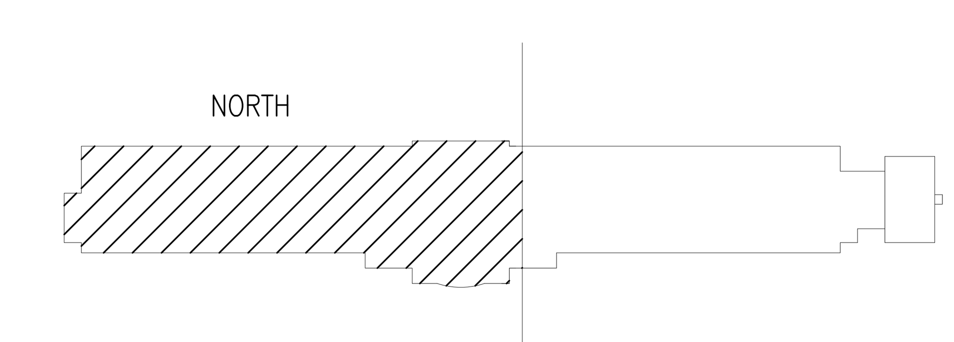
TRUE NORTH

PLAN NORTH

GRAPHIC SCALE

1/8" = 1'-0" 0 4' 8' 16'

KEY PLAN



SYM	DESCRIPTION	DATE	APPR

SEAL

A/E INFO

APPROVED

David W. Greenfield

FOR COMMANDER NAVFAC

ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO | DATE 06/18/2012

DES JLS | DRW JLS | CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NORTHEAST IPT

NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA

NAVSUBASE NEW LONDON | GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL PENTHOUSE DEMOLITION FLOOR PLANS

SCALE: AS NOTED

EPROJECT NO.: 1127117

CONSTR. CONTR. NO. N40085-12-R-1715

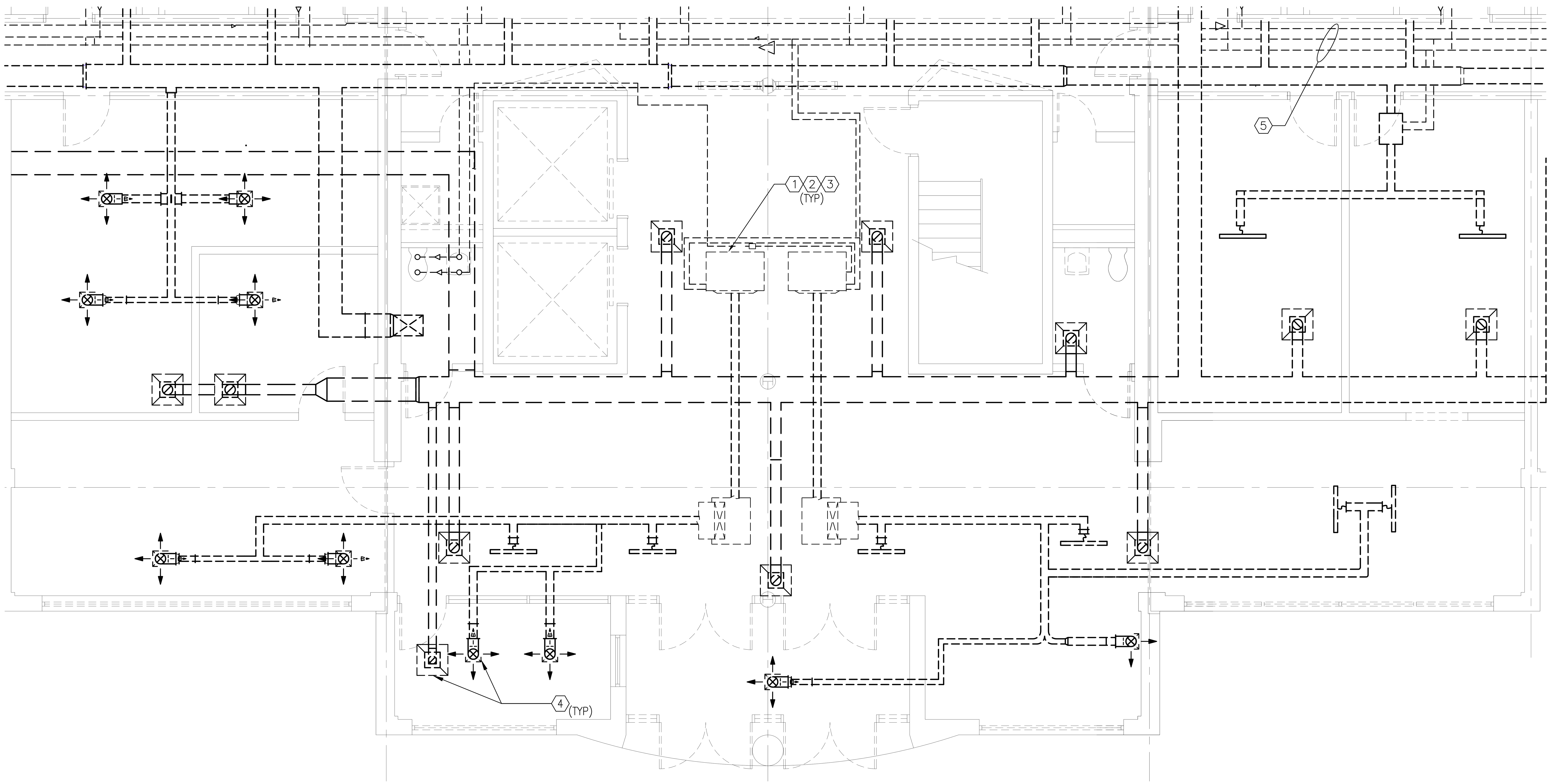
NAVFAC DRAWING NO. 12624706

SHEET 133 OF 206

MD109

DRAWFORM REVISION: 10 MARCH 2009

FILE NAME: P:\Connecticut\New_London\Buildings\B-534\2013_1127117_RM10-7303_Repair Sub A School BQ 534\B_Design\Drawings\Working\Mechanical\BLD0534-1127117-MD-109.dwg LAYOUT NAME: MD-109 PLOTTED: Thursday, July 12, 2012 - 10:44am USER: joseph.l.smith4



DEMOLITION FIRST FLOOR ENLARGED LOBBY PLAN

SCALE: 1/4" = 1'-0"   MD101   D3

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

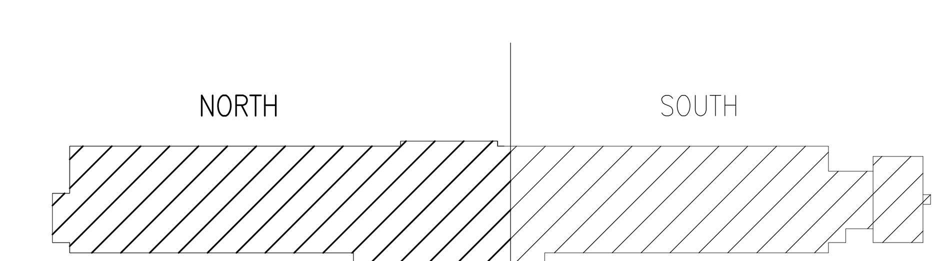
DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
4. REMOVE ALL DUCTWORK, DAMPERS, GRILLES, AND DIFFUSERS.
5. SEE DRAWING MD-403 FOR PIPING ORIGIN.

GRAPHIC SCALE

1/4" = 1'-0"   0 2' 4' 8'

KEY PLAN



APPROVED David Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON

SATISFACTORY TO   DATE 06/18/2012

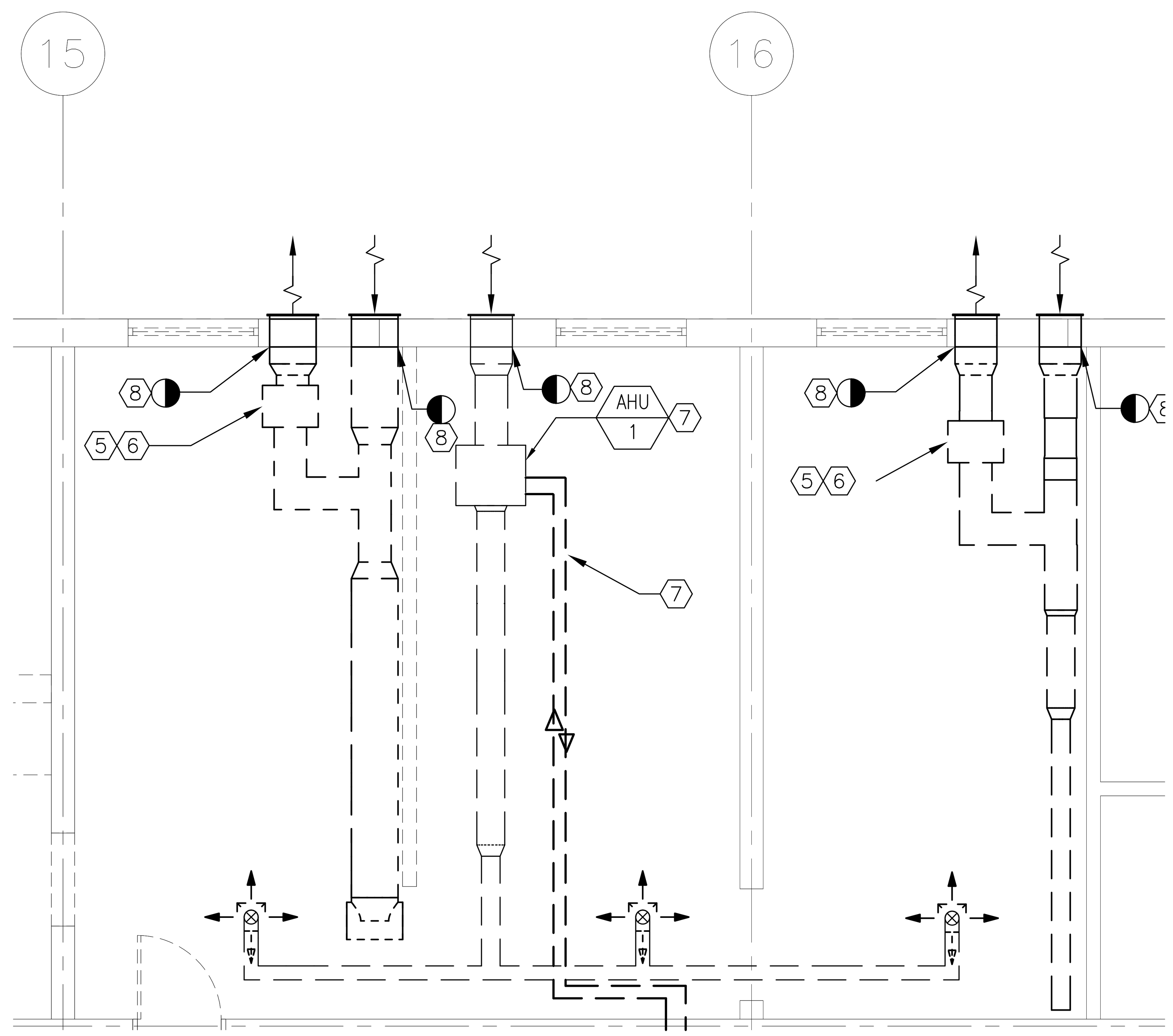
DES	JLS	DRW	MDS	CHK	JM

<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION SLW

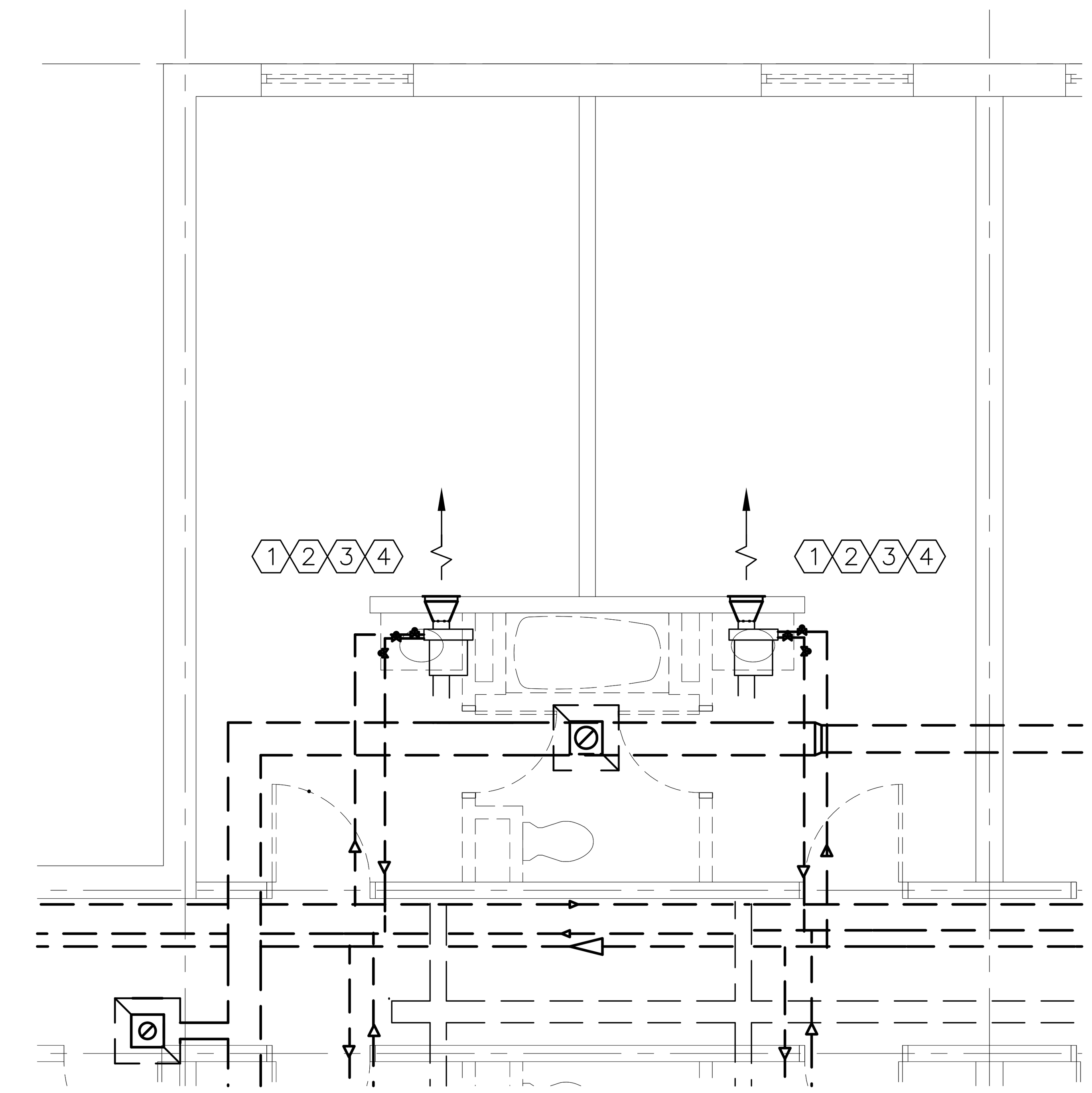
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NORTHEAST IPT   NAVAL STATION – NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON   GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
DEMOLITION PLAN – FIRST FLOOR – ENLARGED LOBBY

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624707
SHEET 134 OF 206
MD401

DRAWFORM REVISION: 10 MARCH 2009



DEMOLITION FIRST FLOOR ENLARGED LAUNDRY RM PLAN
SCALE: 1/4" = 1'-0" MD102 D1



DEMOLITION FIRST FLOOR SUITE PLAN (TYP)
SCALE: 1/4" = 1'-0" MD101 THRU MD108 D4

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.

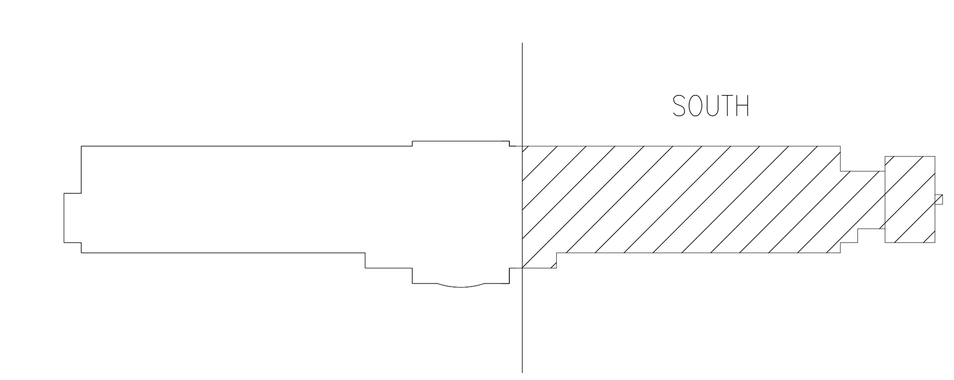
DEMOLITION KEYNOTES

1. REMOVE ALL RUN-OUT PIPING TO FAN COIL UNIT (TYPICAL)
2. FAN COIL UNIT AND PIPING ENCLOSURE TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING (TYP). ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
4. SEE DRAWING MD-403 FOR PIPING ORIGIN.
5. REMOVE ALL RUN-OUT PIPING TO INLINE EXHAUST FAN (TYPICAL)
6. INLINE FAN UNIT AND ENCLOSED PIPING TO BE REMOVED (TYP). ALL WALL AND FLOOR OPENINGS NOT TO BE REUSED TO BE PATCHED, SEALED, AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
7. REMOVE AHU, PIPING, AND DUCTWORK.
8. REMOVE EXISTING LOUVER. WALL PENETRATIONS TO BE REUSED.

GRAPHIC SCALE

1/4" = 1'-0" 0 2' 4' 8'

KEY PLAN



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FOR COMMANDER NAVFAC
ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON
SATISFACTORY TO DATE 06/18/2012
DES JLS DRW MDS CHK JM
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NORTHEAST IPT NAVAL STATION - NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
DEMOLITION PLAN - FIRST FLOOR - ENLARGED LAUNDRY RM & SUITES

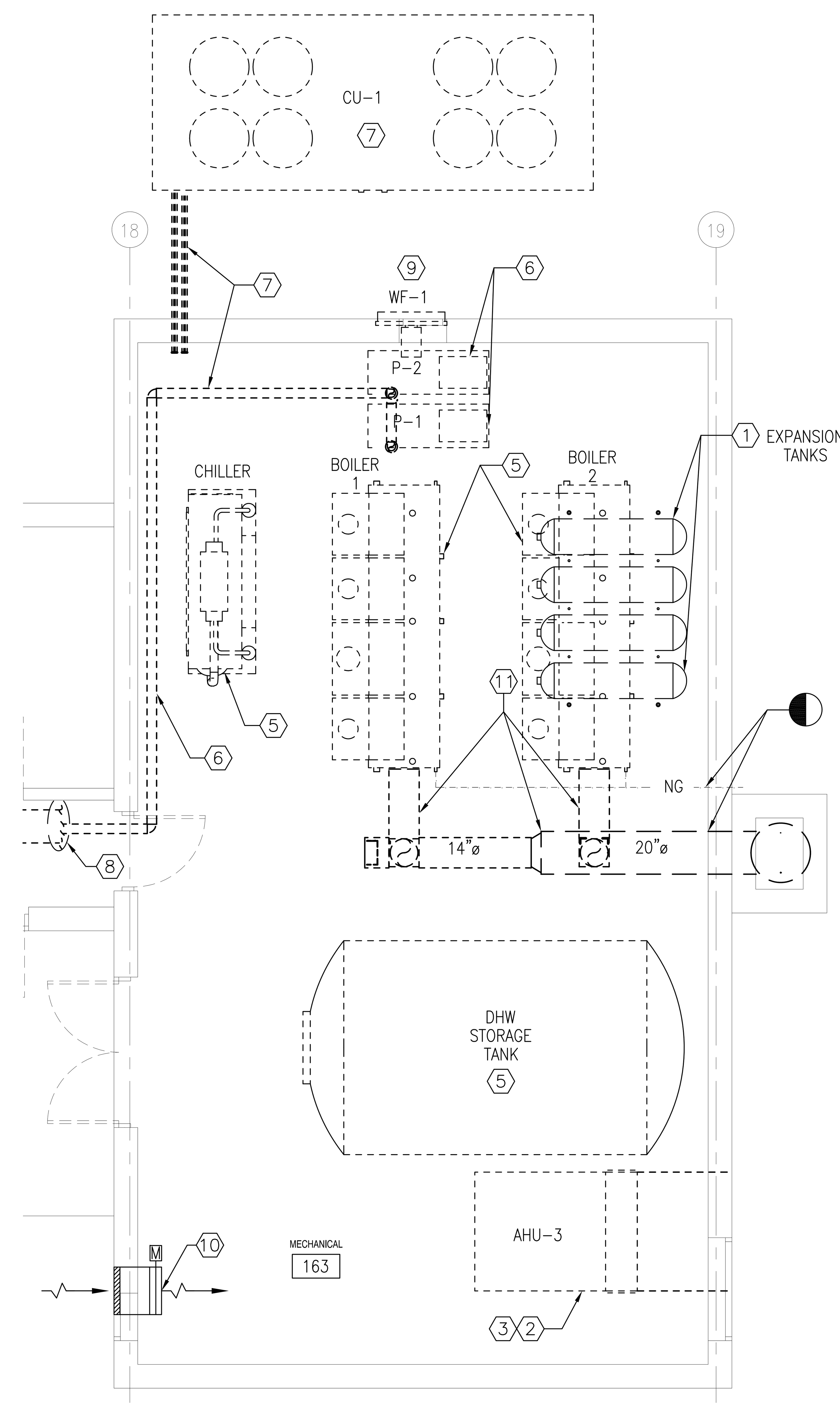
SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624708
SHEET 135 OF 206
MD402
DRAWFORM REVISION: 10 MARCH 2009

GENERAL SHEET NOTES

1. REMOVE ALL BRACKETS, HANGERS AND MOUNTING HARDWARE FOR MECHANICAL EQUIPMENT AND DUCTWORK TO BE REMOVED. PATCH AND PAINT BUILDING SURFACES TO REMAIN TO MATCH ADJACENT FINISHED SURFACES.
2. REMOVE ALL EXISTING FAN COIL UNITS AND ALL ASSOCIATED PIPING, DUCTWORK, AND ANY OTHER RELATED EQUIPMENT FOR THE OPERATION OF THE FAN COIL UNITS.
3. REMOVE ALL EXISTING DUCTWORK, GRILLES AND DIFFUSERS.

DEMOLITION KEYNOTES

1. REMOVE EXPANSION TANKS AND ALL RUN-OUT PIPING TO EXPANSION TANKS.
2. AIR HANDLER UNIT AND PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
3. REMOVE CONDENSATE DRAIN PIPING. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
4. EXISTING INSULATED HEATING/COOLING SUPPLY AND RETURN PIPING TO BE REMOVED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
5. REMOVE CHILLER, BOILERS, HOT WATER STORAGE TANK AND ALL OF THE ASSOCIATED PIPING. ALL CONCRETE PADS TO BE REUSED. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING ADJACENT FINISHES.
6. REMOVE ALL PUMPS AND THERE ASSOCIATED PIPING AND VALVES. CONCRETE PADS TO BE REUSED.
7. REMOVE ALL CONDENSER AND THE EXISTING CONCRETE PAD. REMOVE ALL ASSOCIATED PIPING. ALL WALL AND FLOOR OPENINGS NOT REUSED TO BE PATCHED, SEALED AND PAINTED TO MATCH EXISTING FINISHES.
8. PIPING CONTINUATION FROM SHEETS MD-101 THRU MD-108.
9. REPLACE EXISTING WF-1 WITH NEW WALL MOUNTED FAN, SEE M-502.
10. EXISTING 24" X 24" MOTOR OPERATED LOUVER AND DAMPER TO REMAIN. RESET FOR 1700 CFM.
11. REMOVE COMBUSTION DUCTWORK.



DEMOLITION PLAN FIRST FLOOR ENLARAGED MECHANICAL RM

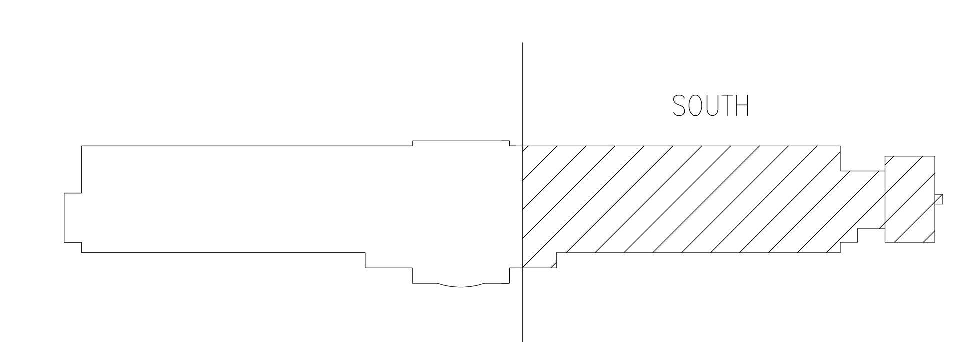
SCALE: 1/4" = 1'-0"

MD102 A8

GRAPHIC SCALE

1/4" = 1'-0" 0 2' 4' 8'

KEY PLAN



SYM | DESCRIPTION | DATE | APPR

SEAL

A/E INFO

APPROVED: David W. Grunfeld

FOR COMMANDER NAVFAC

ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO | DATE 06/18/2012

DES JLS | DRW MDS | CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON
GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

DEMOLITION PLAN - FIRST FLOOR - ENLARGED MECHANICAL RM

SCALE: AS NOTED

EPROJECT NO.: 1127117

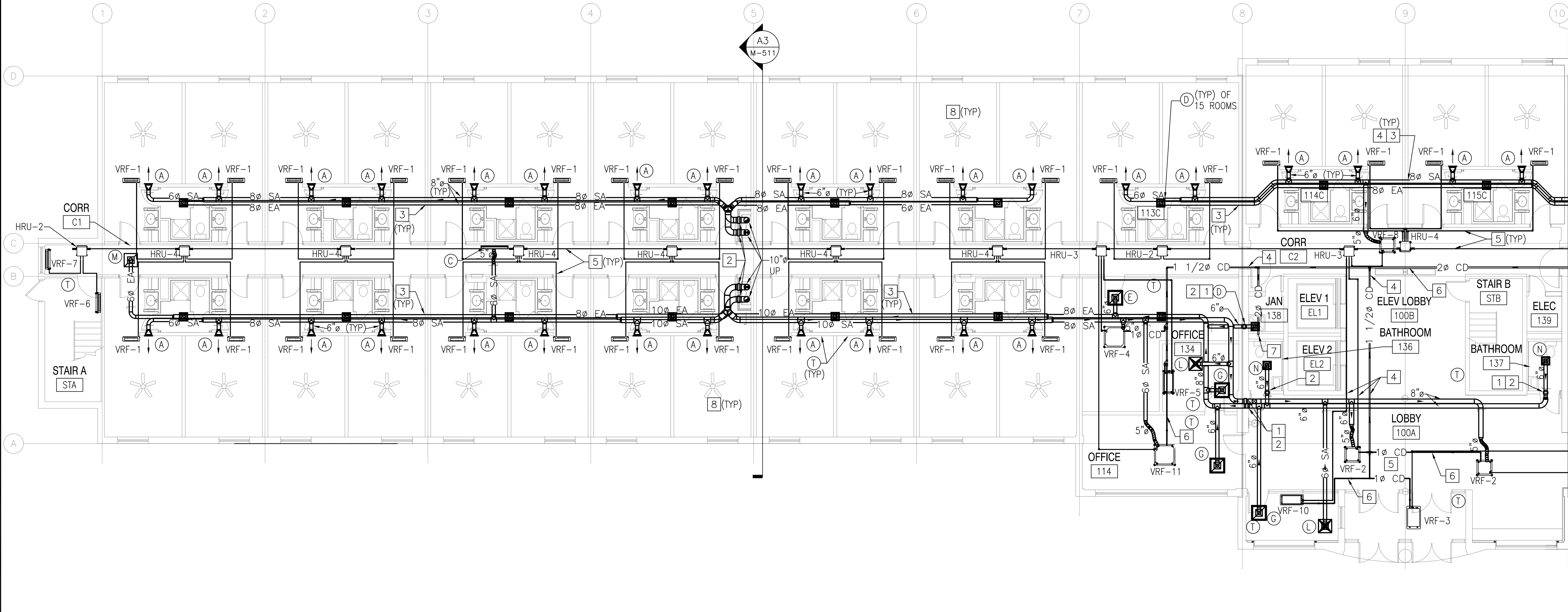
CONSTR. CONTR. NO.: N40085-12-R-1715

NAVFAC DRAWING NO.: 12624709

SHEET 136 OF 206

MD403

DRAWFORM REVISION: 10 MARCH 2009



DUCTWORK PLAN - FIRST FLOOR - NORTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
4. REUSE ALL EXISTING WALL OPENINGS, UNLESS OTHERWISE NOTED. REPAIR AND SEAL ALL OPENINGS NOT USED.
5. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
6. THIS IS A 3-PIPE VRF REFRIGERANT SYSTEM WHICH REQUIRES A MANUFACTURE CERTIFIED CONTRACTOR FOR INSTALLATION. PROVIDE AND INSTALL VRF PIPING PER MANUF'S. SPECIFICATIONS. INSTALL ALL SOLENOID VALVE KITS ON ALL PIPING PER MANUFACTURES RECOMMENDATIONS.
7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

NEW WORK KEYNOTES

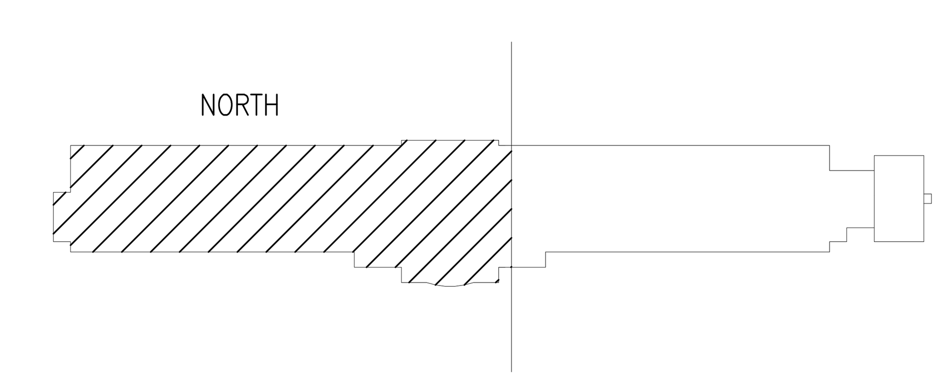
1. DUCTWORK IN THESE AREAS TO BE ROUTED BELOW THE STEEL BEAMS.
2. PROVIDE FIRE DAMPERS AND SEAL WALL PENETRATIONS TO FIRE RATING STATUS PER CODE. SEE DETAIL ON SHEET M-503 DETAIL B1.
3. WALL AREA TO BE CHECKED FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM. DUCTWORK TO BE STACKED.
4. WALL PENETRATIONS IN THIS AREA NEED TO BE SEALED WITH A 2 HR FIRE RATING SEALANT.
5. ALL VRF INSULATED REFRIGERANT PIPING TO BE ROUTED HIGH IN CEILING CAVITY.
6. ALL VRF CONDENSATE PIPING TO SLOPE 1/8".
7. CONDENSATE PIPE TO CONNECT TO SANITARY PIPING.
8. PROVIDE CEILING FAN AT 8 FEET ABOVE FINISHED FLOOR. SEE SHEET M-601. TYPICAL OF ALL FLOORS.

GRAPHIC SCALE

1/8" = 1'-0"

0 4' 8' 16'

KEY PLAN



APPROVED	David W. Greenfield
FOR COMMANDER NAVFAC	
ACTIVITY	PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON
SATISFACTORY TO	DATE 06/18/2012
DES JLS	DRW JLS / CHK JM
<<PM/DM>>	RS/NW
BRANCH MANAGER	
CHIEF ENG/ARCH	DWG
FIRE PROTECTION	SLW

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NORTHEAST IPT
NAVSUBASE NEW LONDON

NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND – MID-ATLANTIC
NAVAL STATION – NORFOLK, VIRGINIA
GROTON, CT

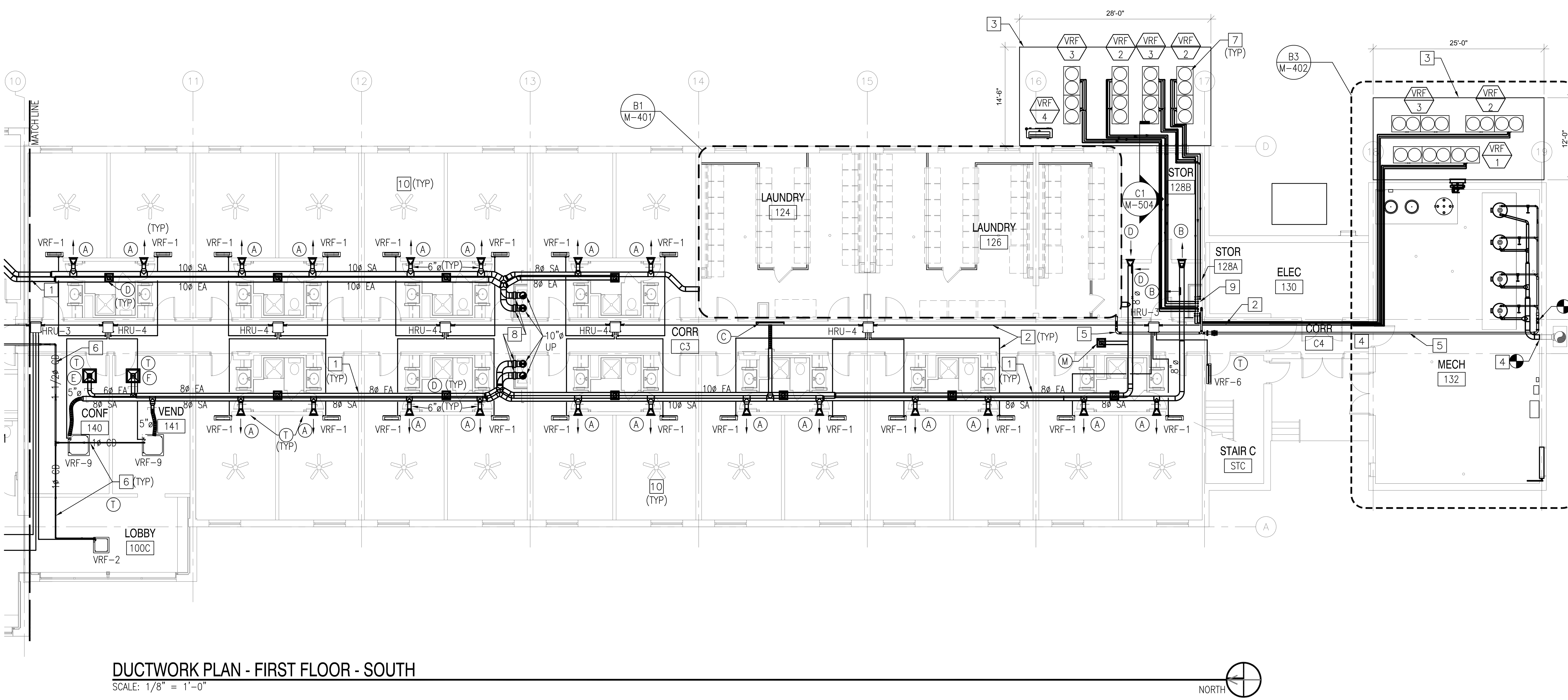
REPAIRS TO SUBMARINE A SCHOOL BQ 534

PIPING & DUCTWORK PLAN – FIRST FLOOR – NORTH

SCALE:	AS NOTED
EPROJECT NO.:	1127117
CONSTR. CONTR. NO.	N40085-12-R-1715
NAVFAC DRAWING NO.	12624710
SHEET	137 OF 206

M-101

DRAWFORM REVISION: 10 MARCH 2009



DUCTWORK PLAN - FIRST FLOOR - SOUTH

SCALE: 1/8" = 1'-0"

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
4. REUSE ALL EXISTING WALL OPENINGS, UNLESS OTHERWISE NOTED. REPAIR AND SEAL ALL OPENINGS NOT USED.
5. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
6. THIS IS A 3-PIPE VRF REFRIGERANT SYSTEM WHICH REQUIRES A MANUFACTURE CERTIFIED CONTRACTOR FOR INSTALLATION. PROVIDE AND INSTALL VRF PIPING PER MANUF'S. SPECIFICATIONS. INSTALL ALL SOLENOID VALVE KITS ON ALL PIPING PER MANUFACTURES RECOMMENDATIONS.
7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

NEW WORK KEYNOTES

1. WALL AREA TO BE CHECK FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM. DUCTWORK TO BE STACKED.
2. ALL VRF INSULATED REFRIGERANT PIPING TO BE ROUTED HIGH IN CEILING CAVITY.
3. PROVIDE NEW 4 INCH CONCRETE SLAB.
4. NEW GAS VENT WITH TYPE B INSULATION.
5. NEW GAS PIPING TO BE ROUTED HIGH IN CEILING CAVITY.
6. VRF CONDENSATE DRAIN PIPING TO BE SLOPED DOWN AT 1/8" PER FOOT.
7. SEE DETAIL A2 ON SHEET M-502 FOR VRF OUTDOOR UNITS
8. PROVIDING FIRE DAMPERS AND SEAL WALL PENETRATIONS TO FIRE RATING STATUS PER CODE. SEE DETAIL ON SHEET M-503 DETAIL B1.
9. WALL PENETRATIONS IN THIS AREA NEED TO BE SEALED WITH A 2 HR FIRE RATING SEALANT.
10. PROVIDE CEILING FAN AT 8 FEET ABOVE FINISHED FLOOR, SEE SHEET M-601, (TYPICAL OF ALL FLOORS).

GRAPHIC SCALE

1/8" = 1'-0" 0 4' 8' 16'

KEY PLAN

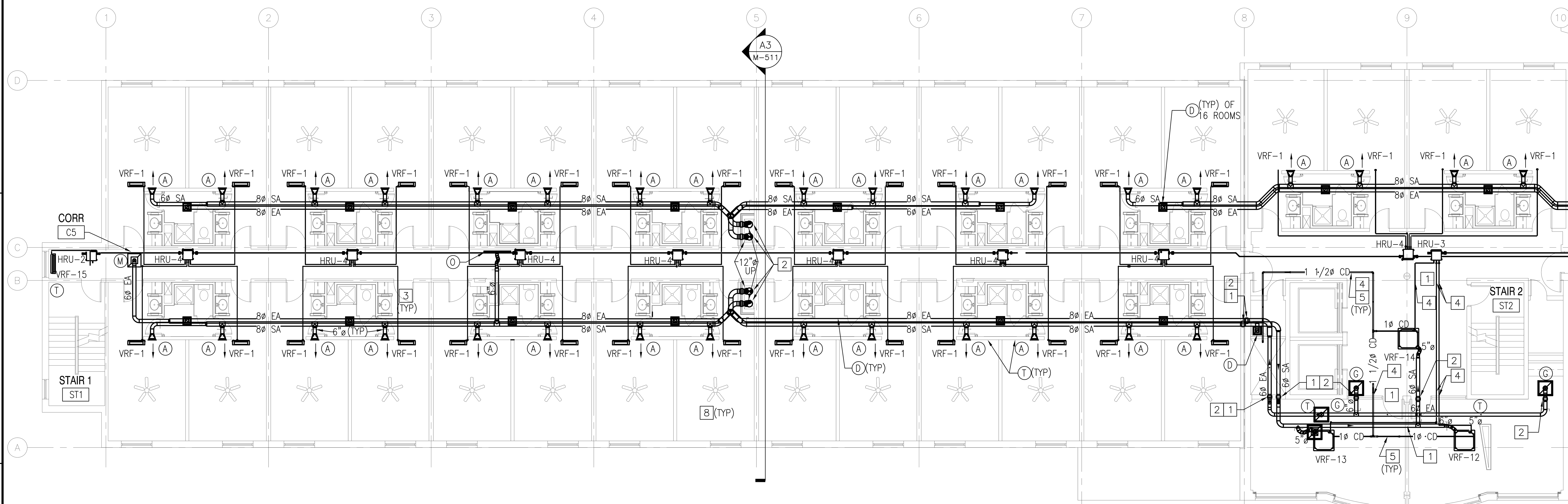
SOUTH

APPROVED: David W. Greenfield
FOR COMMANDER NAVFAC
ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON
SATISFACTORY TO — DATE 06/18/2012
DES JLS | DRW JLS | CHK JM
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NORTHEAST IPT — NAVAL STATION - NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON — GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
PIPING & DUCTWORK PLAN - FIRST FLOOR - SOUTH

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624711
SHEET 138 OF 206
M-102

DRAWFORM REVISION: 10 MARCH 2009



DUCTWORK PLAN - SECOND FLOOR - NORTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
4. REUSE ALL EXISTING WALL OPENINGS, UNLESS OTHERWISE NOTED. REPAIR AND SEAL ALL OPENINGS NOT USED.
5. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
6. THIS IS A 2-PIPE VRF REFRIGERANT SYSTEM WHICH REQUIRES A MANUFACTURE CERTIFIED CONTRACTOR FOR INSTALLATION. PROVIDE AND INSTALL VRF PIPING PER MANUF'S. SPECIFICATIONS. INSTALL ALL SOLENOID VALVE KITS ON ALL PIPING PER MANUFACTURES RECOMMENDATIONS.
7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-?.

NEW WORK KEYNOTES

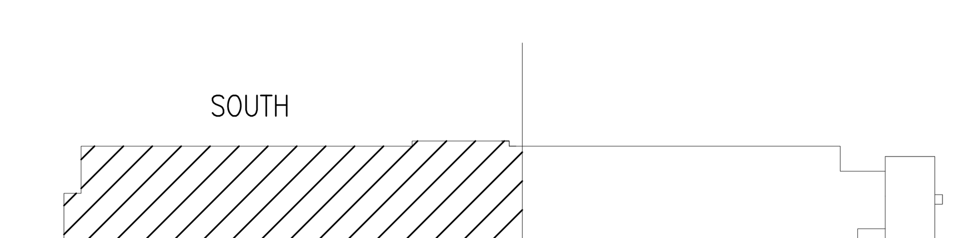
1. DUCTWORK IN THESE AREAS TO BE ROUTED BELOW THE STEEL BEAMS.
2. PROVIDE FIRE DAMPERS AND SEAL WALL PENETRATIONS TO FIRE RATING STATUS PER CODE. SEE DETAIL ON SHEET M-503.
3. WALL AREA TO BE CHECKED FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM. DUCTWORK TO BE STACKED.
4. WALL PENETRATIONS IN THIS AREA NEED TO BE SEALED WITH A 2 HR FIRE RATING SEALANT.
5. VRF CONDENSATE PIPING TO BE SLOPED DOWN 1/8" PER FOOT.
6. PROVIDE CEILING FAN AT 8 FEET ABOVE FINISHED FLOOR. SEE SHEET M-601. TYPICAL OF ALL FLOORS.

GRAPHIC SCALE

1/8" = 1'-0"

0 4' 8' 16'

KEY PLAN



APPROVED FOR COMMANDER NAVFAC

ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO — DATE 06/18/2012

DES JLS | DRW JLS | CHK JLS

PM/DM RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON
GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
PIPING & DUCTWORK PLAN - SECOND FLOOR - NORTH

SCALE: AS NOTED

EPROJECT NO.: 1127117

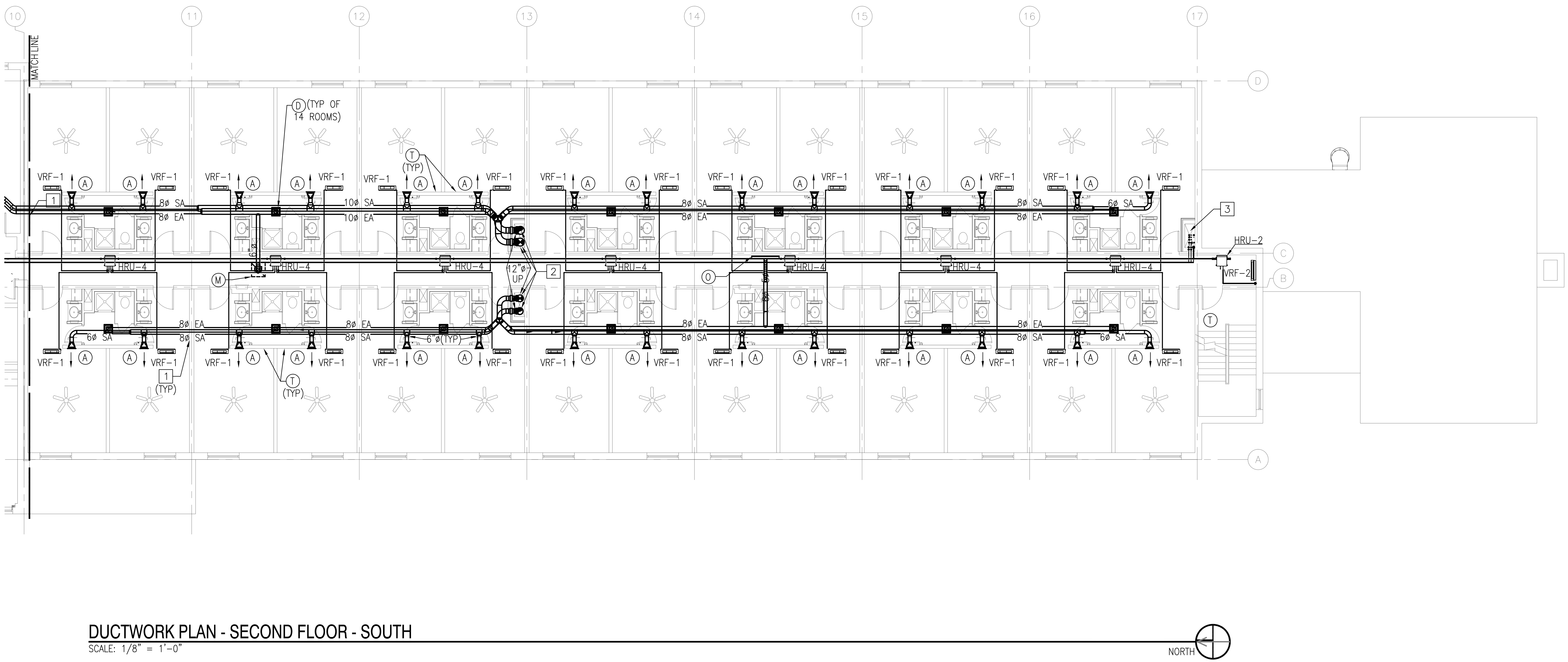
CONSTR. CONTR. NO.: N40085-12-R-1715

NAVFAC DRAWING NO.: 12624712

SHEET 139 OF 206

M-103

DRAWFORM REVISION: 10 MARCH 2009



DUCTWORK PLAN - SECOND FLOOR - SOUTH

SCALE: 1/8" = 1'-0"

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
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7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

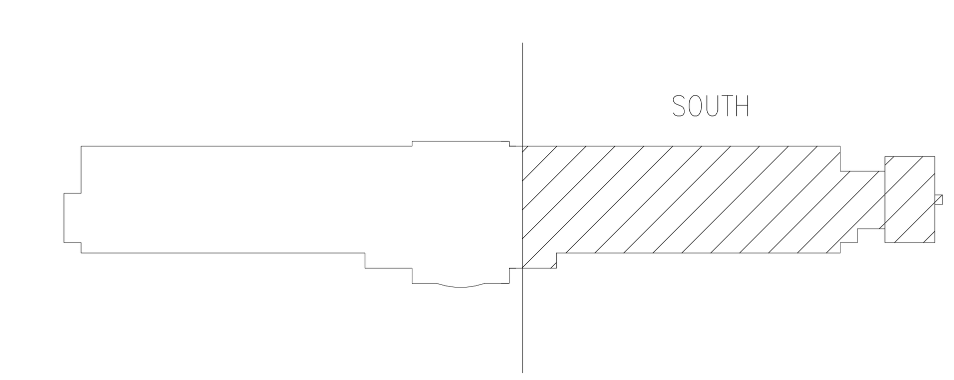
NEW WORK KEYNOTES

1. WALL AREA TO BE CHECK FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM. DUCTWORK TO BE STACKED.
2. PROVIDE FIRE DAMPERS AND SEAL WALL PENETRATIONS TO FIRE RATING STATUS PER CODE. SEE DETAIL ON SHEET M-503 DETAIL B1.
3. WALL PENETRATIONS IN THIS AREA NEED TO BE SEALED WITH A 2 HR FIRE RATING SEALANT.

GRAPHIC SCALE

1/8" = 1'-0" 0 4' 8' 16'

KEY PLAN

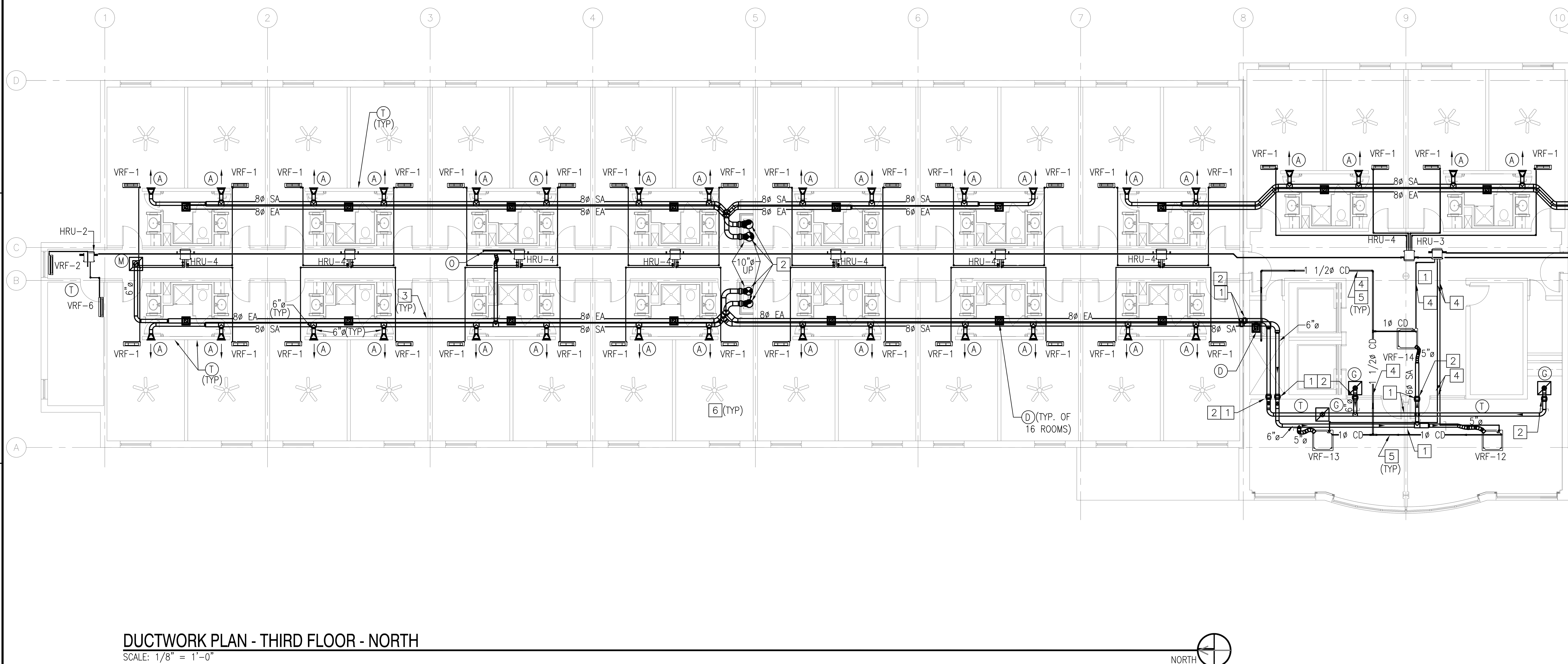


APPROVED: David W. Greenfield
FOR COMMANDER NAVFAC
ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON
SATISFACTORY TO / DATE 06/18/2012
DES JLS | DRW JLS | CHK JLS
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION – NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
PIPING & DUCTWORK PLAN – SECOND FLOOR – SOUTH

SCALE:	AS NOTED
EPROJECT NO.:	1127117
CONSTR. CONTR. NO.	N40085-12-R-1715
NAVFAC DRAWING NO.	12624713
SHEET	140 OF 206
	M-104

DRAWFORM REVISION: 10 MARCH 2009



DUCTWORK PLAN - THIRD FLOOR - NORTH

SCALE: 1/8" = 1'-0"

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
4. REUSE ALL EXISTING WALL OPENINGS, UNLESS OTHERWISE NOTED. REPAIR AND SEAL ALL OPENINGS NOT USED.
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7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

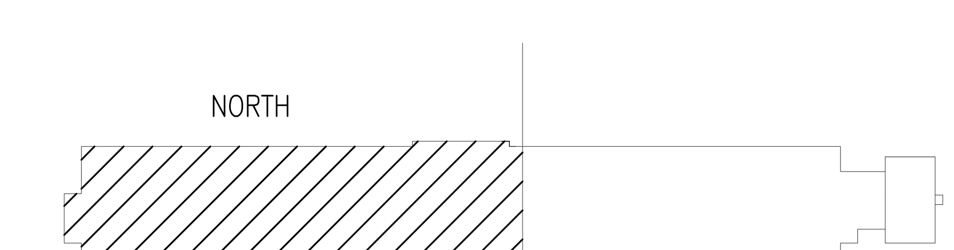
NEW WORK KEYNOTES

1. DUCTWORK IN THESE AREAS TO BE ROUTED BELOW THE STEEL BEAMS.
2. PROVIDE FIRE DAMPERS AND SEAL WALL PENETRATIONS TO FIRE RATING STATUS PER CODE. SEE DETAIL ON SHEET M-503 DETAIL B1.
3. WALL AREA TO BE CHECK FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM. DUCTWORK TO BE STACKED.
4. WALL PENETRATIONS IN THIS AREA NEED TO BE SEALED WITH A 2 HR FIRE RATING SEALANT.
5. VRF CONDENSATE DRAIN PIPING TO BE SLOPED DOWN 1/8" PER FOOT.
6. PROVIDE CEILING FAN AT 8 FEET ABOVE FINISHED FLOOR. SEE SHEET M-601. TYPICAL OF ALL FLOORS.

GRAPHIC SCALE

1/8" = 1'-0" 0 4' 8' 16'

KEY PLAN



APPROVED David Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO DATE 06/18/2012

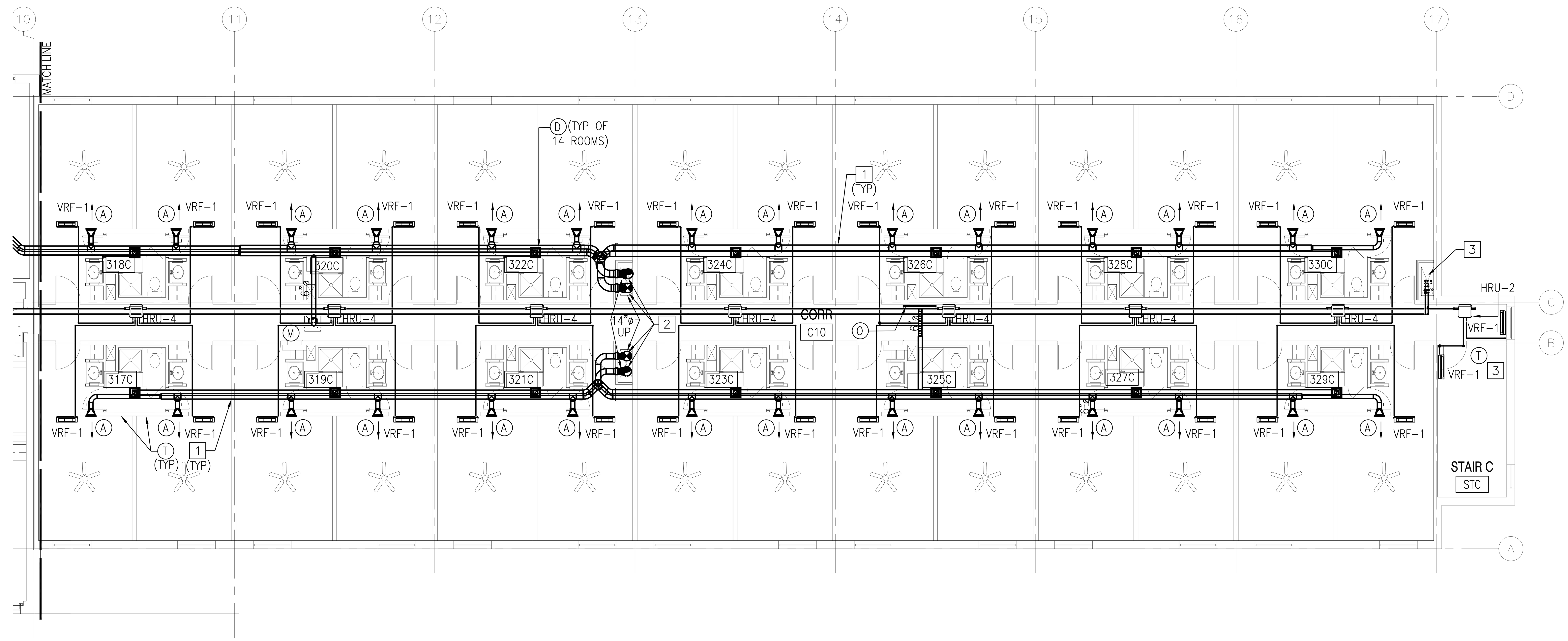
DES	JLS	DRW	JLS	CHK	JM

<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION SLW

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NORTHEAST IPT
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
PIPING & DUCTWORK PLAN - THIRD FLOOR - NORTH

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624714
SHEET 141 OF 206
M-105

DRAWFORM REVISION: 10 MARCH 2009



DUCTWORK PLAN - THIRD FLOOR - SOUTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
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5. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
6. THIS IS A 3-PIPE VRF REFRIGERANT SYSTEM WHICH REQUIRES A MANUFACTURE CERTIFIED CONTRACTOR FOR INSTALLATION. PROVIDE AND INSTALL VRF PIPING PER MANUF'S. SPECIFICATIONS. INSTALL ALL SOLENOID VALVE KITS ON ALL PIPING PER MANUFACTURES RECOMMENDATIONS.
7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

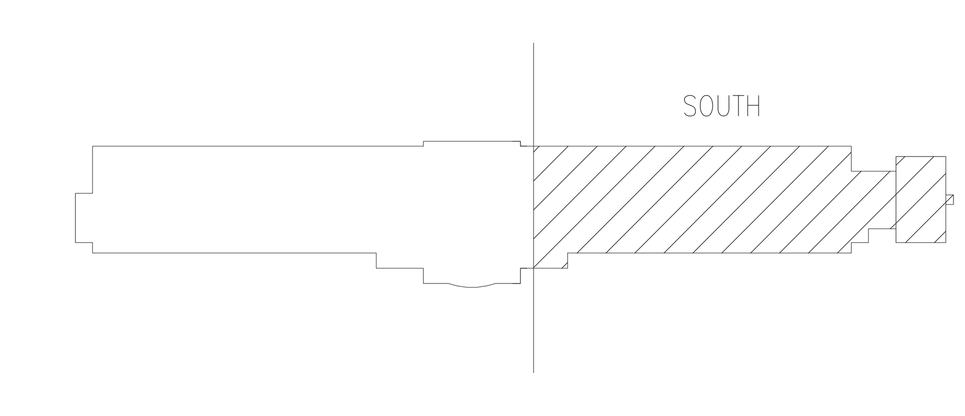
NEW WORK KEYNOTES

1. WALL AREA TO BE CHECK FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM. DUCTWORK TO BE STACKED.
2. PROVIDE FIRE DAMPERS AND SEAL WALL PENETRATIONS TO FIRE RATING STATUS PER CODE. SEE DETAIL ON SHEET M-503 DETAIL B1.
3. WALL PENETRATIONS IN THIS AREA NEED TO BE SEALED WITH A 2 HR FIRE RATING SEALANT.

GRAPHIC SCALE

1/8" = 1'-0" 0 4' 8' 16'

KEY PLAN



SYM | DESCRIPTION | DATE | APPR

NAVFAC

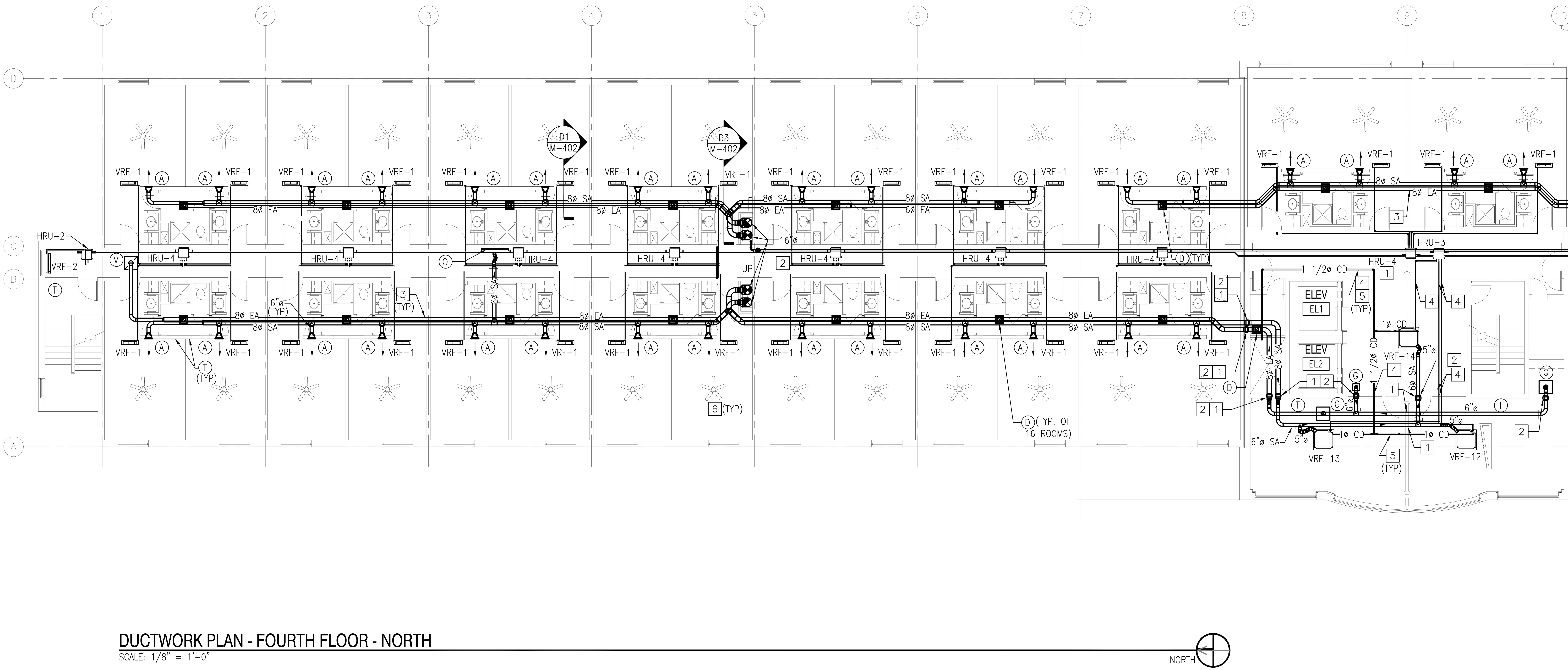
A/E INFO

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON
SATISFACTORY TO DATE 06/18/2012
DES JLS DRW JLS CHK JM
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
PIPING & DUCTWORK PLAN - THIRD FLOOR - SOUTH

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624715
SHEET 142 OF 206
M-106

DRAWFORM REVISION: 10 MARCH 2009



DUCTWORK PLAN - FOURTH FLOOR - NORTH

SCALE: 1/8" = 1'-0"

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
4. REUSE ALL EXISTING WALL OPENINGS, UNLESS OTHERWISE NOTED. REPAIR AND SEAL ALL OPENINGS NOT USED.
5. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
6. THIS IS A 3-PIPE VRF REFRIGERANT SYSTEM WHICH REQUIRES A MANUFACTURES CERTIFIED CONTRACTOR FOR INSTALLATION. PROVIDE AND INSTALL VRF PIPING PER MANUF'S. SPECIFICATIONS. INSTALL ALL SOLENOID VALVE KITS ON ALL PIPING PER MANUFACTURES RECOMMENDATIONS.
7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

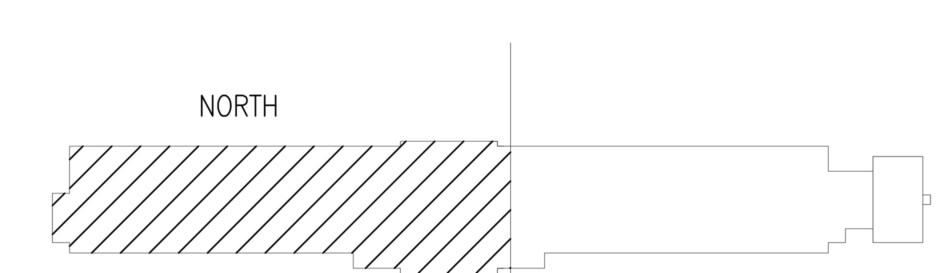
NEW WORK KEYNOTES

1. DUCTWORK IN THESE AREAS TO BE ROUTED BELOW THE STEEL BEAMS.
2. PROVIDE FIRE DAMPERS AND SEAL WALL PENETRATIONS TO FIRE RATING STATUS PER CODE. SEE DETAIL ON SHEET M-503 DETAIL B1.
3. WALL AREA TO BE CHECK FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM. DUCTWORK TO BE STACKED PER DETAIL A6 ON SHEET M-40?.
4. WALL PENETRATIONS IN THIS AREA NEED TO BE SEALED WITH A 2 HR FIRE RATING SEALANT.
5. VRF CONDENSATE DRAIN PIPING TO BE SLOPED DOWN 1/8" PER FOOT.
6. PROVIDE CEILING FAN AT 8 FEET ABOVE FINISHED FLOOR. SEE SHEET M-601. TYPICAL OF ALL FLOORS.

GRAPHIC SCALE

1/8" = 1'-0"   0 4' 8' 16'

KEY PLAN



NAVFAC

APPROVED: David W. Greenfield
FOR COMMANDER NAVFAC

ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON
SATISFACTORY TO — DATE 06/18/2012

DES	JLS	DRW	JLS	CHK	JM
<<PM/DM>>	RS/NW				
BRANCH MANAGER					
CHIEF ENG/ARCH	DWG				
FIRE PROTECTION	S/W				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NORTHEAST IPT — NAVAL STATION - NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON — GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

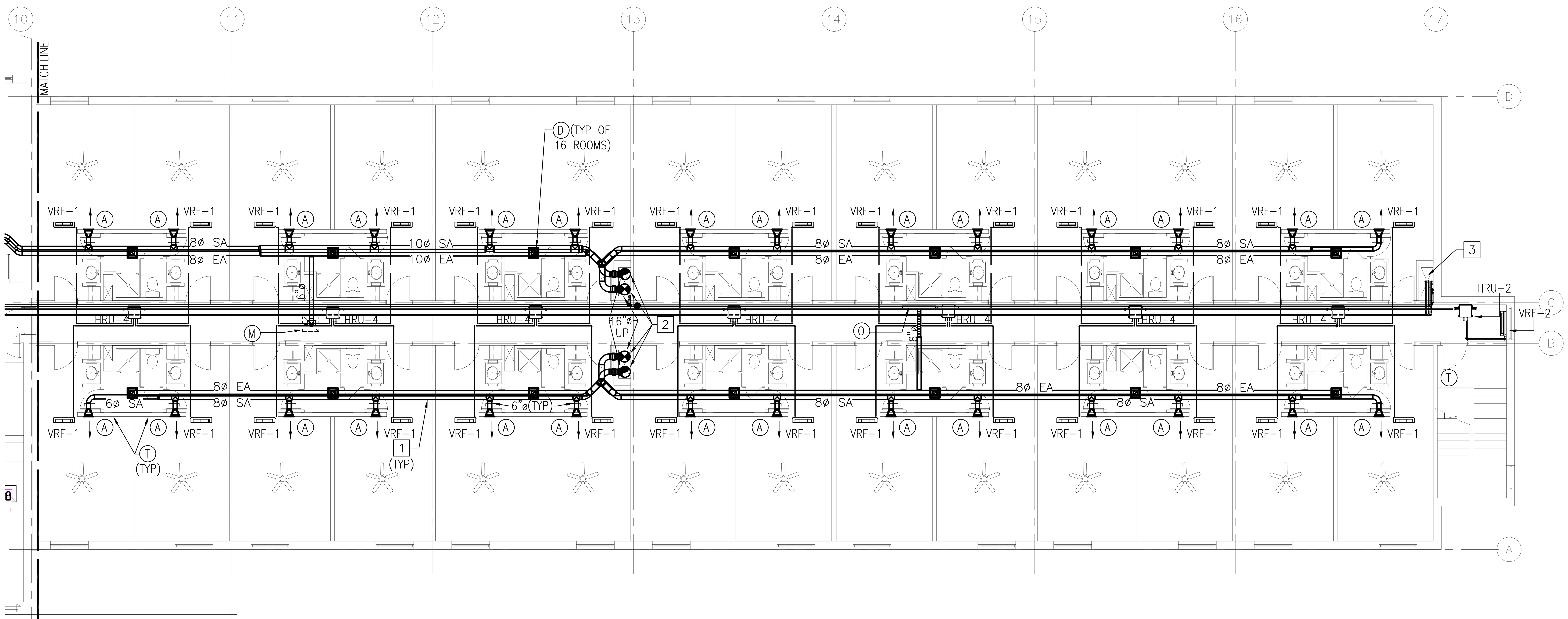
PIPING & DUCTWORK PLAN - FOURTH FLOOR - NORTH

SCALE:	AS NOTED
EPROJECT NO.:	1127117
CONSTR. CONTR. NO.	N40085-12-R-1715
NAVFAC DRAWING NO.	12624716
SHEET	143 OF 206

M-107

DRAWFORM REVISION: 10 MARCH 2009

FILE NAME: P:\Connecticut\New_London\Buildings\B-534\2013_1127117_RM10-7303_Repair Sub A School BQ 534\B_Design\Drawings\Working\Mechanical\BLDG534-1127117-M-107.dwg LAYOUT NAME: M-107 PLOTTED: Wednesday, July 11, 2012 - 10:06am USER: marcus.d.smith



DUCTWORK PLAN - FOURTH FLOOR - SOUTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
4. REUSE ALL EXISTING WALL OPENINGS, UNLESS OTHERWISE NOTED. REPAIR AND SEAL ALL OPENINGS NOT USED.
5. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
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7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

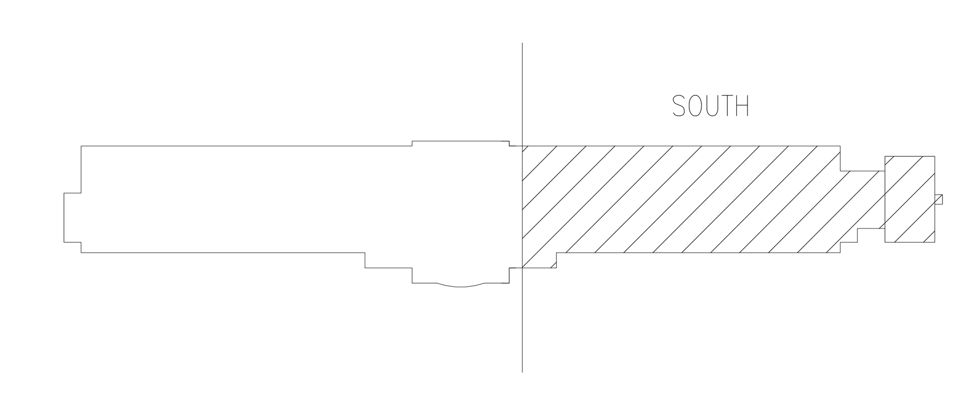
NEW WORK KEYNOTES

1. WALL AREA TO BE CHECK FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM. DUCTWORK TO BE STACKED.
2. PROVIDE FIRE DAMPERS AND SEAL WALL PENETRATIONS TO FIRE RATING STATUS PER CODE. SEE DETAIL ON SHEET M-503 DETAIL B1.
3. WALL PENETRATIONS IN THEIS AREA NEED TO BE SEALED WITH A 2 HR FIRE RATING SEALANT.

GRAPHIC SCALE

1/8" = 1'-0" — 0 4' 8' 16'

KEY PLAN



APPROVED: David W. Greenfield
FOR COMMANDER NAVFAC

ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO — DATE 06/18/2012

DES JLS | DRW JLS | CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

PIPING & DUCTWORK PLAN - FOURTH FLOOR - SOUTH

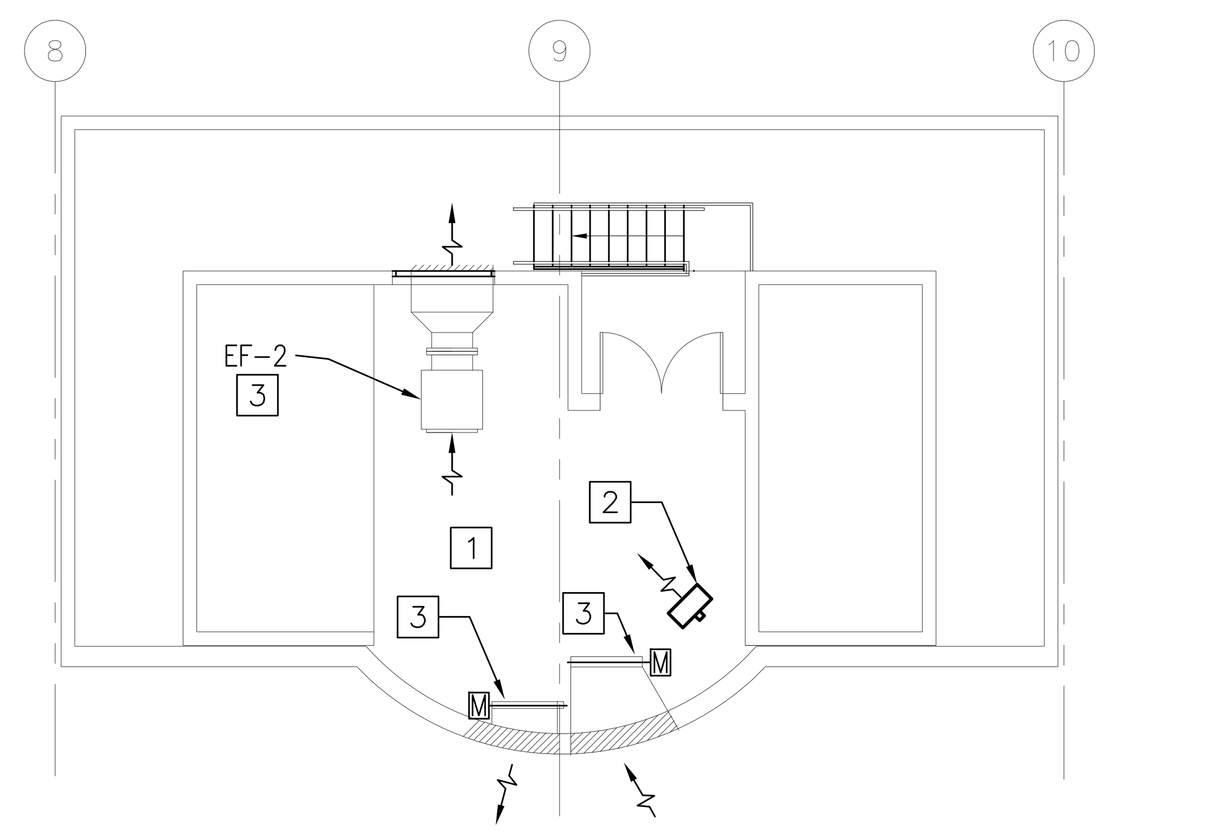
SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO.: N40085-12-R-1715
NAVFAC DRAWING NO.: 12624717
SHEET 144 OF 206

M-108

DRAWFORM REVISION: 10 MARCH 2009

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
4. REUSE ALL EXISTING WALL OPENINGS, UNLESS OTHERWISE NOTED. REPAIR AND SEAL ALL OPENINGS NOT USED.
5. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
6. THIS IS A 3-PIPE VRF REFRIGERANT SYSTEM WHICH REQUIRES A MANUFACTURE CERTIFIED CONTRACTOR FOR INSTALLATION. PROVIDE AND INSTALL VRF PIPING PER MANUF'S. SPECIFICATIONS. INSTALL ALL SOLENOID VALVE KITS ON ALL PIPING PER MANUFACTURES RECOMMENDATIONS.
7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. MANUFACTURE PROVIDE A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.



ELEVATOR MACHINE ROOM FLOOR PLAN

SCALE: 1/8" = 1'-0"

NORTH

NEW WORK KEYNOTES

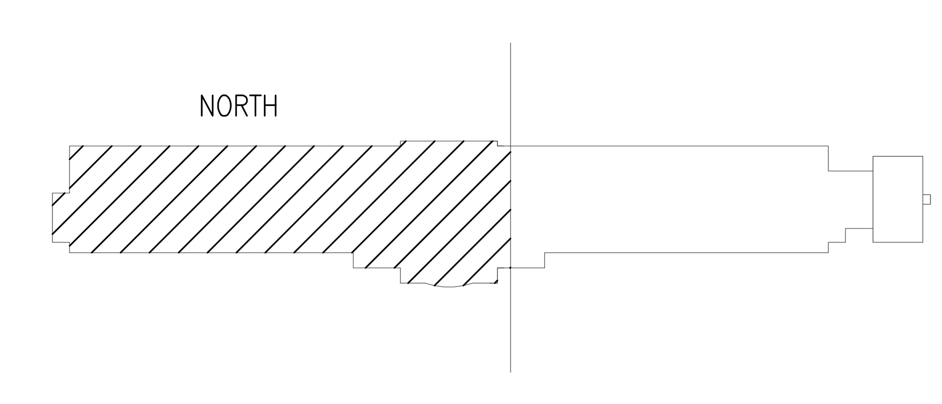
1. ALL EXISTING EQUIPMENT, DUCTWORK, AND ASSOCIATED PIPING TO BE REMOVED. SEE SHEET MD-109 AND ARCHITECTURAL FOR REFERENCE.
2. NEW ELECTRIC UNIT HEATER. SEE SHEET M-601 AND SHEET M-503
3. CHECK ALL MOTORIZED DAMPERS AND INLINE EXHAUST FAN FOR PROPER WORKING ORDER AND RESET CONTROLS TO NEW REQUIREMENTS. SEE SHEET M-703.

GRAPHIC SCALE

1/8" = 1'-0"

0 4' 8' 16'

KEY PLAN



SYM	DESCRIPTION	DATE	APPR

SEAL

A/E INFO

APPROVED David W. Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO DATE 06/18/2012

DES JLS DRW JLS CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION SLW

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NORTHEAST IPT
NAVSUBASE NEW LONDON
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
GROTON, CT

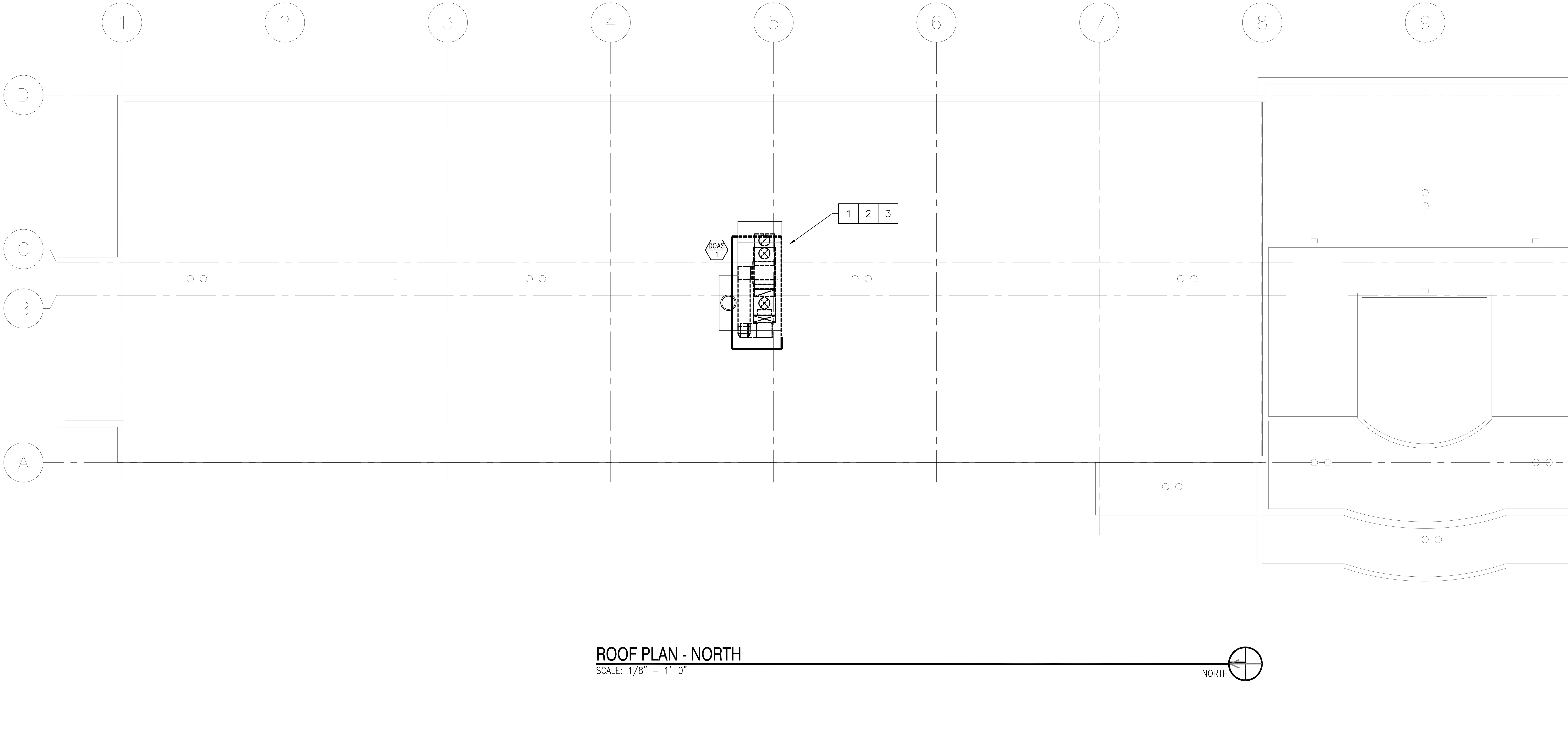
REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL PENTHOUSE FLOOR PLANS

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624718
SHEET 145 OF 206

M-109

DRAWFORM REVISION: 10 MARCH 2009



ROOF PLAN - NORTH

SCALE: 1/8" = 1'-0"

NORTH

GENERAL SHEET NOTES

1. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS AT ALL LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
2. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
3. PROVIDE DISCONNECTS FOR CONDENSER UNIT, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.

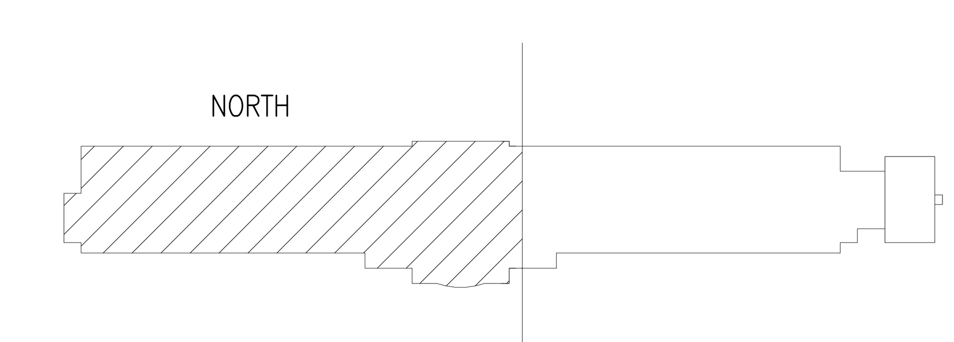
NEW WORK KEYNOTES

1. DOAS-1 TO BE MOUNTED ON A CUSTOM FITTED CURB BY THE MANUFACTURER. SHEET S-503 DETAIL C2 FOR DUCTWORK SHAFT OPENING.
2. DOAS-1 SHALL HAVE COASTAL PROTECTIVE COATING PER THE SPECIFICATIONS.
3. INSULATE EXTERIOR OF ALL DUCTWORK.

GRAPHIC SCALE

1/8" = 1'-0"   0   4'   8'   16'

KEY PLAN



SYM | DESCRIPTION | DATE | APPR

NAVFAC

SEAL

A/E INFO

APPROVED David W. Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO   DATE 06/18/2012

DES JLS   DRW JLS   CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON   GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

ROOF PLAN - NORTH

SCALE: AS NOTED

EPROJECT NO.: 1127117

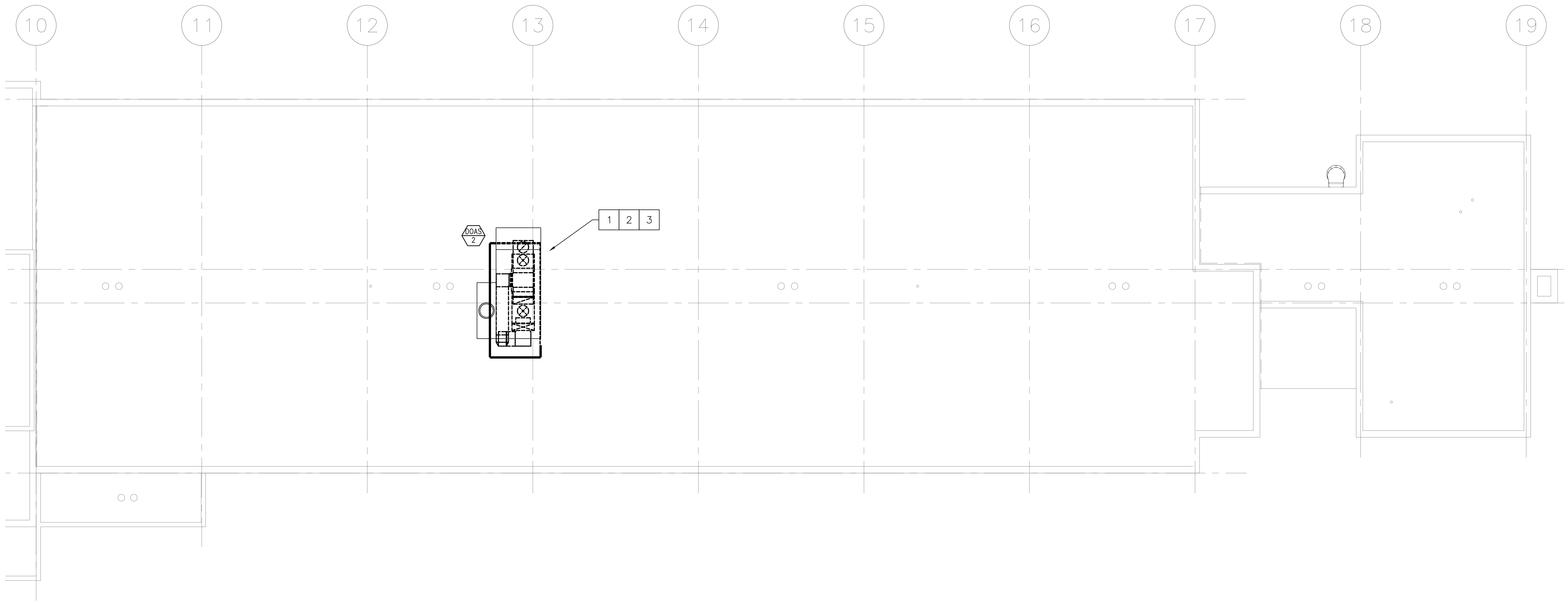
CONSTR. CONTR. NO. N40085-12-R-1715

NAVFAC DRAWING NO. 12624719

SHEET 146 OF 206

M-110

DRAWFORM REVISION: 10 MARCH 2009



ROOF PLAN - SOUTH

SCALE: 1/8" = 1'-0"

GENERAL SHEET NOTES

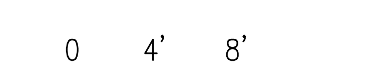
1. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS AT ALL LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
2. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
3. PROVIDE DISCONNECTS FOR CONDENSER UNIT, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.

NEW WORK KEYNOTES

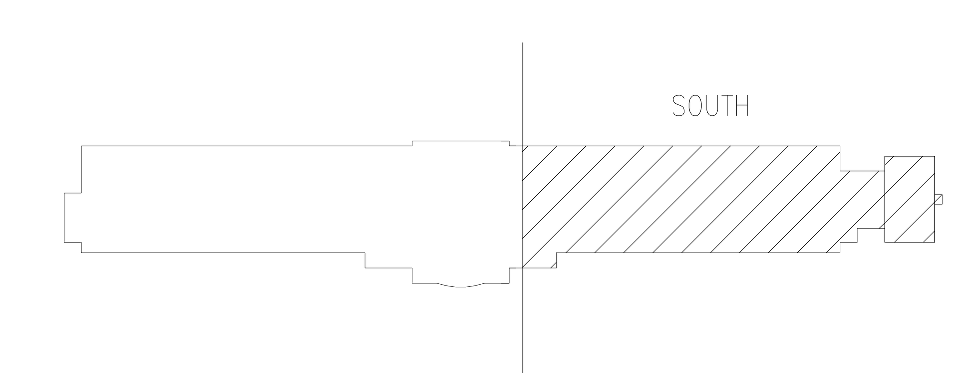
1. DOAS-1 TO BE MOUNTED ON A CUSTOM FITTED CURB BY THE MANUFACTURER. SHEET S-503 DETAIL C2 FOR DUCTWORK SHAFT OPENING.
2. DOAS-2 SHALL HAVE COASTAL PROTECTIVE COATING PER THE SPECIFICATIONS.
3. INSULATE EXTERIOR OF ALL DUCTWORK IN THE EXTERIOR.

GRAPHIC SCALE

1/8" = 1'-0"



KEY PLAN



SYM | DESCRIPTION | DATE | APPR

NAVFAC

SEAL

A/E INFO

APPROVED
David Greenfield
David W Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO DATE 06/18/2012

DES JLS	DRW JLS	CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION SLW

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NORTHEAST IPT
NAVAL STATION - NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON
GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

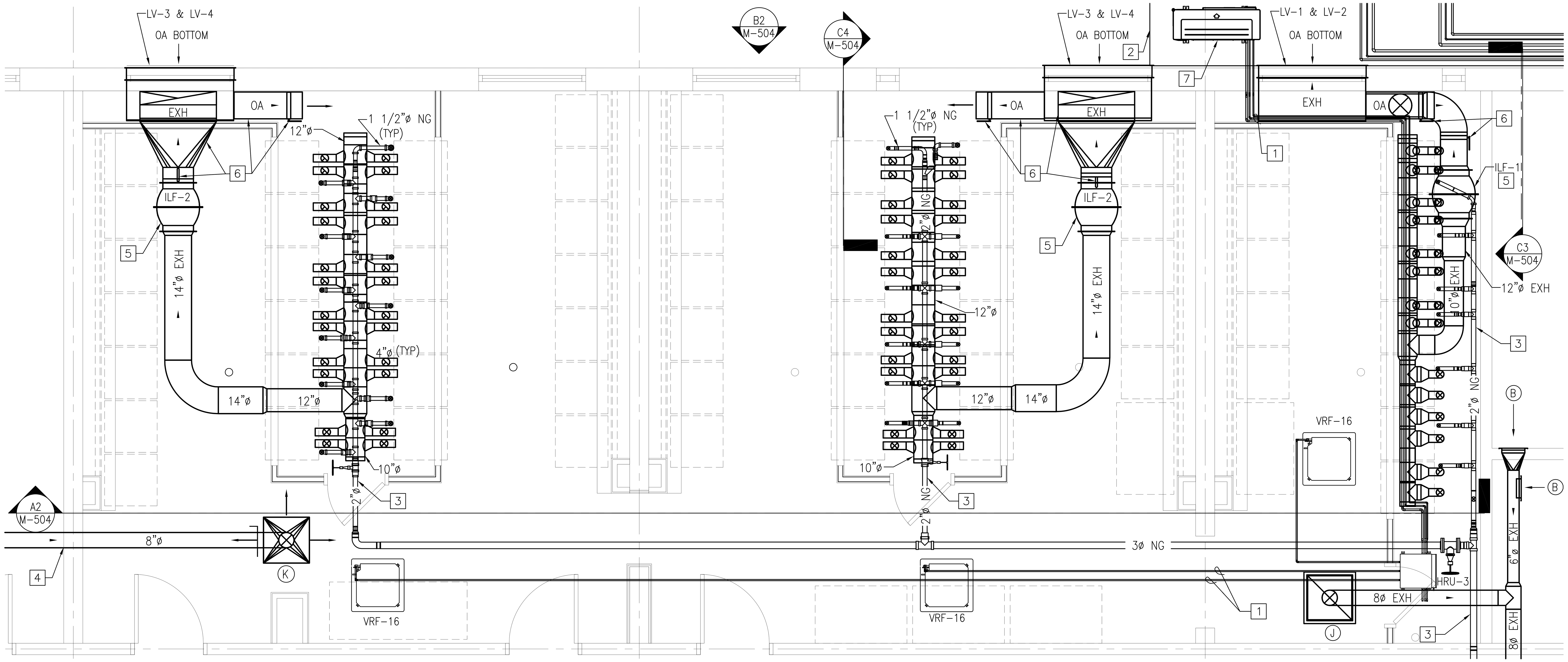
ROOF PLAN - SOUTH

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624720
SHEET 147 OF 206

M-111

DRAWFORM REVISION: 10 MARCH 2009

FILE NAME: P:\Connecticut\New_London\Buildings\B-534\2013_1127117_RM10-7303_Repair Sub A School BQ 534\B_Design\Drawings\Working\Mechanical\BLD0534-1127117-M-111.dwg LAYOUT NAME: M-111 PLOTTED: Wednesday, July 11, 2012 - 10:08am USER: marcus.d.smith



ENLARGED LAUNDRY ROOM EQUIPMENT PLAN

SCALE: NTS

M-102 B1

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
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7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

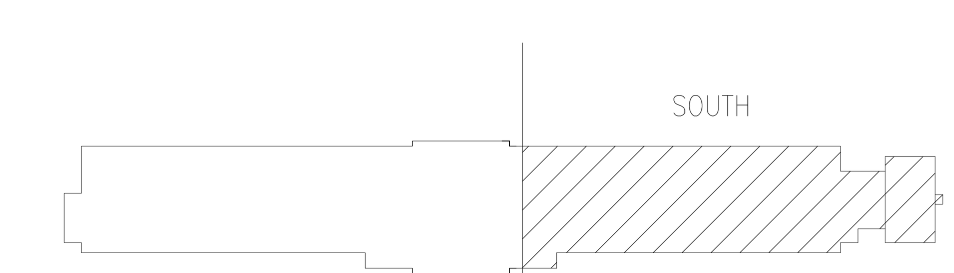
NEW WORK KEYNOTES

1. VRF INSULATED REFRIGERANT PIPING TO BE SIZED PER MANUFACTURER'S REQUIREMENTS. ALL PIPING INSIDE THE BUILDING TO BE ROUTED TIGHT TO THE CEILING DECKING AS POSSIBLE. SEE DETAIL C4 ON SHEET M-503 FOR PIPE SUPPORT DETAILS.
2. NEW 4 INCH EQUIPMENT CONCRETE PAD. SEE SHEET M-102.
3. GAS PIPING TO BE ROUTED CLOSE TO THE CEILING DECKING AS POSSIBLE. SEE DETAIL C4 ON SHEET M-503 FOR PIPE SUPPORT DETAILS.
4. WALL AREA TO BE CHECKED FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM. SEE DETAIL A5 ON SHEET M-503 FOR WALL PENETRATION DETAILS.
5. PROVIDE NEW INLINE EXHAUST FANS PER SCHEDULE ON SHEET M-601. FANS TO BE HUNG FROM CEILING DECK AS TIGHT AS POSSIBLE AND THE BELT DRIVE MOTOR TO BE PLACED TO THE SIDE.
6. EXHAUST LOUVER DUCTWORK TO BE THE UPPER PORTION OF THIS SPLIT OUT-TAKE AND INTAKE LOUVER SYSTEM. THE LOUVER INTAKE DUCTWORK WILL BE THE LOWER PORTION. BOTH EXHAUST AND INTAKE DUCTWORK WILL HAVE MOTORIZED DAMPERS.
7. NEW LAUNDRY ROOM VRF OUTDOOR UNIT, SEE SCHEDULE SHEET M-602.

GRAPHIC SCALE

3/8" = 1'-0" 0 1'-4" 2'-8" 5'-4"

KEY PLAN



PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

DATE 06/18/2012

DES JLS DRW JLS CHK JM

RS/NW

DWG

S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

ENLARGED LAUNDRY ROOM EQUIPMENT PLAN

SCALE: AS NOTED

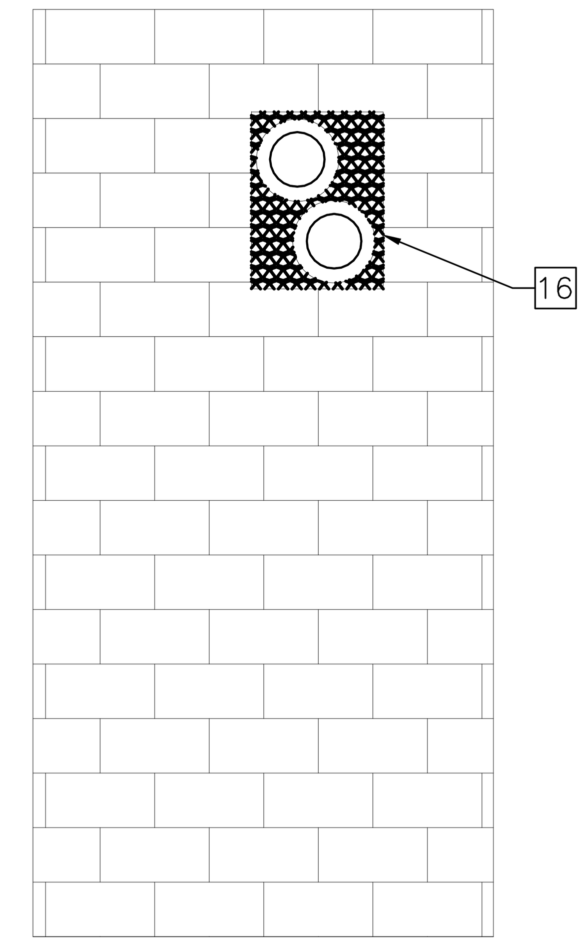
EPROJECT NO: 1127117

CONSTR. CONTR. NO. N40085-12-R-1715

NAVFAC DRAWING NO. 12624721

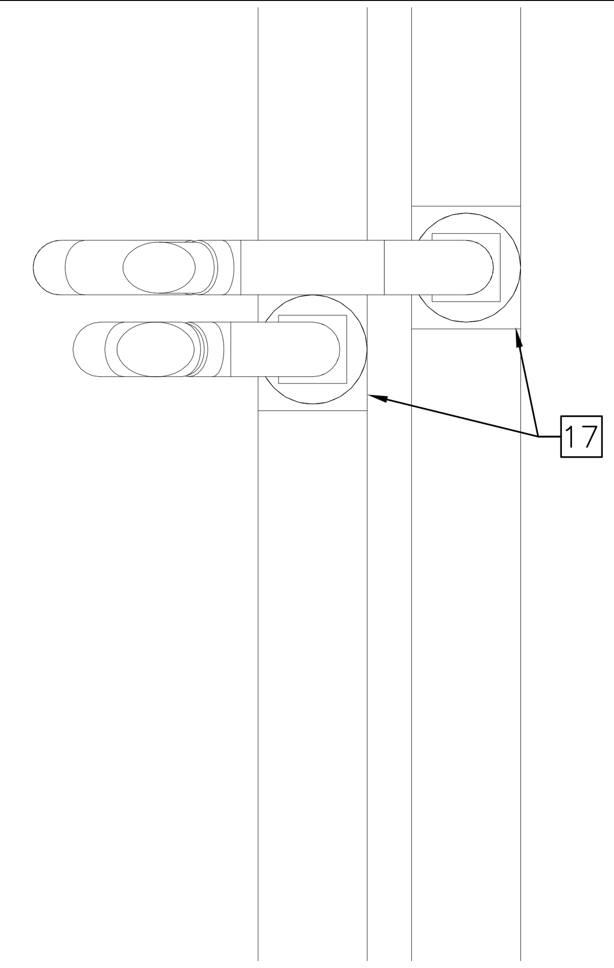
SHEET 148 OF 206

M-401



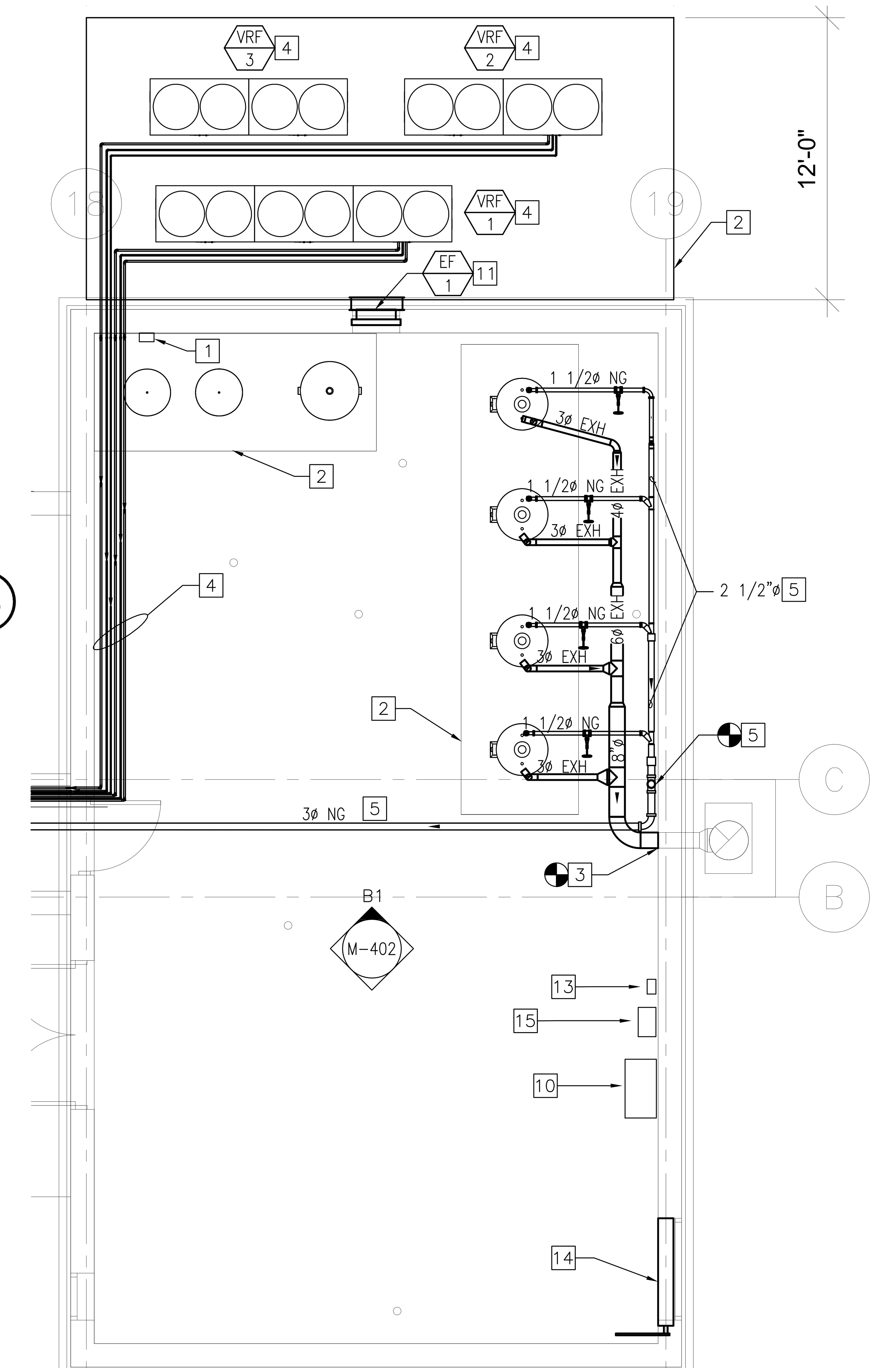
ENLARGED DUCTWORK WALL PENETRATION ELEVATION PLAN (D1)

SCALE: 1/2" = 1'-0" M-107



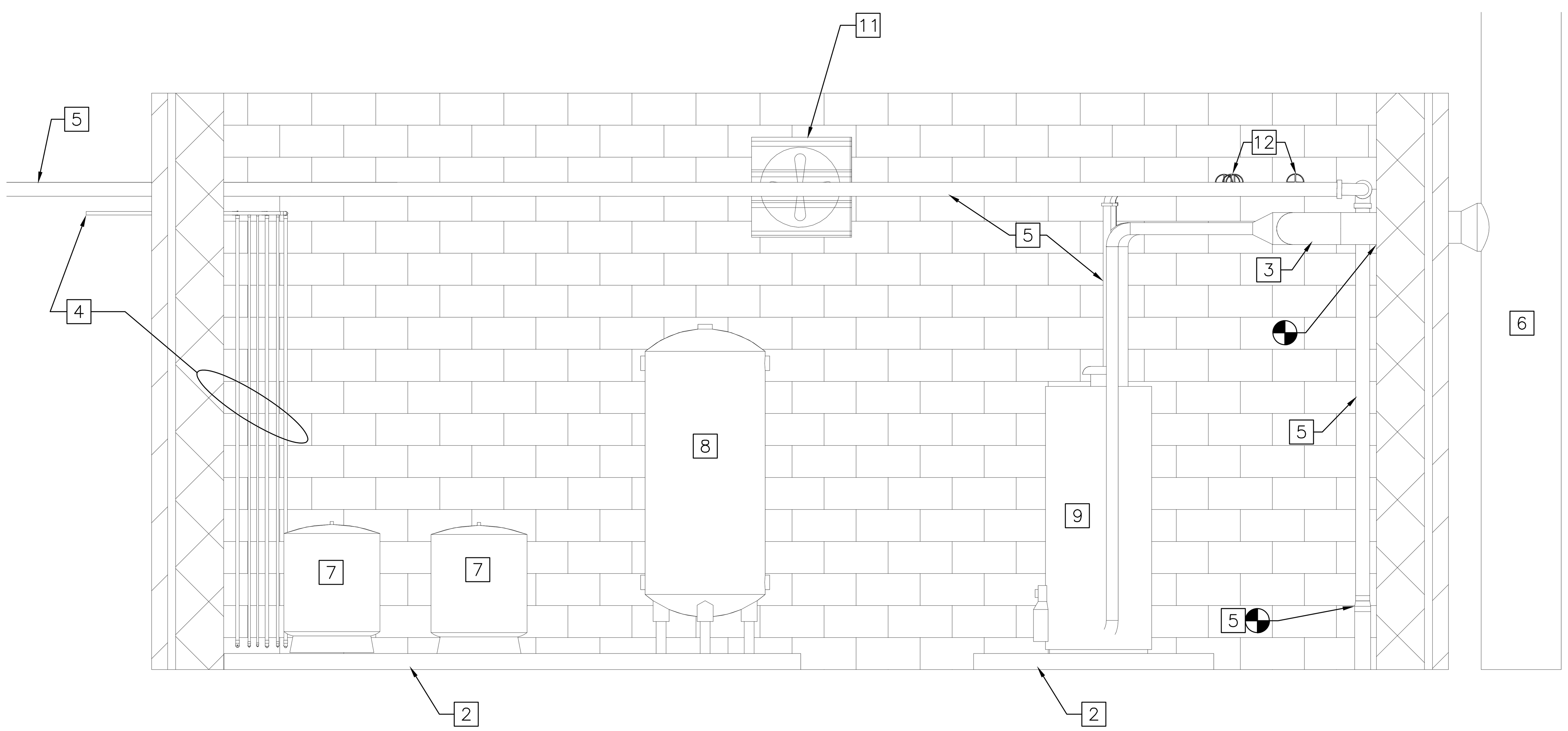
ENLARGED DUCTWORK THRU WALL ELEVATION PLAN (D3)

SCALE: 1/2" = 1'-0" M-107



ENLARGED MECHANICAL ROOM EQUIPMENT PLAN (B4)

SCALE: 1/4" = 1'-0" M-102



ENLARGED MECHANICAL ROOM EQUIPMENT ELEVATION PLAN (B1)

SCALE: 1/2" = 1'-0" M-402

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
4. REUSE ALL EXISTING WALL OPENINGS, UNLESS OTHERWISE NOTED. REPAIR AND SEAL ALL OPENINGS NOT USED.
5. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
6. THIS IS A 3-PIPE VRF REFRIGERANT SYSTEM WHICH REQUIRES A MANUFACTURER CERTIFIED CONTRACTOR FOR INSTALLATION. PROVIDE AND INSTALL VRF PIPING PER MANUF'S. SPECIFICATIONS. INSTALL ALL SOLENOID VALVE KITS ON ALL PIPING PER MANUFACTURES RECOMMENDATIONS.
7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

NEW WORK KEYNOTES

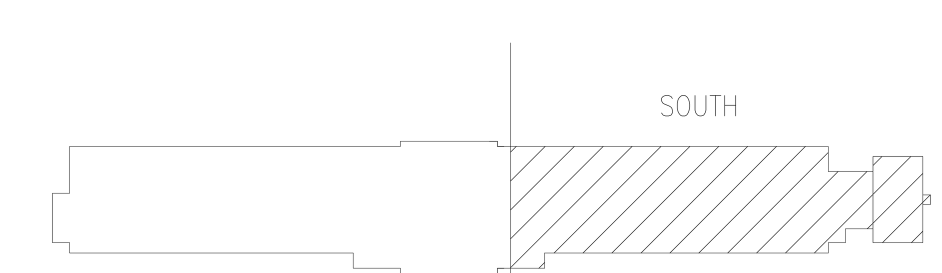
1. PROVIDE REFRIGERANT LEAKAGE SENSOR.
2. NEW 4 INCH CONCRETE PAD.
3. NEW GAS VENT DUCT WITH TYPE B INSULATION.
4. VRF INSULATED REFRIGERANT PIPING TO BE SIZE PER MANUFACTURER'S REQUIREMENTS. PIPING INSIDE OF THE BUILDING TO BE ROUTED TIGHT TO THE CEILING DECKING AS POSSIBLE.
5. GAS PIPING TO BE ROUTED CLOSE TO CEILING DECKING AS POSSIBLE.
6. EXISTING FLUE STACK.
7. NEW EXPANSION TANK.
8. NEW HOT WATER STORAGE TANK.
9. NEW WATER GAS HEATERS.
10. NEW DDC LOCATION.
11. NEW EXHAUST FAN, SEE SHEET M-601.
12. NEW VALVES FOR THE GAS PIPING.
13. PROVIDE A CARBON MONOXIDE DETECTOR.
14. NEW MOTORIZED DAMPER, SEE DETAIL C1 ON SHEET M-501.
15. NEW ELECTRIC UNIT HEATER, SEE DETAIL A3 ON SHEET M-503.
16. TYPICAL DUCTWORK THRU WALL PENETRATIONS FROM ROOM TO ROOM, SEE SHEET AD-402 DETAIL A2 FOR STRUCTURAL DETAILS.
17. DUCTWORK CONNECTIONS AT VERTICAL RISER TO THE ROOFTOP UNITS.

GRAPHIC SCALE

1/4" = 1'-0" 0 2' 4' 8'

1/2" = 1'-0" 0 1' 2' 4'

KEY PLAN



NAVFAC

APPROVED FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO DATE 06/18/2012

DES JLS DRW JLS CHK JM

PM/DM RS/NW

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NORTHEAST IPT NAVAL STATION - NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

ENLARGED MECHANICAL ROOM EQUIPMENT PLAN

SCALE: AS NOTED

EPROJECT NO.: 1127117

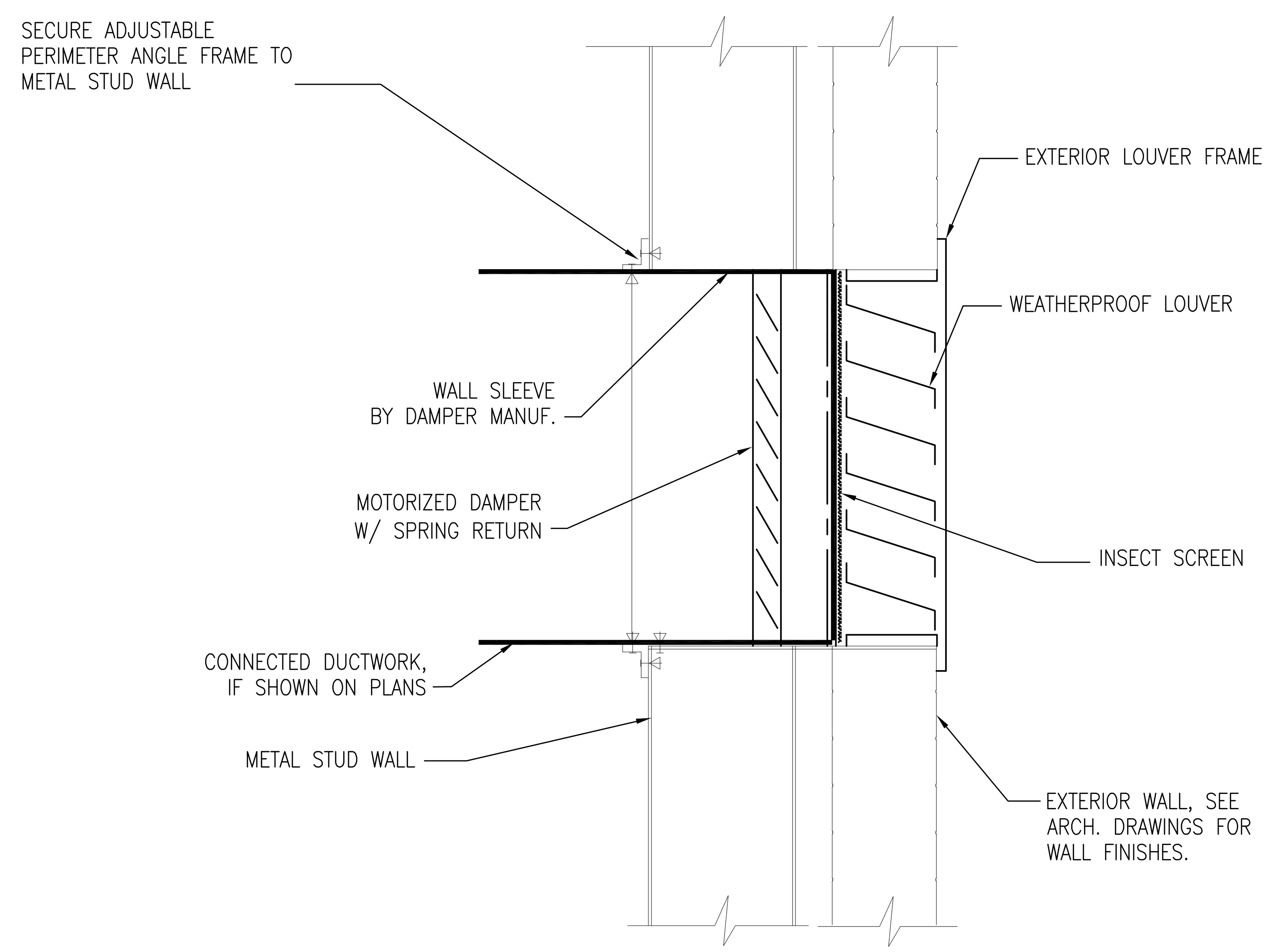
CONSTR. CONTR. NO. N40085-12-R-1715

NAVFAC DRAWING NO. 12624722

SHEET 149 OF 206

M-402

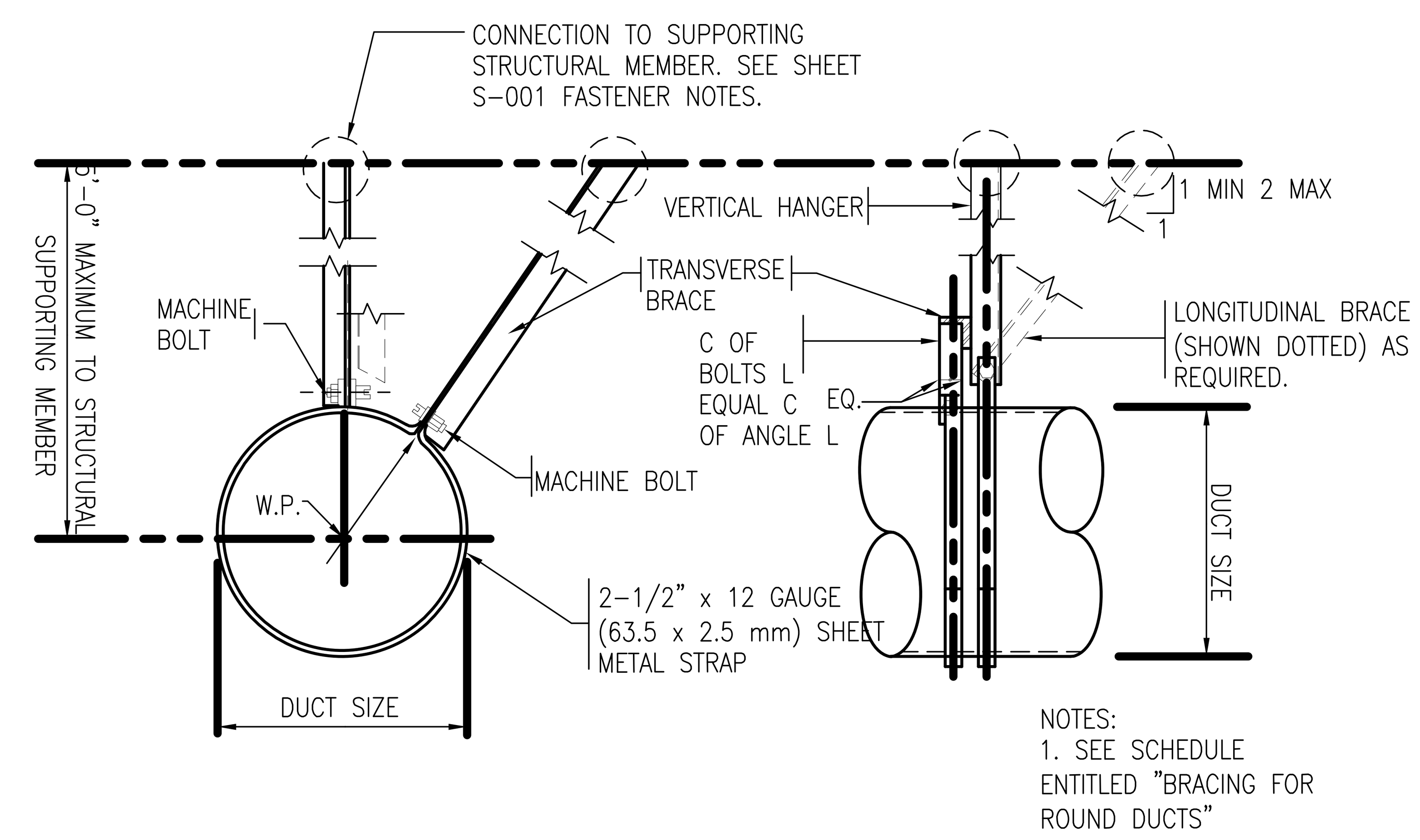
DRAWFORM REVISION: 10 MARCH 2009



WALL MOUNTED MOTORIZED DAMPER DETAIL (C1)

NO SCALE

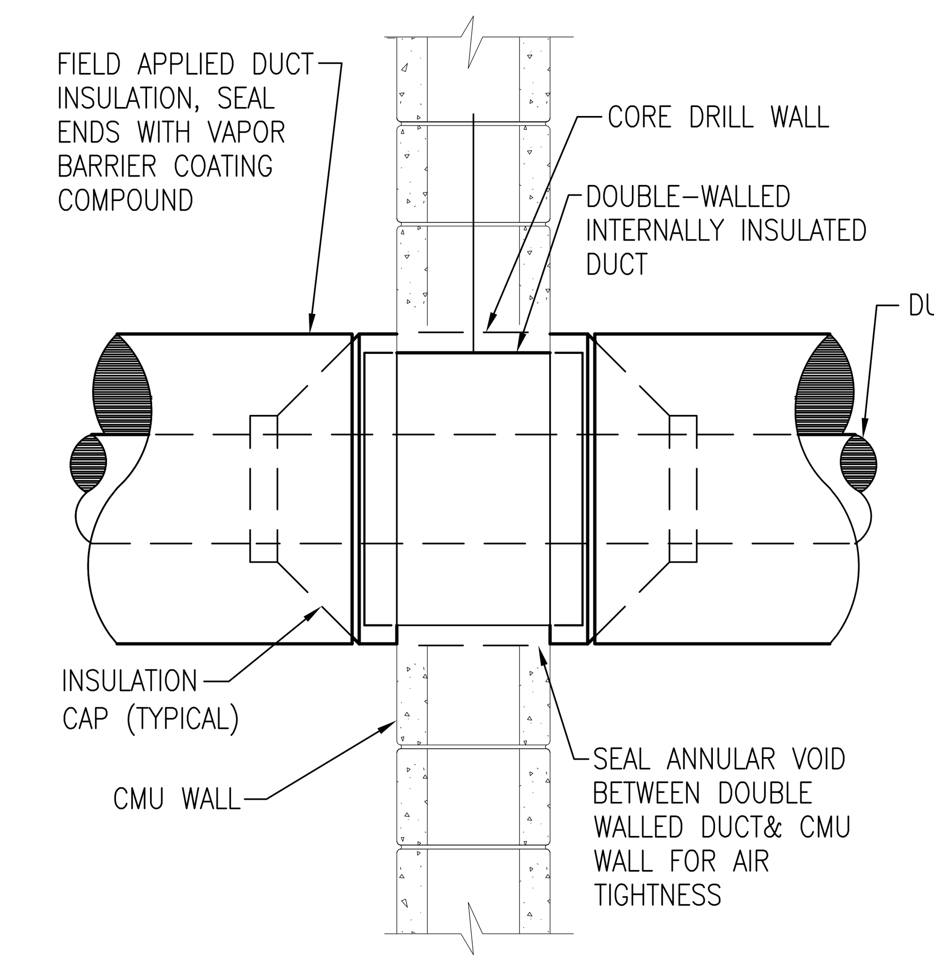
M-402



SINGLE HANGER BRACING FOR ROUND DUCTS DETAIL (C3)

NOT TO SCALE

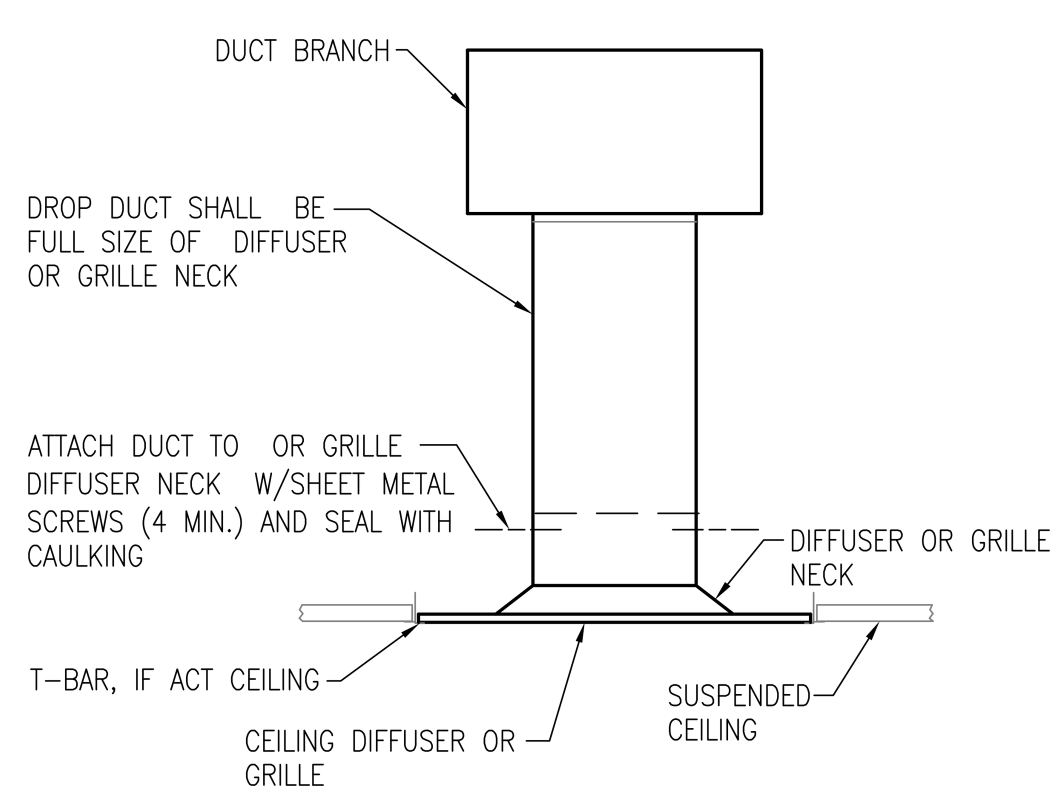
M-101 THRU- M-108



ROUND DUCT WALL PENETRATION DETAIL (C5)

NO SCALE

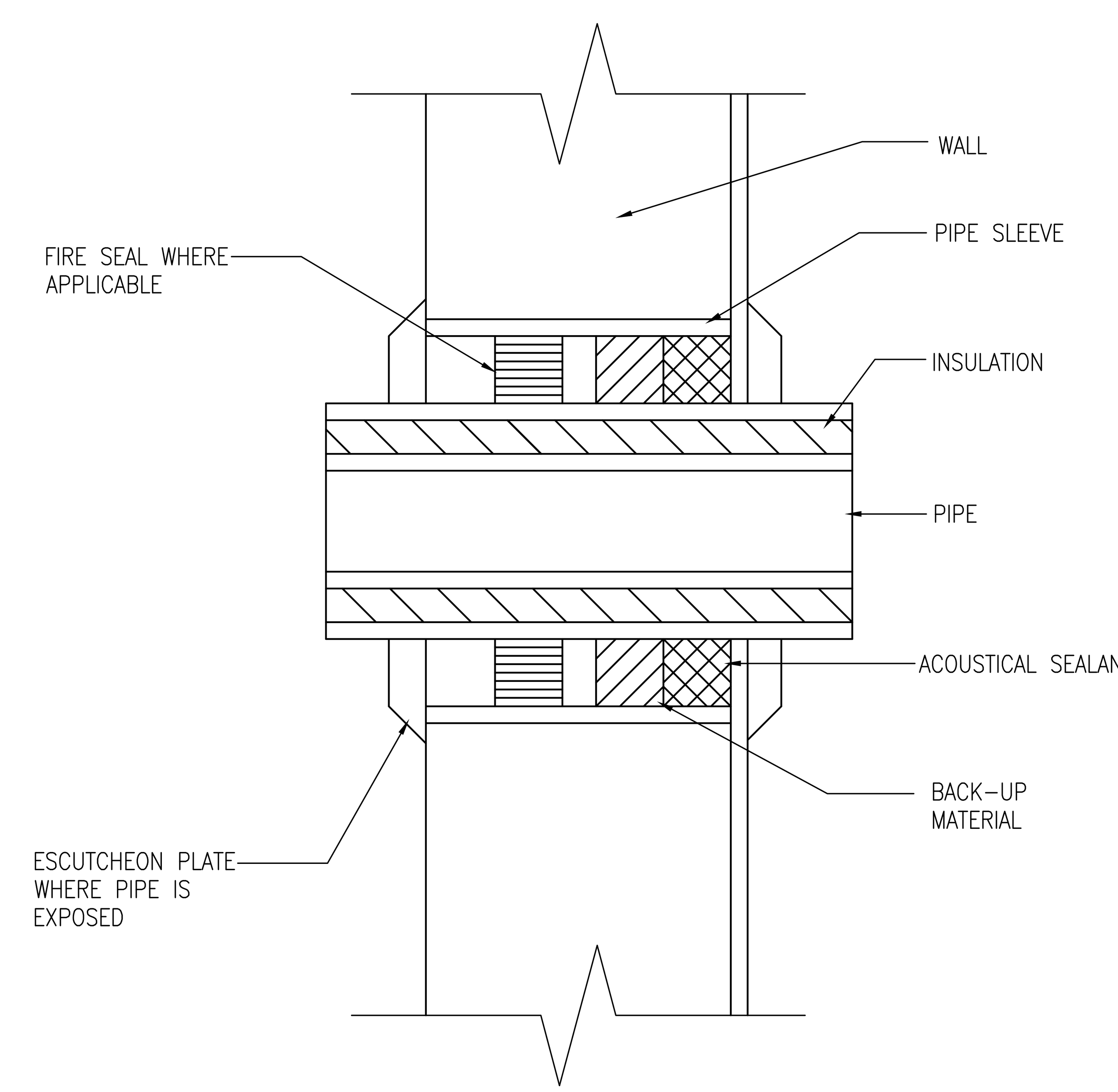
M-101 THRU M-108



DUCT TO DIFFUSER/GRILLE CONNECTION DETAIL (A2)

NO SCALE

M-101 THRU M-108



WALL PIPE SLEEVE DETAIL (A3)

NOT TO SCALE

M-101 THRU M-108

SYM | DESCRIPTION | DATE | APPR

SEAL

A/E INFO

APPROVED: David W. Greenfield
FOR COMMANDER NAVFAC

ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO | DATE 06/18/2012

DES JLS | DRW JLS | CHK JLS

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC

NORTHEAST IPT NAVAL STATION - NORFOLK, VIRGINIA

NAVSUBASE NEW LONDON GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL DETAILS

SCALE: AS NOTED

EPROJECT NO.: 1127117

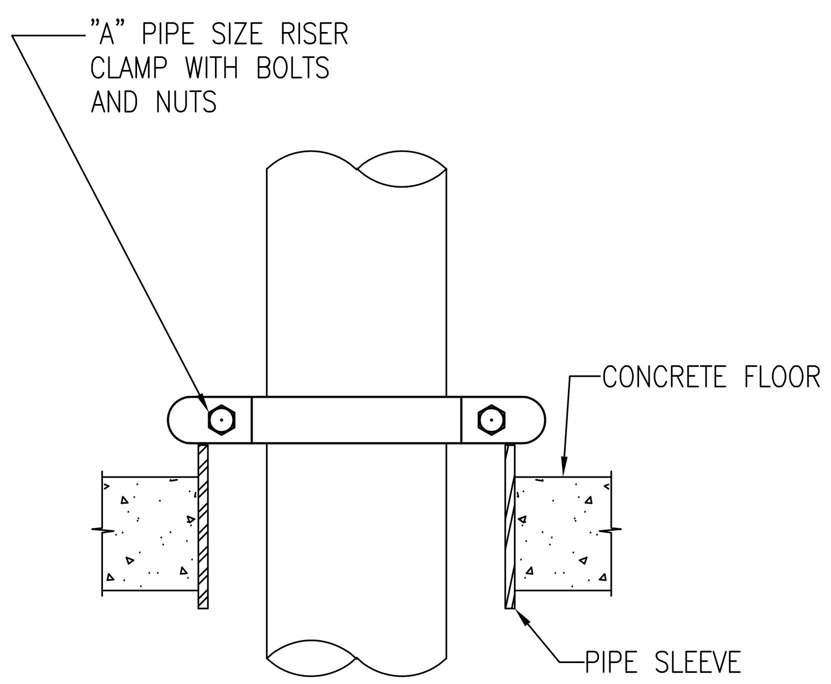
CONSTR. CONTR. NO.: N40085-12-R-1715

NAVFAC DRAWING NO.: 12624723

SHEET 150 OF 206

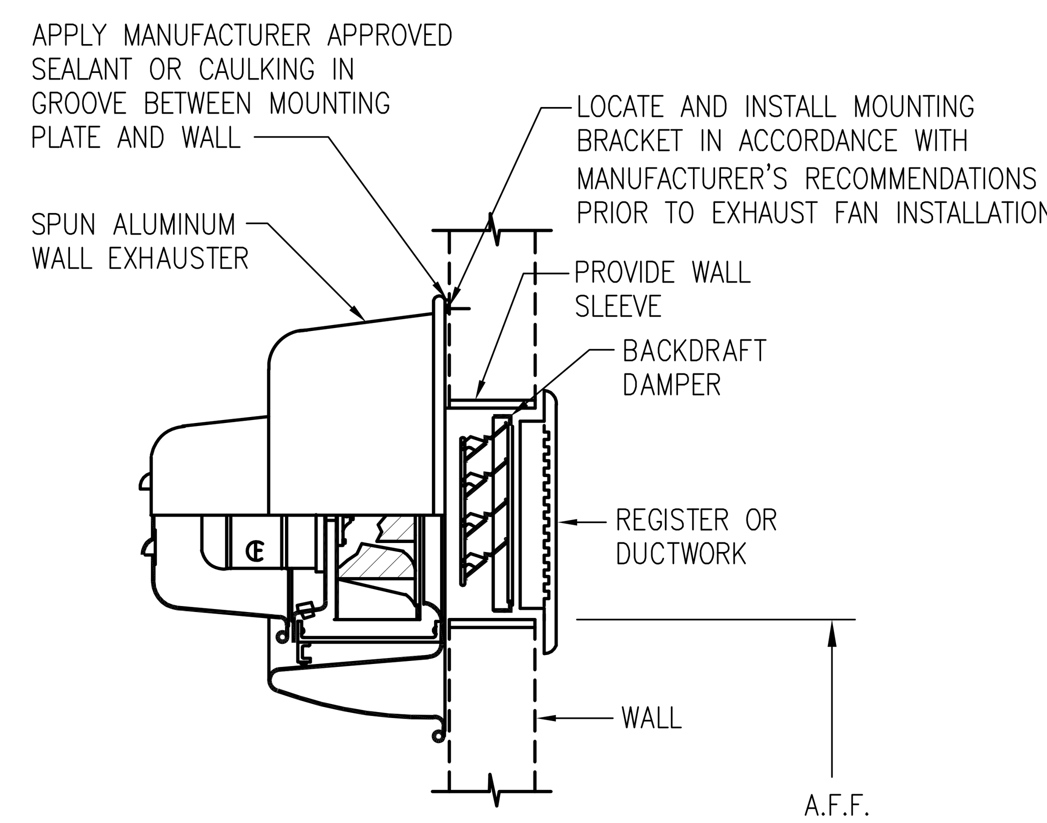
M-501

DRAWFORM REVISION: 10 MARCH 2009



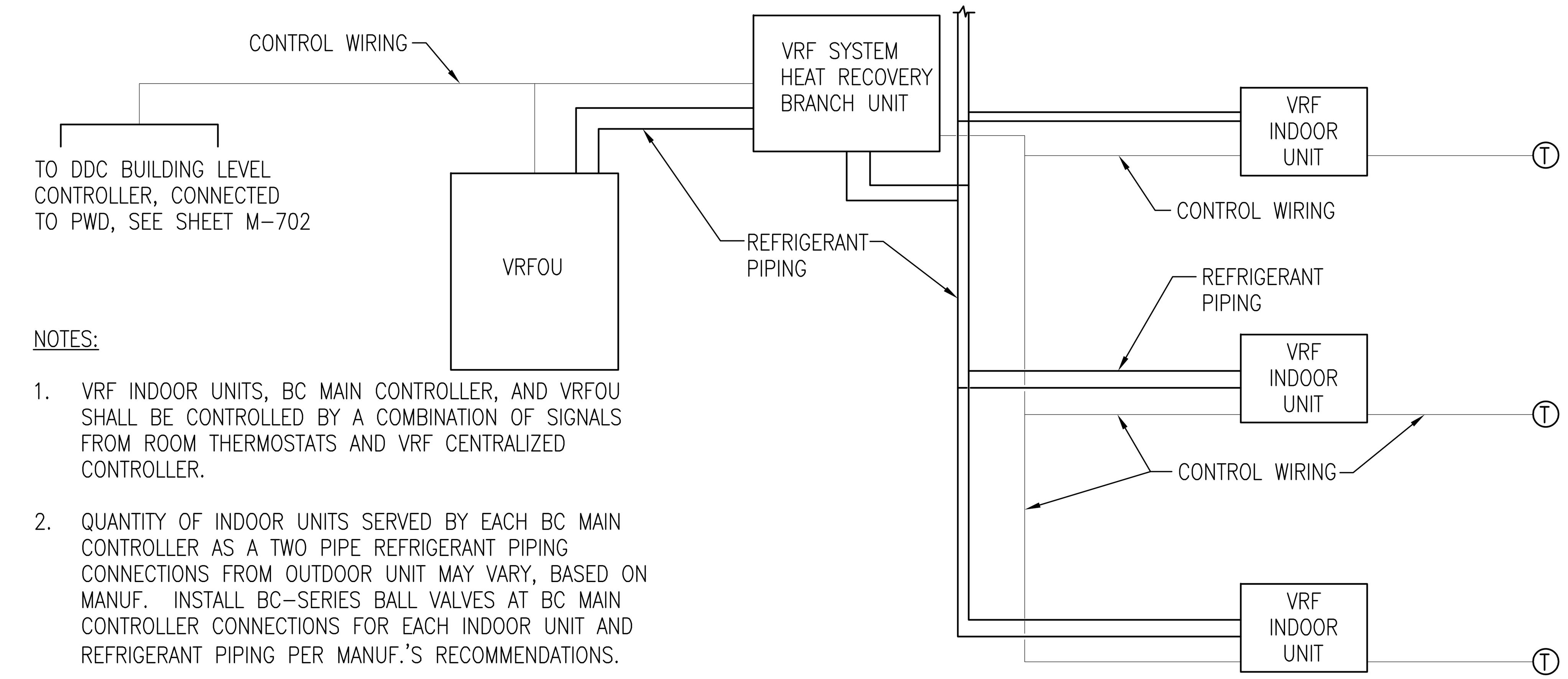
REFRIGERANT PIPING SUPPORT DETAIL (C1)

NO SCALE M-101 THRU M-108



WALL MOUNTED EXHAUST FAN DETAIL (C2)

NO SCALE M-402

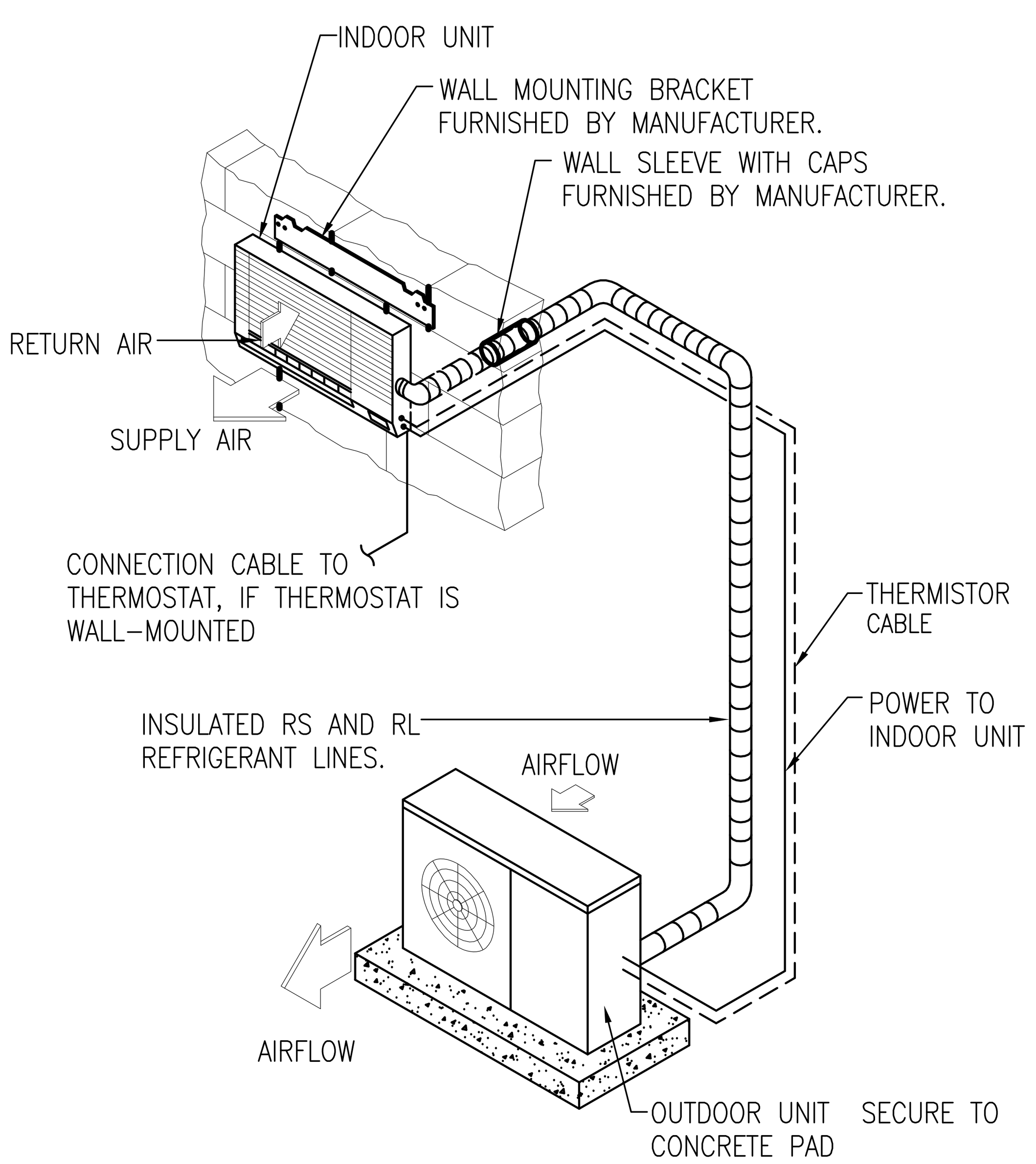


NOTES:

1. VRF INDOOR UNITS, BC MAIN CONTROLLER, AND VRFOU SHALL BE CONTROLLED BY A COMBINATION OF SIGNALS FROM ROOM THERMOSTATS AND VRF CENTRALIZED CONTROLLER.
2. QUANTITY OF INDOOR UNITS SERVED BY EACH BC MAIN CONTROLLER AS A TWO PIPE REFRIGERANT PIPING CONNECTIONS FROM OUTDOOR UNIT MAY VARY, BASED ON MANUF. INSTALL BC-SERIES BALL VALVES AT BC MAIN CONTROLLER CONNECTIONS FOR EACH INDOOR UNIT AND REFRIGERANT PIPING PER MANUF.'S RECOMMENDATIONS.

TYPICAL VRF REFRIGERANT PIPING AND CONTROL DIAGRAM (C4)

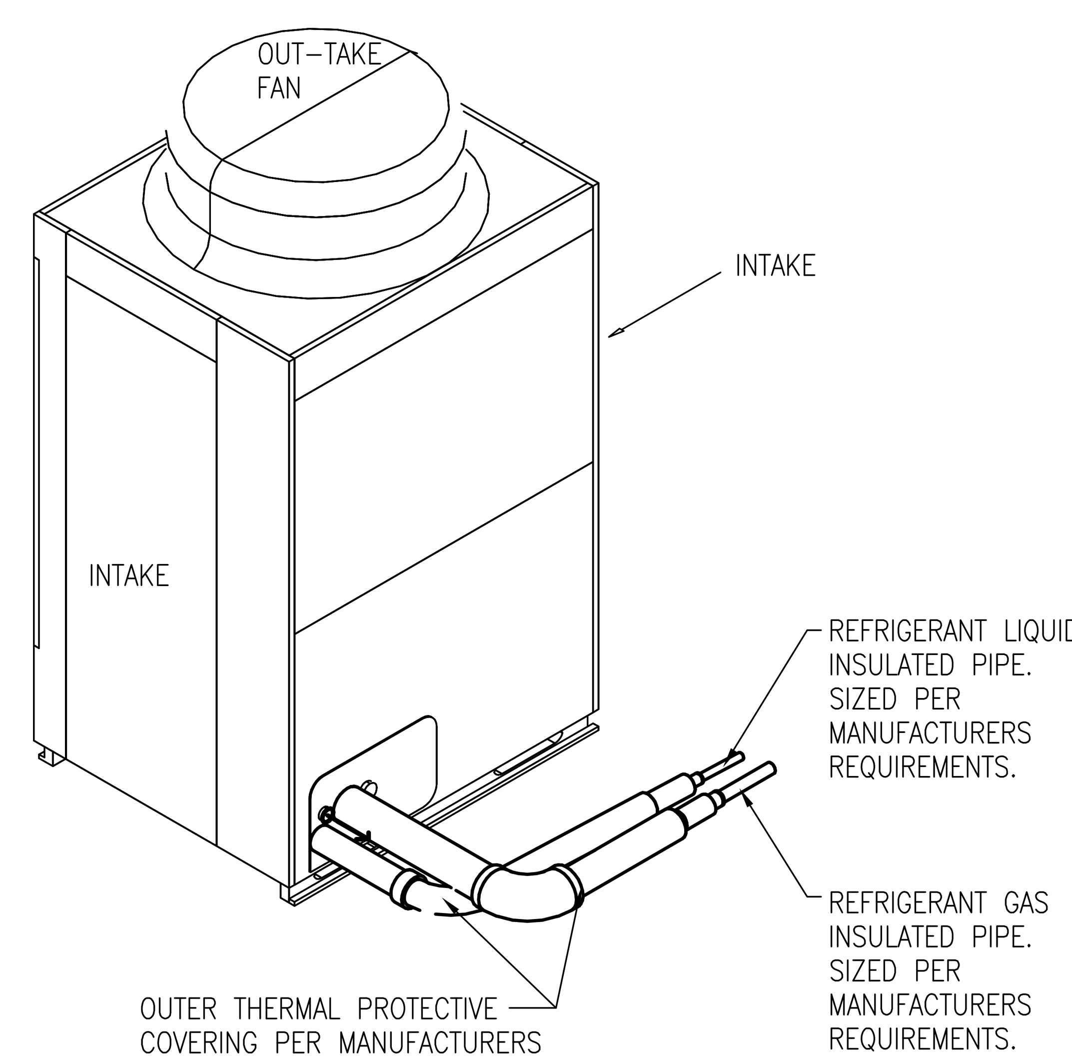
NO SCALE M-101 THRU M-108



NOTE: INSTALL INDOOR WALL UNIT AS HIGH AS POSSIBLE, UNLESS OTHERWISE SPECIFIED.

VRF WALL MOUNTED DUCTLESS UNIT DETAIL (A1)

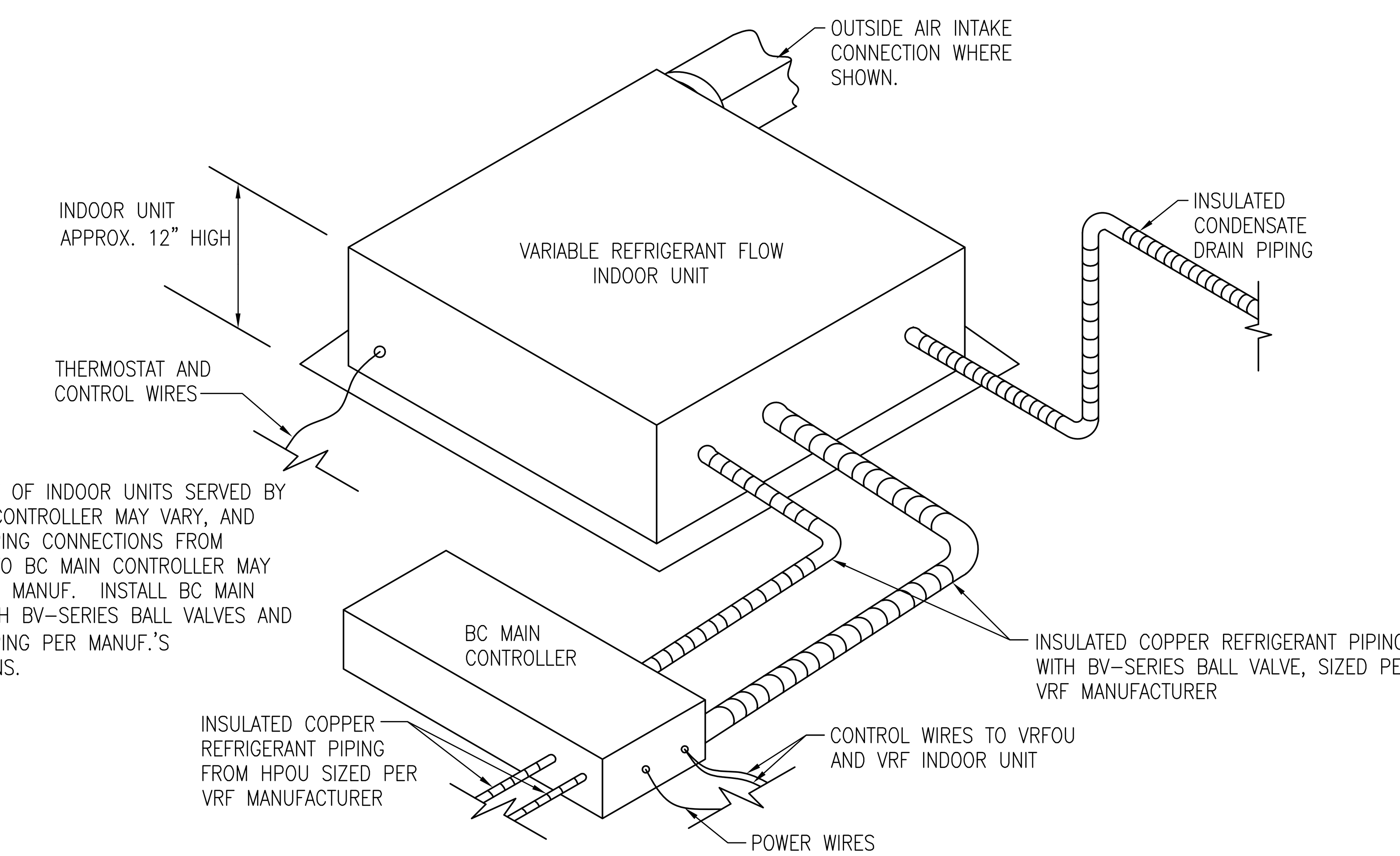
NO SCALE M-502



NOTE: SEE MANUFACTURERS REQUIREMENTS FOR ALL PIPING PENETERATIONS THRU WALLS, EXTERIOR AND INTERIOR.

VRF OUTDOOR UNIT DETAIL (A2)

NO SCALE M-102



NOTES: QUANTITY OF INDOOR UNITS SERVED BY EACH BC MAIN CONTROLLER MAY VARY, AND REFRIGERANT PIPING CONNECTIONS FROM OUTDOOR UNIT TO BC MAIN CONTROLLER MAY VARY, BASED ON MANUF. INSTALL BC MAIN CONTROLLER WITH BV-SERIES BALL VALVES AND REFRIGERANT PIPING PER MANUF.'S RECOMMENDATIONS.

TYPICAL VRF INDOOR UNIT INSTALLATION DETAIL (A4)

NO SCALE M-101 THRU M-108

NAVFAC

APPROVED David Greenfield

FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON

SATISFACTORY TO DATE 06/18/2012

DES JLS DRW JLS CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION SLW

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC

NORTHEAST IPT NAVAL STATION – NORFOLK, VIRGINIA

NAVSUBASE NEW LONDON GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL DETAILS

SCALE: AS NOTED

EPROJECT NO.: 1127117

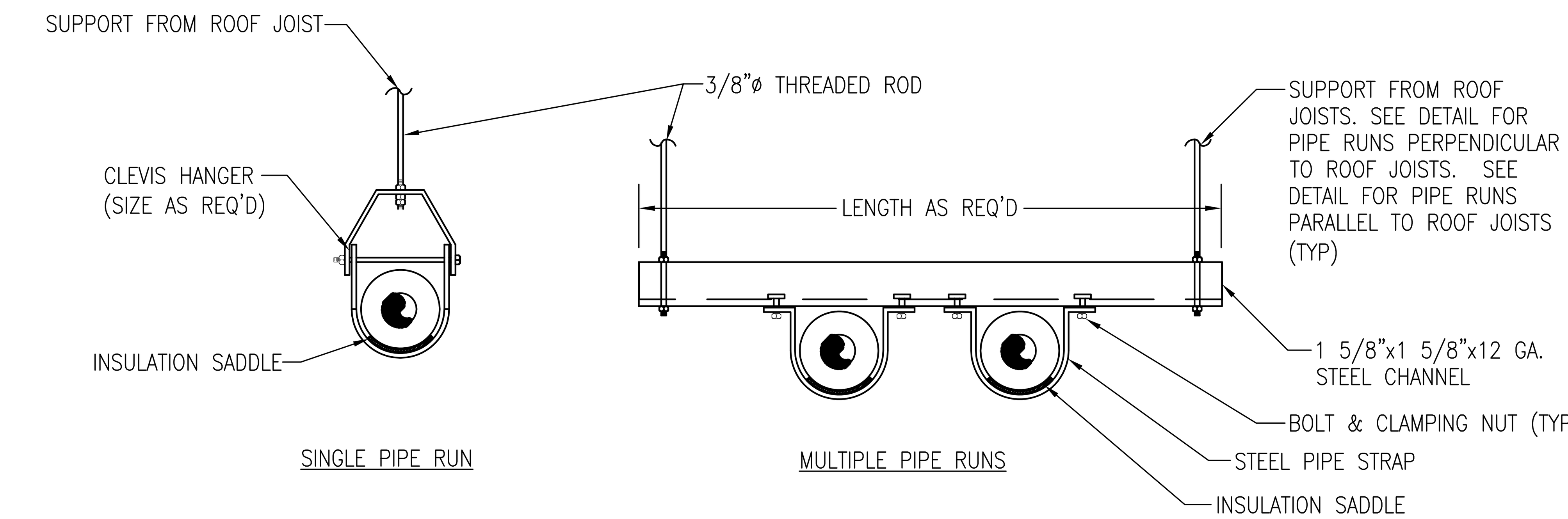
CONSTR. CONTR. NO. N40085-12-R-1715

NAVFAC DRAWING NO. 12624724

SHEET 151 OF 206

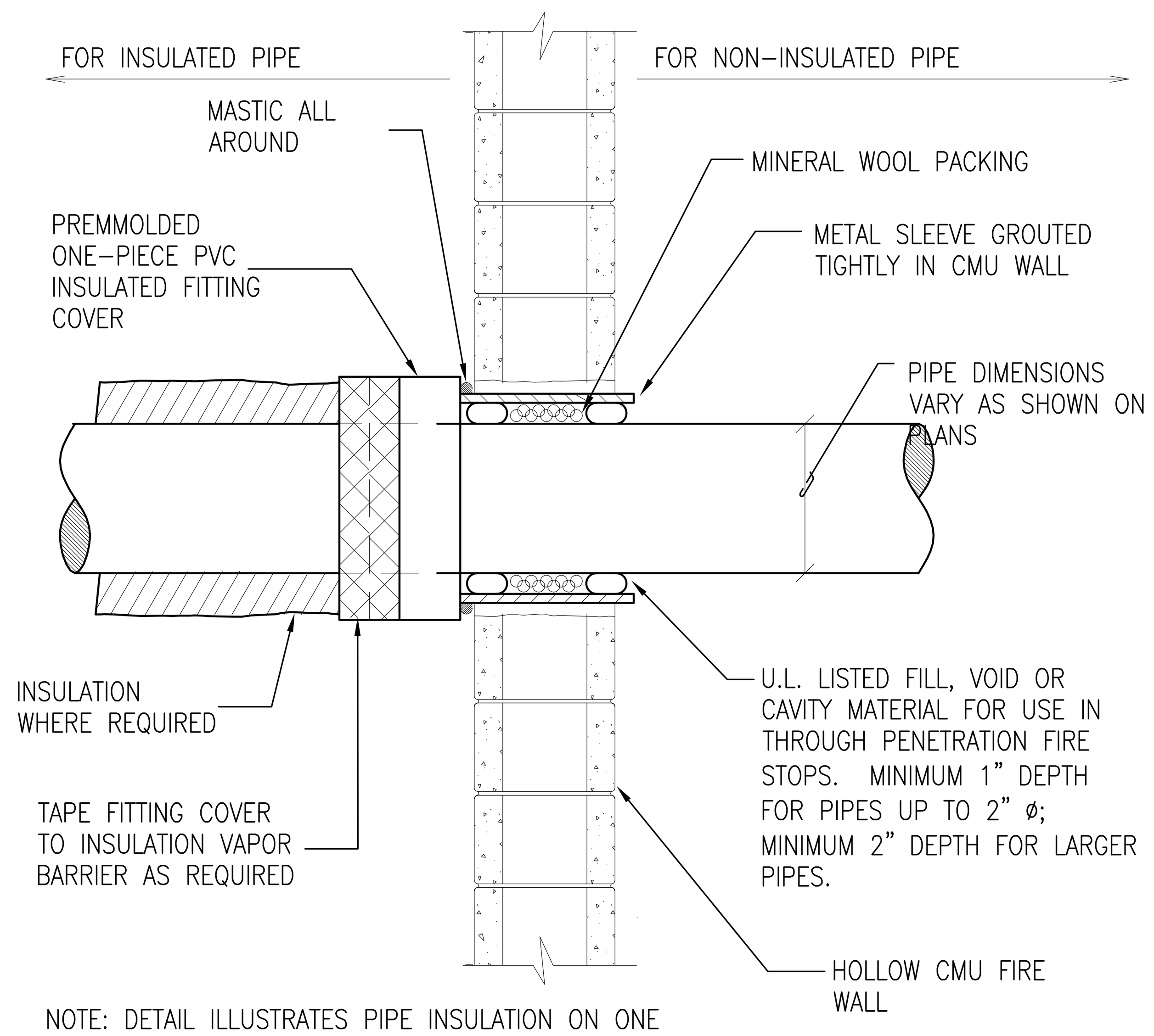
M-502

DRAWFORM REVISION: 10 MARCH 2009



HANGING PIPE SUPPORT DETAILS (C4)

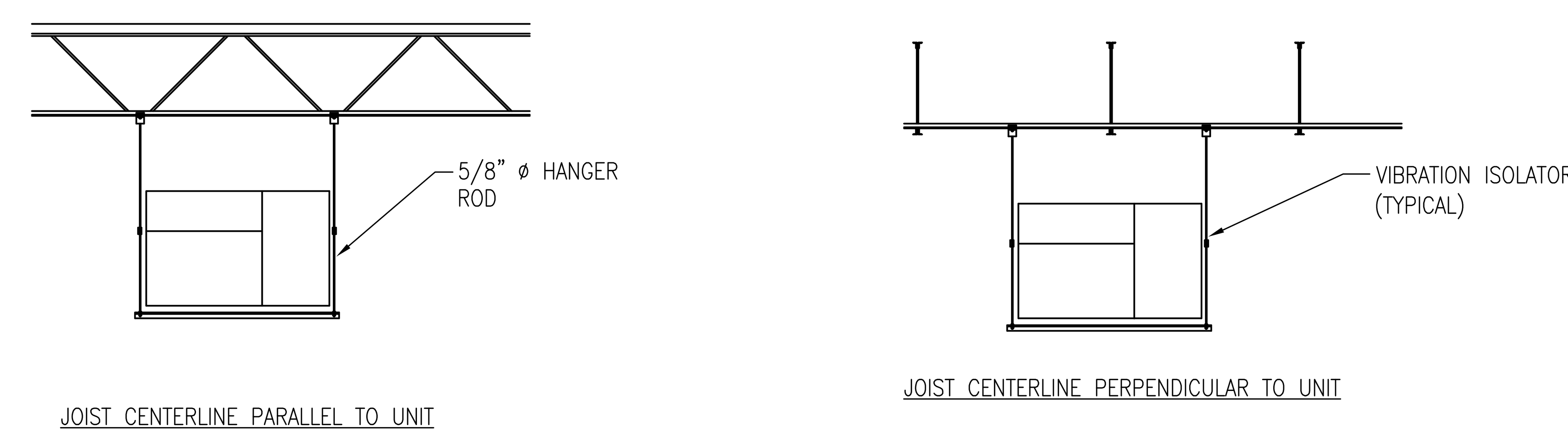
NO SCALE — M-101 THRU M-108



NOTE: DETAIL ILLUSTRATES PIPE INSULATION ON ONE SIDE OF FIRE RATED WALL. WHERE PIPE IS REQUIRED TO BE INSULATED ON BOTH SIDES, THE INSULATION END TREATMENT SHALL BE DUPLICATED ON BOTH SIDES.

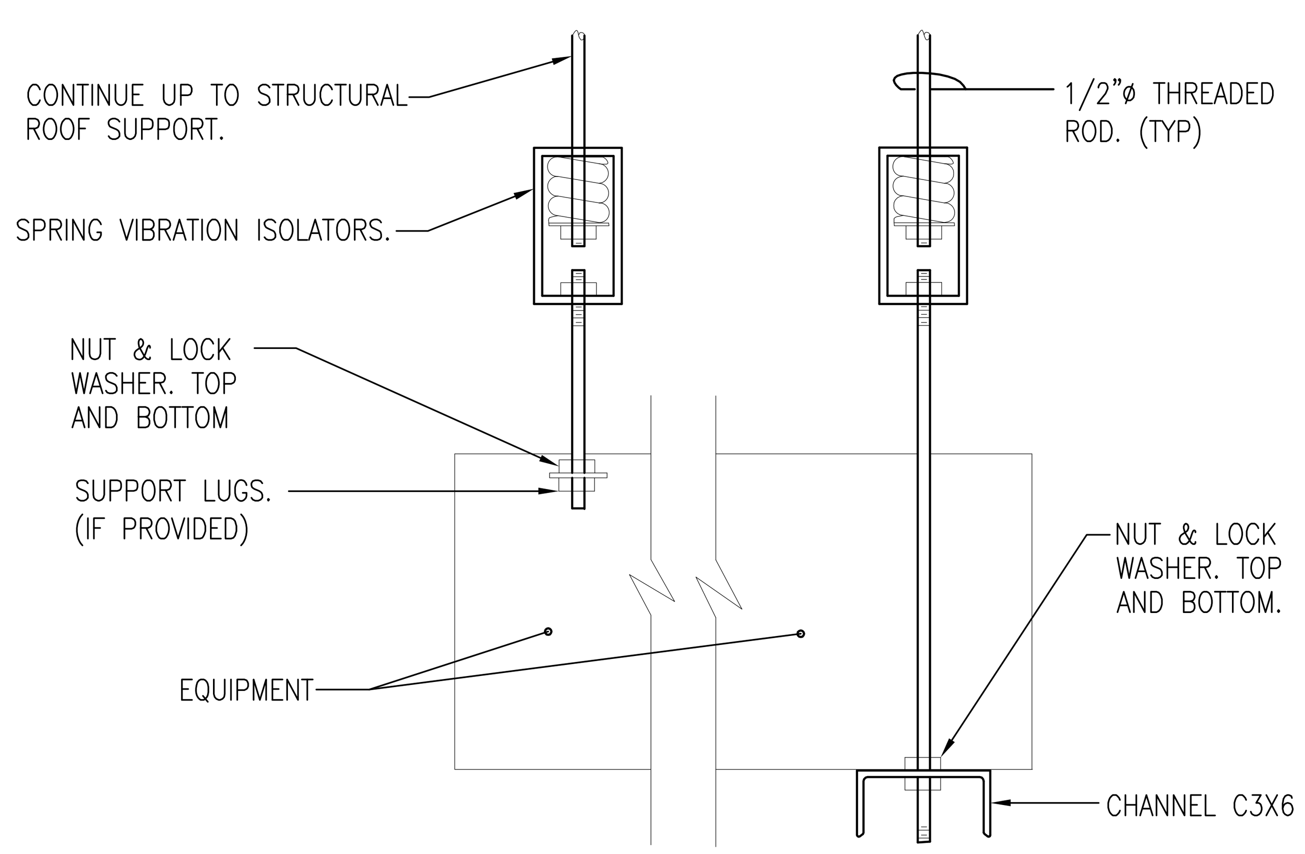
FIRE WALL PENETRATION DETAIL (B1)

NO SCALE — M-101, M-103, M-105, M-107



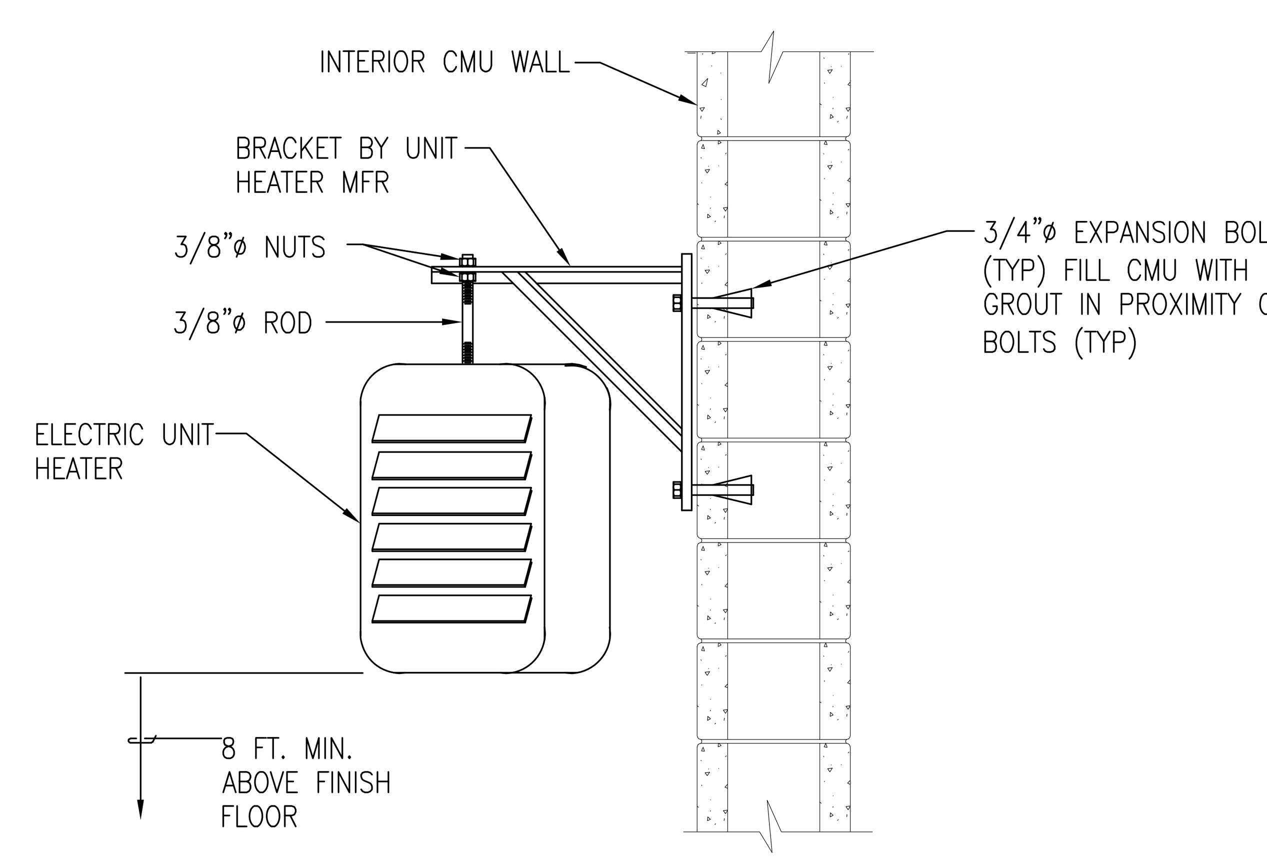
HANGING UNIT SUPPORT DETAILS (B4)

NO SCALE — M-503



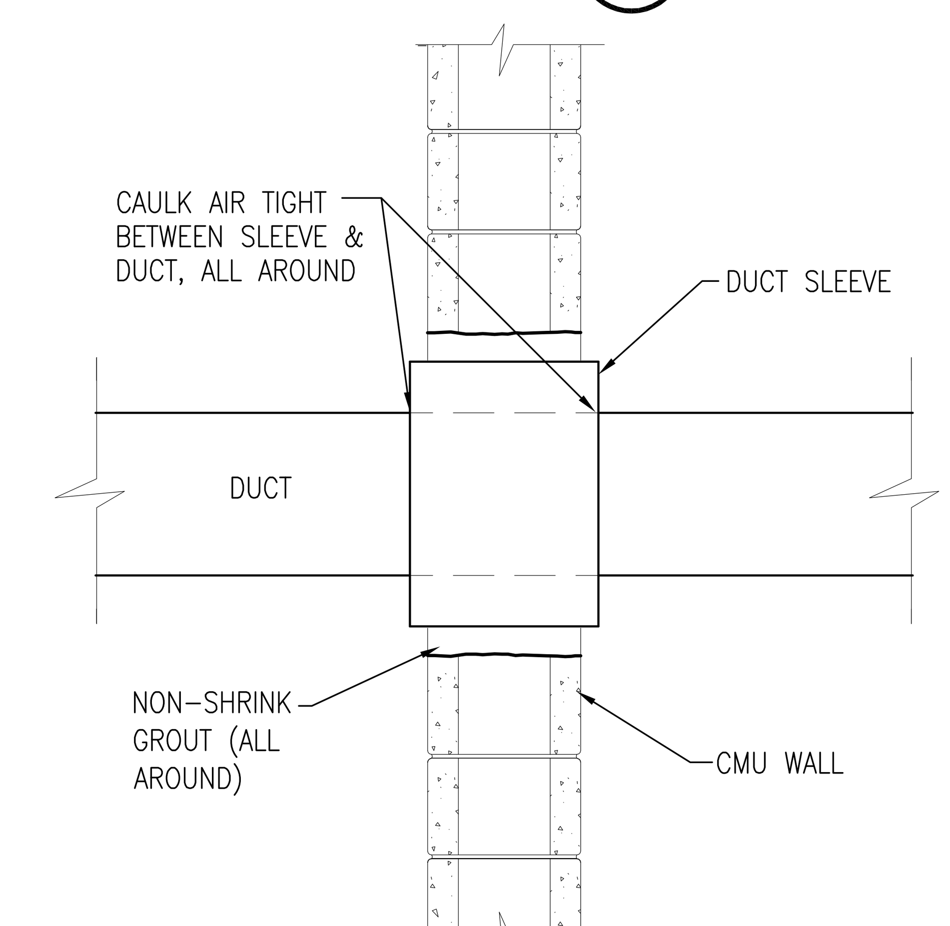
SUSPENDED EQUIPMENT SUPPORT DETAIL (A1)

NO SCALE — M-101 THRU M-108



UNIT HEATER SUPPORT DETAIL (A3)

NO SCALE — M-109, M-402



ROUND DUCT WALL PENETRATION DETAIL (A5)

NO SCALE — M-101 THRU M-108

SYM | DESCRIPTION | DATE | APPR

NAVFAC

SEAL

A/E INFO

APPROVED David W. Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO — DATE 06/18/2012

DES JLS | DRW JLS | CHK JLS

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

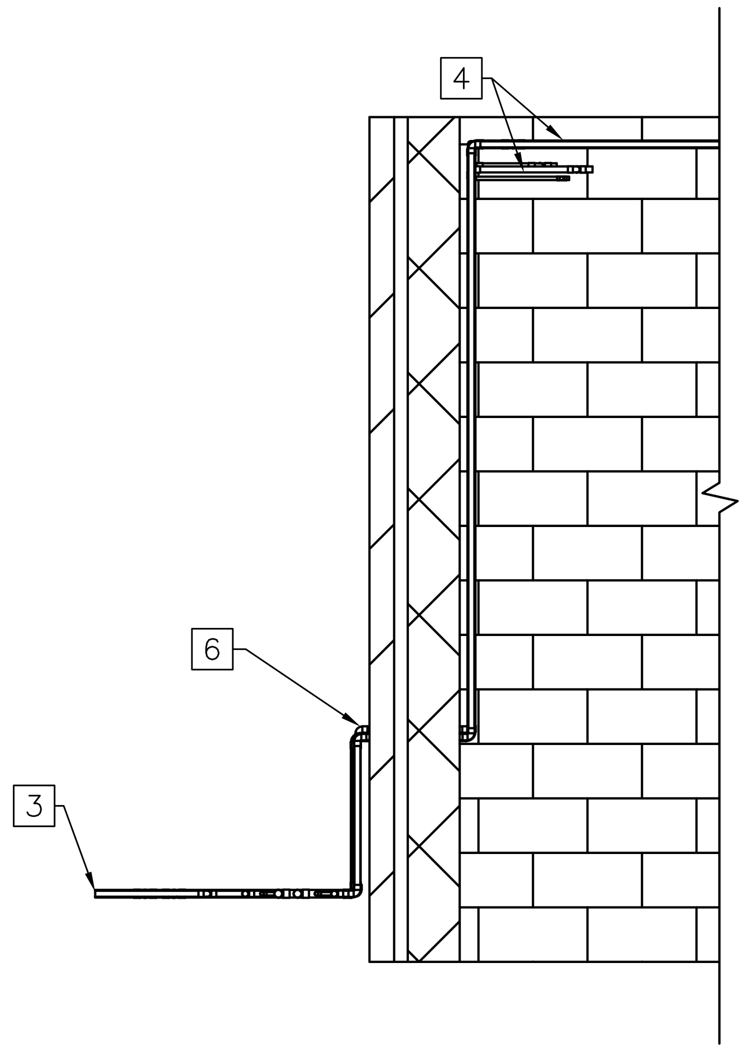
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NORTHEAST IPT
NAVSUBASE NEW LONDON — GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
MECHANICAL DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624725
SHEET 152 OF 206

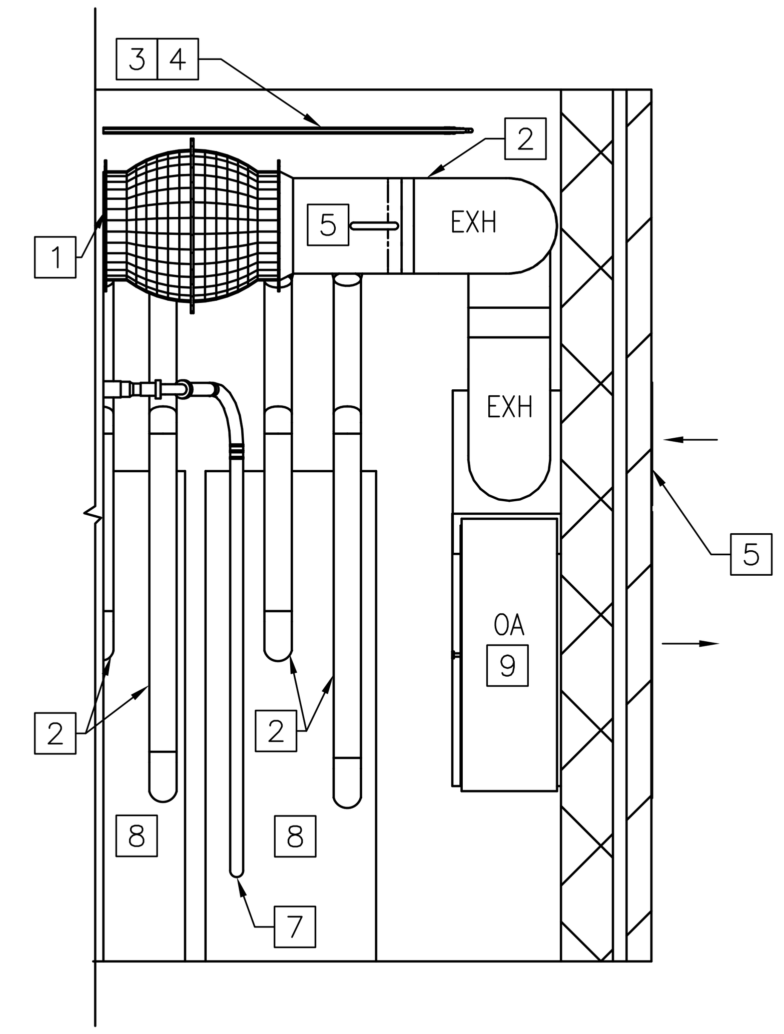
M-503

DRAWFORM REVISION: 10 MARCH 2009

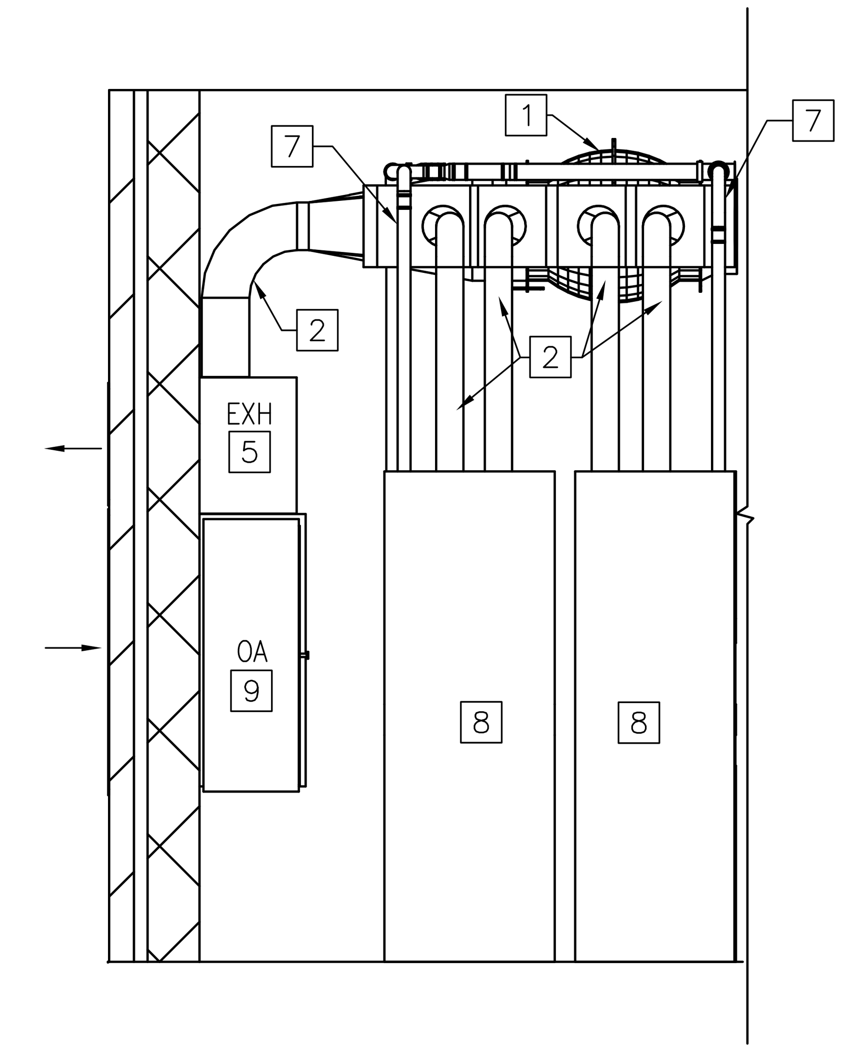
FILE NAME: P:\Connecticut\New_London\Buildings\B-534\2013_1127117_RM10-7303_Repair Sub A School BQ 534\B_Design\Drawings\Working\Mechanical\BLDG534-1127117-M-503.dwg LAYOUT NAME: M-503 PLOTTED: Wednesday, July 11, 2012 - 10:10am USER: marcus.d.smith



STORAGE RM 159 SECTION ELEVATION
SCALE: 1/2" = 1'-0" M-102 C1



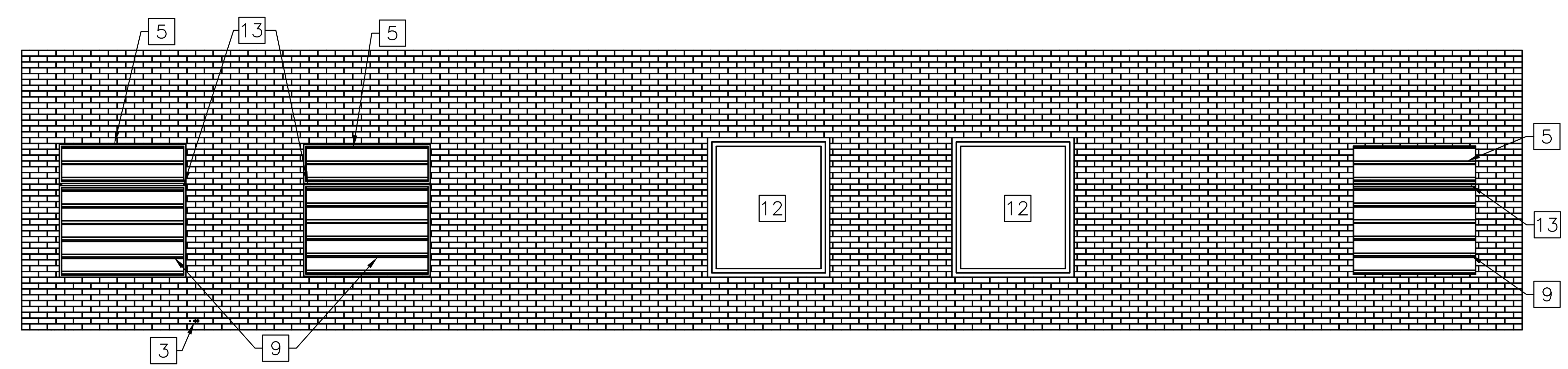
LAUNDRY ROOM SECTION ELEVATION
SCALE: 1/2" = 1'-0" M-401 C3



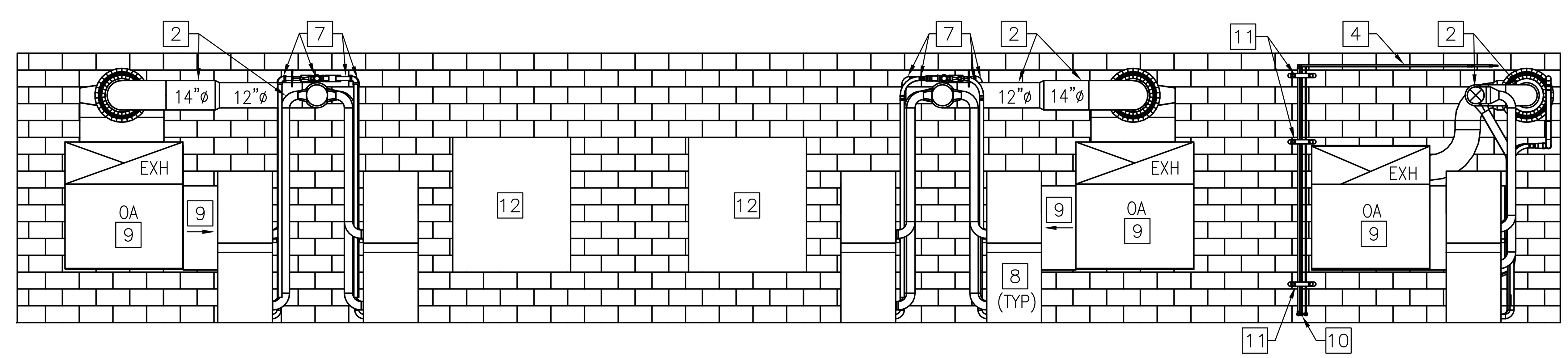
LAUNDRY ROOM SECTION ELEVATION
SCALE: 1/2" = 1'-0" M-401 C4

NEW WORK KEYNOTES

1. NEW INLINE EXHAUST FAN, SEE SHEET M-401 AND M-601 FOR DETAILED INFORMATION. ALL FANS NEED TO BE MOUNTED TIGHT TO THE CEILING DECK AS POSSIBLE. BELT DRIVE MOTOR TO BE SET TO THE SIDE OF THE UNIT.
2. PROVIDE NEW DRYER EXHAUST DUCTWORK. SEE DETAIL C3 ON SHEET M-501 FOR MAIN DUCT HANGING SUPPORT DETAIL.
3. REFRIGERANT PIPING CONTINUATION, SEE SHEET M-102 FOR MORE DETAILED INFORMATION.
4. ALL REFRIGERANT PIPING NEEDS TO BE MOUNTED TIGHT TO THE CEILING DECK, AS POSSIBLE. SEE DETAIL C4 ON SHEET M-503 FOR PIPE SUPPORT DETAILS.
5. EXHAUST LOUVER WITH MOTORIZE DAMPER. SEE SHEET M-401 AND M-601 FOR DETAILED INFORMATION.
6. REFRIGERANT PIPING PASSING THRU EXISTING WINDOW OPENING.
7. NATURAL GAS PIPING GOING INTO THE BACK OF THE DOUBLE STACK DRYERS. SEE DETAIL C4 ON SHEET M-503 FOR PIPE SUPPORT DETAILS.
8. STACKED DRYER PROVIDED BY THE FACILITY.
9. PROVIDE NEW OUTSIDE AIR LOUVER WITH DUCTWORK PLENUM AND MOTORIZED DAMPER BEFORE OPENING.
10. WALL PIPE SLEEVE, SEE DETAIL A3 ON SHEET M-501.
11. PROVIDE VRF REFRIGERANT PIPING WALL SUPPORT BRACKET, SEE DETAIL C1 ON SHEET M-502.
12. NEW WINDOW, SEE ARCHITECTURAL SHEETS.
13. PROVIDE INSULATED LOUVER FASCIA COVER THAT MATCHES THE LOUVER MATERIAL AND COLOR, SEAL FOR WATER TIGHT FINISH.



LAUNDRY ROOM EXTERIOR SECTION ELEVATION
SCALE: 1/4" = 1'-0" M-401 B2



LAUNDRY ROOM INTERIOR SECTION ELEVATION
SCALE: 1/4" = 1'-0" M-401 A2

GRAPHIC SCALE

1/4" = 1'-0" 0 2' 4' 8'

1/2" = 1'-0" 0 1' 2' 4'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

SEAL

A/E INFO

APPROVED
David Greenfield
David W Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON
SATISFACTORY TO DATE 06/18/2012

DES	JLS	DRW	JLS	CHK	JM

<<PM/DM>> RS/NW
BRANCH MANAGER [signature]
CHIEF ENG/ARCH DWG
FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NORTHEAST IPT NAVAL STATION – NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON GROTON, CT

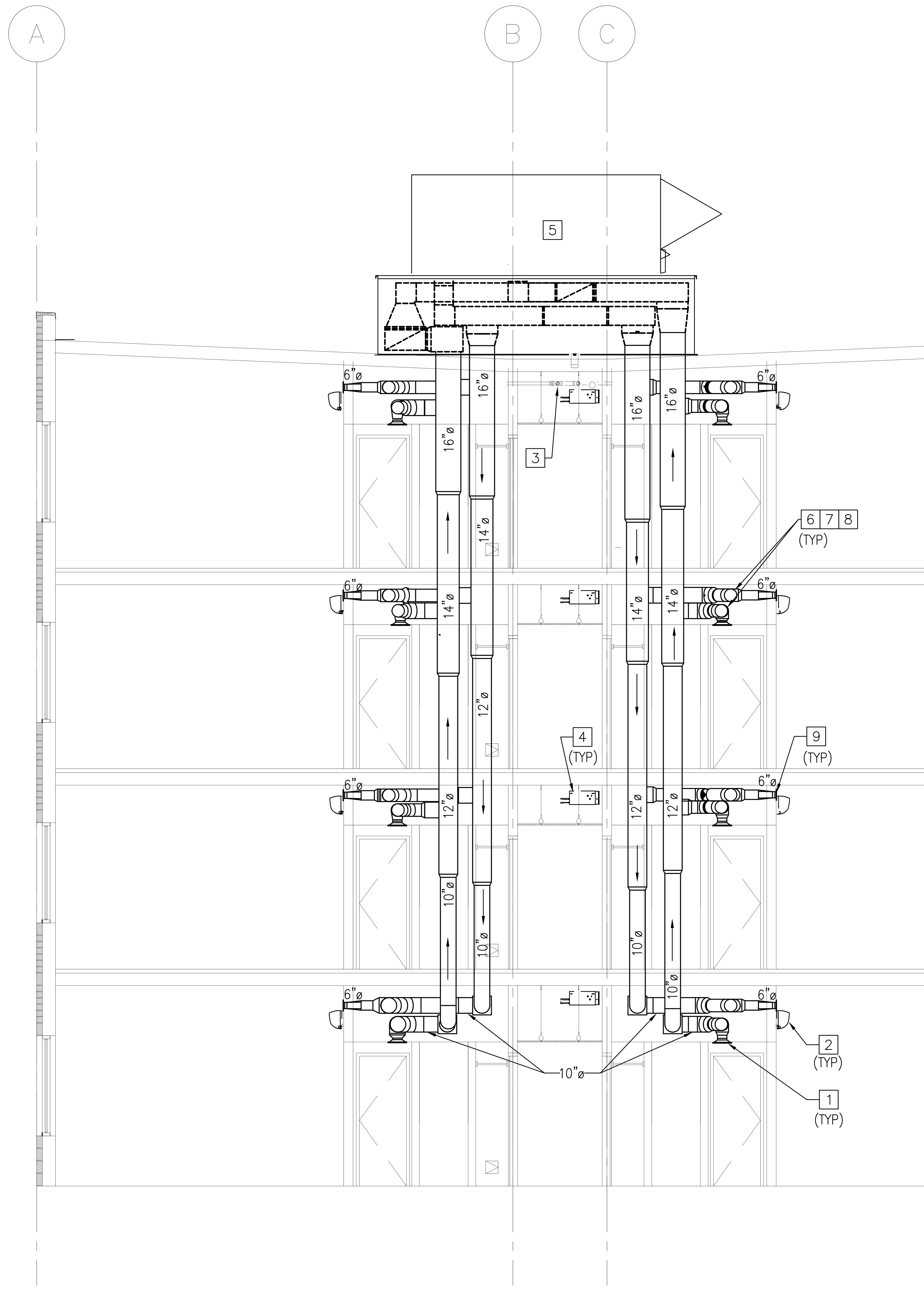
REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL – ELEVATION SECTION
LAUNDRY & STORAGE RM 128

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624726
SHEET 153 OF 206

M-504

DRAWFORM REVISION: 10 MARCH 2009



MECHANICAL - ELEVATION BUILDING SECTION

1/4" = 1'-0"

M-101 A3

GENERAL SHEET NOTES

1. INSTALL DUCTWORK AS HIGH AS POSSIBLE AND TRANSITION TO MAINTAIN ELEVATION OF TOP OF DUCT, UNLESS OTHERWISE NOTED.
2. BALANCE AIRFLOWS FROM DOAS WITH ALL DOORS CLOSED.
3. COORDINATE ALL MECHANICAL WORK WITH ELECTRICAL AND FIRE PROTECTION WORK. VERIFY SPACE REQUIREMENTS ABOVE ALL DROP CEILING LOCATIONS FOR ALL DISCIPLINES WORK BEFORE STARTING INSTALLATION.
4. REUSE ALL EXISTING WALL OPENINGS, UNLESS OTHERWISE NOTED. REPAIR AND SEAL ALL OPENINGS NOT USED.
5. SEAL ALL WALL AND FLOOR PENETRATIONS WITH ACOUSTICAL SEALANT. PROVIDE WALL DUCT SLEEVE AT EVERY PENETRATION BY DUCTWORK.
6. THIS IS A 3-PIPE VRF REFRIGERANT SYSTEM WHICH REQUIRES A MANUFACTURER CERTIFIED CONTRACTOR FOR INSTALLATION. PROVIDE AND INSTALL VRF PIPING PER MANUF'S. SPECIFICATIONS. INSTALL ALL SOLENOID VALVE KITS ON ALL PIPING PER MANUFACTURES RECOMMENDATIONS.
7. PROVIDE FUSED POWERED DISCONNECTS FOR VRF SYSTEM, PER MANUFACTURES RECOMMENDATIONS. IF MANUFACTURE RECOMMENDS A DISCONNECT FOR EACH INDOOR UNIT, INSTALL DISCONNECT CLOSE TO UNIT. SEE ELECTRICAL DRAWINGS FOR POWER WIRING.
8. INSTALL ALL HEAT PUMPS PER MANUFACTURES RECOMMENDATIONS AND DETAIL SHEET M-502.

NEW WORK KEYNOTES

1. NEW EXHAUST GRILLE.
2. NEW VRF WALL MOUNTED UNIT.
3. EXISTING ROOF DRAIN PIPE WITH CLEANOUT AND EXISTING FIRE PROTECTION PIPING.
4. NEW HEAT RECOVERY UNIT.
5. NEW DOAS WITH CURB PROVIDED BY MANUFACTURER. CONTRACTOR TO WORK WITH MANUFACTURE ON SIZE AND DUCTWORK FROM THE UNIT FOR THE LOWEST POSSIBLE HEIGHT OF CURB. UNIT AND CURB MUST BE SECURED AND RATED FOR HURRICANE FORCE WINDS.
6. DUCTWORK TO BE STACKED AT THRU WALL PENETRATIONS PER DETAIL A2 SHEET AD-402.
7. WALL AREA TO BE CHECK FOR STRUCTURAL CORE FILL. RUN DUCTWORK BETWEEN CORE FILL AREAS AND UNDER BOND BEAM.
8. ALL DUCTWORK AND PIPING WALL PENETRATIONS NEED TO BE SEALED PER DETAIL B3 AND A3 ON SHEET M-501, UNLESS OTHERWISE NOTED.
9. NEW OUTSIDE AIR SIDEWALL SUPPLY GRILLE.

GRAPHIC SCALE

1/4" = 1'-0" 0 2' 4' 8'

SYM | DESCRIPTION | DATE | APPR

NAVFAC

SEAL

A/E INFO

APPROVED: David W. Greenfield
FOR COMMANDER NAVFAC

ACTIVITY: PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC - SUBASE NEW LONDON

SATISFACTORY TO / DATE 06/18/2012

DES JLS | DRW JLS | CHK JLS

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION SLW

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NORTHEAST IPT
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
MECHANICAL - ELEVATION BUILDING SECTION

SCALE: AS NOTED

EPROJECT NO.: 1127117

CONSTR. CONTR. NO.: N40085-12-R-1715

NAVFAC DRAWING NO.: 12624727

SHEET 154 OF 206

M-511

DRAWFORM REVISION: 10 MARCH 2009

100% DEDICATED OUTSIDE AIR UNIT SCHEDULE

ODU # MARK	LOCATION	SUPPLY FAN DATA: SUPPLY AIR CFM	SUPPLY FAN DATA: OUTSIDE AIR CFM	SUPPLY FAN DATA: ESP (IN. WG)	SUPPLY FAN DATA: HP	EXHAUST FAN DATA: EXHAUST AIR CFM	EXHAUST FAN DATA: ESP (IN. WG)	EXHAUST FAN DATA: HP	ENERGY RECOVERY SECTION: EFFECTIVENESS	NATURAL GAS HEATER: OUTPUT MBH	NATURAL GAS HEATER: EFF. MBH	NATURAL GAS HEATER: EAT °FDB	NATURAL GAS HEATER: LAT °FDB	COOLING COIL DATA: TOTAL MBH	COOLING COIL DATA: SENSIBLE MBH	COOLING COIL DATA: EAT °FDB	COOLING COIL DATA: EAT °FWB	COOLING COIL DATA: LAT °FDB	COOLING COIL DATA: LAT °FWB	CONDENSING SECTION: EAT °FDB	HOT GAS REHEAT COIL: MIN. MBH	HOT GAS REHEAT COIL: EAT °FDB	HOT GAS REHEAT COIL: LAT °FDB	ELECTRICAL DATA: FLA	ELECTRICAL DATA: VOLTS	ELECTRICAL DATA: PHASE	ELECTRICAL DATA: HERTZ	ELECTRICAL DATA: MCA	ELECTRICAL DATA: MOP	FILTER TYPE	DIRTY FILTER ALLOWANCE (IN. WG)	FILTER AREA SF	REMARKS
DOAS–1	ROOF	4065	4065	2.00	5	4065	2.0	2	79%	160	0.80	50.1	86.4	187.6	118.8	80.1	67.5	53.2	52.6	50.1	93.5	50.1	86.4	–	208	3	60	97.1	125.0	PLEATED	0.23	–	1,2,3,4,5
DOAS–2	ROOF	4065	4065	2.00	5	4065	2.0	2	79%	160	0.80	50.1	86.4	187.6	118.8	80.1	67.5	53.2	52.6	50.1	93.5	50.1	86.4	–	208	3	60	97.1	125.0	PLEATED	0.23	–	1,2,3,4,5

NOTES:

1. UNIT SELECTION BASED ON VALENT MANUFACTURER'S RECOMMENDATIONS OR APPROVED EQUAL.
2. PROVIDE UNIT WITH INDIRECT–FIRED NATURAL–GAS HEAT, STAINLESS–STEEL HEAT EXCHANGER TO PREVENT CORROSION FROM CONDENSATION, DOUBLE WALL CONSTRUCTION, SEA–COAST COATING IN THE UNIT, FILTER SECTION, PLEATED 2" MERV 8 THROWAWAY FILTERS, A 2ND COMPLETE SET OF FILTERS, FACTORY–INSTALLED VARIABLE FREQUENCY DRIVES AND DDC CONTROLS INCLUDING SUPPLY TEMPERATURE SENSOR, SINGLE POINT POWER CONNECTION, FACTORY MOUNTED AND WIRED 120 VOLT RECEPTACLE, UNIT–MOUNTED FUSED DISCONNECT, AND LOW VOLTAGE TRANSFORMERS AS REQUIRED. PROVIDE MANUF.'S NATURAL GAS CONNECTIONS AND INSTALL PER MANUF.'S RECOMMENDATIONS. UNIT FANS ARE CONSTANT VOLUME.
3. UNIT CURB TO BE PROVIDED BY MANUFACTURER TO ACCOMMODATE THE OUTLINE SHOWN ON THE DRAWINGS. CONTRACTOR AND MANUFACTURER TO COORDINATE SIZING OF ROOF CURB FOR THE CONDITIONS.
4. UNIT SELECTION BASED ON A MODULE 100% OA UNIT OR APPROVED EQUAL.
5. TOTAL OPERATING WEIGHT INCLUDING CURB SHALL NOT EXCEED 4000 POUNDS. MAX CURB HEIGHT = 3'–6". MAX PLAN DIMENSIONS OF UIT = 14'–11"X 7'–10". IF THE DIMENSIONS OF EITHER CURB OR UNIT EXCEED THE MAXIMUM DIMENSIONS GIVEN, THE CONTRACTOR SHALL DESIGN, FURNISH, & INSTALL STRUCTURAL REINFORCEMENT FOR THE EXISTING ROOF STRUCTURE, AT NO ADDITIONAL COST TO THE GOVERNMENT.

INLINE FAN SCHEDULE

MARK	SERVICE	AREA OR EQUIP SERVED	TYPE	CFM	ESP (IN. WG)	FAN DATA: RPM	FAN DATA: TIP SPEED FPM	FAN DATA: MAX. SONES	ELECTRICAL DATA: HP	ELECTRICAL DATA: VOLTS	ELECTRICAL DATA: PHASE	REMARKS
ILF–1	EXHAUST	LAUNDRY RM 124	BELT DRIVE	2250	0.5	2423	––	24	1	208	3	1,2,3
ILF–2	EXHAUST	LAUNDRY RM 126	BELT DRIVE	4200	0.5	1766	––	21	2	208	3	1,2,3

NOTES:

1. PROVIDE (FROM MANUF., WHERE POSSIBLE): VIBRATION ISOLATORS, AND MOTORIZED DAMPER ACTUATORS FOR THE LOUVER INTAKE & EXHAUST CONTROL DAMPERS.
2. NEW INLINE FANS TO BE CONTROLLED WITH NEW CONTROL SYSTEM.
3. NEW INLINE FAN TO USE NEW WALL LOUVER. SEAL TIGHT FOR NO AIR GAPS.

FAN SCHEDULE

MARK	SERVICE	AREA OR EQUIP SERVED	TYPE	CFM	ESP (IN. WG)	FAN DATA: RPM	FAN DATA: TIP SPEED FPM	FAN DATA: MAX. SONES	ELECTRICAL DATA: HP	ELECTRICAL DATA: VOLTS	ELECTRICAL DATA: PHASE	REMARKS
EF–1	EXHAUST	MECH ROOM	DIRECT DRIVE	1700	0.10	860	––	5.0	1/8	120	1	1,3
EF–2	EXHAUST	ELEVATOR MACHINE RM	DIRECT DRIVE	5000	0.20	–––	––	––	1/2	208	3	1,2
CF–1	CIRC.	SLEEPING ROOMS	DIRECT DRIVE	6900	–––	–––	––	––	91W	120	1	4

NOTES:

1. PROVIDE (FROM MANUF., WHERE POSSIBLE): WALL–MOUNTED THERMOSTAT, FAN IS VARIABLE AIR VOLUME; PROVIDE VFD (WITH INTEGRAL DISCONNECT), VIBRATION ISOLATORS, AND MOTORIZED DAMPER ACTUATORS FOR THE INTAKE & EXHAUST CONTROL DAMPERS.
2. EXISTING EXHAUST FANS TO BE CHECKED AND CONTROLLED WITH NEW CONTROL SYSTEM.
3. NEW SIDEWALL EXHAUST FAN TO USE EXISTING WALL OPENING. SEAL TIGHT FOR NO AIR GAPS.
4. PROVIDE 54" FAN, 3 SPEED, MAX 32 LBS. PROVIDE CEILING MOUNTING RODS TO ALLOW FAN INSTALLATION 8' ABOVE FINISHED FLOOR. PROVIDE 3 SPEED FAN CONTROL WALL SWITCH.

DUCT CONSTRUCTION AND LEAK TEST SCHEDULE

MARK	DUCT PRESSURE CLASS (IN. WC): SUPPLY DUCT	DUCT PRESSURE CLASS (IN. WC): RETURN DUCT	DUCT PRESSURE CLASS (IN. WC): EXHAUST DUCT	DUCT PRESSURE CLASS (IN. WC): OA DUCT	SUPPLY ROUND/OVAL: SEAL CLASS	SUPPLY ROUND/OVAL: LEAK CLASS	SUPPLY RECTANGLE: SEAL CLASS	SUPPLY RECTANGLE: LEAK CLASS	RETURN/EXHAUST/OUTSIDE AIR: SEAL CLASS	RETURN/EXHAUST/OUTSIDE AIR: LEAK CLASS	DUCT TEST PRESSURE (IN. WC)	NOTES
DOAS–1&2	2	–1	–1	–1	A	3	–	A	A	12	2	1
	2	–1	–1	–1	A	3	–	A	A	12	2	1
INL–1,2,3	2	–1	–1	–1	A	3	A	12	A	12	2	1

NOTES:

1. TEST IN ACCORDANCE WITH HVAC TESTING/ADJUSTING/BALANCING SPECIFICATIONS AND THE PROCEDURES IN SMACNA HVAC AIR DUCT LEAKAGE TEST MANUAL. CONSTRUCT AND SEAL PER SMACNA HVAC DCS, LATEST EDITION.
2. DUCT TEST PRESSURE SHALL BE 100% OF DUCT PRESSURE CLASS.

DESIGN CONDITIONS °F

	COOLING DB	COOLING WB	HEATING DB
OUTSIDE	85	72	9
MECH ROOM	VENTILATED		55
INSIDE ALL OTHER SPACES	76	63.4	72

UNIT HEATER SCHEDULE

MARK	HEATING (MBH)	NO. OF STAGES	CFM	ELECTRICAL: KW	ELECTRICAL: VOLTS/PH	ELECTRICAL: AMPS	ELECTRICAL: MOTOR HP	MOUNTING	FILTER TYPE	REMARKS
UH–1	17	1	350	5.0	208/1	24.0	1/100	CEILING	NONE	1,2,3,4,5
UH–2	25.6	1	650	7.5	208/1	36.0	1/30	CEILING	NONE	1,2,3,4,5

NOTES:

1. ELECTRIC UNIT HEATER
2. UNIT–MOUNTED THERMOSTAT, WITH INTERNAL DISCONNECT.
3. CABINET UNIT HEATER WITH FAN.
4. INSTALL WITH MANUFACTURER'S RECOMMENDED CLEARANCES ON ALL SIDES.
5. HORIZONTAL FAN–POWERED UNIT HEATER. INSTALL AS HIGH AS POSSIBLE WITHOUT OBSTRUCTION TO AIRFLOW.
6. WALL–MOUNTED THERMOSTAT

LOUVER SCHEDULE

MARK	TYPE	LOCATION	SIZE (IN.) WIDTH	SIZE (IN.) HEIGHT	AIRFLOW (CFM)	MAX. PD (IN. WG)	MIN. NET FREE AREA (SF)	REMARKS
LV–1	INTAKE	LAUNDRY ROOM	56	40	2300	<0.07	2.90	1,3,4
LV–2	EXHAUST	LAUNDRY ROOM	56	24	2300	<0.01	1.36	1,3,4
LV–3	INTAKE	LAUNDRY ROOM	56	40	4200	<0.34	2.06	1,3,4
LV–4	EXHAUST	LAUNDRY ROOM	56	24	4200	<0.70	3.53	1,3,4
LV–5	INTAKE	ELEVATOR MACH.	–	–	5000	<0.01	–	2
LV–6	INTAKE	MECH ROOM	24	24	1700	<0.01	–	2

NOTES

1. PROVIDE FLANGED–FRAME DRAINABLE LOUVERS TO MOUNT IN BLOCK WALLS. PROVIDE INSECT SCREEN ON BUILDING–INTERIOR SIDE.
2. EXISTING LOUVER MOTORIZED DAMPERS TO BE CHECKED FOR WORKING ORDER AND CONNECTED TO THE NEW DDC CONTROL SYSTEM. PROVIDE NEW BIRDSCREEN.
3. PROVIDE NEW MOTORIZED BACKDRAFT DAMPER.
4. LOUVERS TO BE SPLIT WITH A SPACE AND BRACING IN BETWEEN THEM. THIS SPACING TO BE INSULATED AND HAVE LOUVER MATCHING FACEPLATE COVER.

GRILLE, REGISTER, AND DIFFUSER SCHEDULE

MARK	SERVICE	CFM RANGE	MAX. NC	MAX. PD IN. H2O	NECK SIZE IN. SQ. OR DIA.	SHELL SIZE	THROW FT (AT 50 FPM TERMINAL VEL.)	REMARKS
A	SUPPLY	30	<10	0.013	5ø	8X6	10	1,3,8,10
B	SUPPLY	60	<10	0.013	5ø	10X6	15	1,3,8,10
C	SUPPLY	45	<10	0.031	6ø	48"	12	1,2,11
D	RETURN	60	<10	0.032	5ø	6X6	–	1,3,9
E	RETURN	60	<10	0.02	6ø	24X24	–	1,2,9
F	RETURN	40	<10	0.022	6ø	24X24	–	1,2,9
G	RETURN	30	<10	0.02	6ø	24X24	–	1,2,9
H	RETURN	60	<10	0.02	6ø	24X24	–	1,2,9
J	RETURN	200	<10	0.02	8ø	24X24	–	1,2,9
K	SUPPLY	200	<10	0.002	8ø	24X24	20	1,2,5
L	SUPPLY	30	<10	0.002	6ø	24X24	10	1,2,5
M	RETURN	55	<10	0.002	6ø	24X24	–	1,2,9
N	RETURN	50	<10	0.002	6ø	24X24	–	1,2,9
O	SUPPLY	55	<10	0.031	6ø	6X6	–	1,3,9

NOTES:

1. ALL ALUMINUM CONSTRUCTION
2. LAY–IN TYPE
3. SURFACE MOUNT TYPE
4. 4–WAY THROW, SQUARE
5. 3–WAY THROW, SQUARE
6. 2–WAY THROW, OPPOSITE SIDES, SQUARE
7. 2–WAY THROW, CORNER, SQUARE
8. 1–WAY THROW, SQUARE
9. DESIGN BASIS: RETURN GRILLES
10. DESIGN BASIS: SIDEWALL SUPPLY GRILLE, SIZE FOR CFM
11. DESIGN BASIS: SLOT DIFFUSER (4 FOOT LONG)

A/E INFO

APPROVED David Greenfield
David W Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON
SATISFACTORY TO DATE 06/18/2012
DES ??? DRW ??? CHK ???
<<PM/DM>> RS/NW
BRANCH MANAGER
CHIEF ENG/ARCH DWG
FIRE PROTECTION SLW

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NORTHEAST IPT NAVAL STATION – NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
MECHANICAL SCHEDULES

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085–12–R–1715
NAVFAC DRAWING NO. 12624728
SHEET 155 OF 206
M-601

DRAWFORM REVISION: 10 MARCH 2009

VARIABLE REFRIGERANT FLOW SYSTEM WITH HEAT RECOVERY UNITS

INDOOR UNIT

MARK	FAN DATA						COIL PERFORMANCE (1)							FILTER TYPE	REMARKS	LOCATION
	TOTAL CFM EA	OA CFM EA	ESP (IN. WG)	ELECTRICAL			COOLING				HEATING					
				MCA	VOLTS/PH	NEMA STARTER	TTL (BTU/H)	SEN. (BTU/H)	EAT (°FDB/°FWB)	LAT (°FDB/°FWB)	DESIGN (BTU/H)	SUPP. HEAT (KW)	EAT (°FDB)			
VRF-1	150	30	N/A	0.8	208/1	N/A	8000	7232	79.4/65.5	63.0/59.3	8690	N/A	63.0	PER MANUFACTURER	1,3,8,10	ALL BEDROOMS EACH
VRF-2	335	40	N/A	0.8	208/1	N/A	22275	20450	78.0/65.0	64.7/60.4	22625	N/A	64.7		1,2,9,10	1ST LOBBY 2, 3, 4 EACH
VRF-3	335	N/A	N/A	0.8	208/1	N/A	–	–	–/–	–/–	22625	N/A	64.7		1,2,7,10	1ST LOBBY VESTIBULE
VRF-4	180	50	N/A	0.8	208/1	N/A	600	0.0	77.7/65.6	70.7/63.3	900	N/A	70.0		1,2,7,10	NEW STORAGE RM 135
VRF-5	60	N/A	N/A	0.8	208/1	N/A	5360	4740	77.1/64.7	60.5/58.9	5780	N/A	60.5		1,2,8,10	SECURITY CO RM 134
VRF-6	60	N/A	N/A	0.8	208/1	N/A	–	–	–/–	–/–	3200	N/A	74.5		1,3,8,10	1ST STAIRWELL N/S
VRF-7	210	N/A	N/A	0.8	208/1	N/A	1250	1050	77.7/65.6	55.7/55.7	5750	N/A	55.7		1,3,8,10	1ST CORR NORTH
VRF-8	210	20	N/A	0.8	208/1	N/A	1250	1050	77.7/65.6	55.7/55.7	5750	N/A	55.7		1,2,6,10	1ST CORR SOUTH
VRF-9	80	60	N/A	0.8	208/1	N/A	2900	2000	77.6/64.9	62.5/59.6	3400	N/A	62.5		1,2,9,10	VENDING/CONF RM
VRF-10	280	N/A	N/A	0.8	208/1	N/A	6700	6300	78.3/65.1	64.1/59.4	7800	N/A	64.1		1,2,9,10	FRONT OFFICE RM 113
VRF-11	500	30	N/A	0.8	208/1	N/A	15500	13500	78.3/65.1	64.1/59.4	22400	N/A	60.5		1,2,8,10	SECURITY RM 114
VRF-12	390	40	N/A	0.8	208/1	N/A	79000	68000	77.8/65.0	61.1/58.9	78000	N/A	61.1		1,2,9,10	GAME RM
VRF-13	560	40	N/A	0.8	208/1	N/A	93000	87000	76.7/65.9	59.1/58.1	84000	N/A	63.2		1,2,9,10	MEDIA RM
VRF-14	280	50	N/A	0.8	208/1	N/A	2900	2600	77.4/64.8	64.2/60.2	4400	N/A	64.2		1,2,6,10	LOBBY FL 2,3, & 4
VRF-15	185	55	N/A	0.8	208/1	N/A	3850	3200	77.7/65.6	55.0/55.0	5600	N/A	63.0		1,4,8,10	CORRIDORS FL 2,3 & 4
VRF-16	280	N/A	N/A	0.8	208/1	N/A	12300	86000	86.0/73.0	80.6/67.1	26670	N/A	68.0		1,2,9,10	LAUNDRY RM 3 UNITS EA
–	–	–	–	–	–	–	–	–	–	–	–	–	---		–	–
–	–	–	–	–	–	–	–	–	–	–	–	–	---		–	–

HEAT RECOVERY CONTROLLER UNIT

MARK	LOCATION AREA	TOTAL BRANCHES	MAX. CONNECTED CAPACITY UNITS	ELECTRICAL				REMARKS
				PS MCA	Hz	VOLTS/PH	WATTS	
HRU-2	ALL FLOORS	2	16	0.1	60	208/230/1	14	1&2
HRU-3	ALL FLOORS	3	24	0.15	60	208/230/1	21	1&2
HRU-4	ALL FLOORS	4	32	0.2	60	208/230/1	29	1&2

NOTES:

1. REFER TO MANUFACTURER'S INSTRUCTIONS FOR INSTALLATION.
2. UNITS NEED TO HAVE HIGH AND LOW GAS LINES FOR REFRIGERANT FOR HEATING AND LIQUID LINE. ALL REFRIGERANT LINES COME FROM OUTDOOR UNIT.

VARIABLE REFRIGERANT FLOW SYSTEM OUTDOOR UNITS

OUTDOOR UNIT

VRF # MARK	NOMINAL TONS COOLING (1)	REFRIG. TYPE	ELECTRICAL					MIN. SYSTEM EFF.		SERVES (FL)	REMARKS
			VOLTS/HZ/PH	MCA	MOP	AMPS	HP	EER	COP		
VRF ODU-1	36	R-410A	208-230/60/3	49.9	70	44.8	42	11.3	3.33	1ST	12
VRF ODU-2	22	R-410A	208-230/60/3	49.9	70	44.8	26	11.3	3.33	2-4	12
VRF ODU-3	18	R-410A	208-230/60/3	49.9	70	44.8	22	11.2	3.53	2-4	12
VRF ODU-4	3	R-410A	208-230/60/1	29	50	22.5	1	11.1	N/A	LAUNDRY RM	12
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NOTES:

1. PROVIDE MANUF.'S WALL-MOUNTED TEMPERATURE SENSOR AND CONTROLLER. PROVIDE MANUF.'S CONDENSATE PUMP. PROVIDE FUSED DISCONNECT, IF RECOMMENDED BY MANUF. PROVIDE THROWAWAY FILTER IF APPROVED BY MANUF., OTHERWISE PROVIDE MANUF.'S FILTER. PROVIDE BV-SERIES BALL VALVES AS REQUIRED BY MANUF. QUANTITY OF INDOOR UNITS SERVED BY EACH BV-SERIES BALL VALVES MAY VARY, BASED ON MANUF.
2. CEILING-CASSETTE TYPE MOUNTED IN SUSPENDED CEILING.
3. WALL-MOUNTED TYPE.
4. FLOOR CABINET MOUNTED TYPE.
5. DUCTED UNIT TYPE (DUCT MOUNTED).
6. 3-WAY SUPPLY AIRFLOW PATTERN, OPPOSITE SIDES.
7. 2-WAY SUPPLY AIRFLOW PATTERN, OPPOSITE SIDES.
8. 1-WAY SUPPLY AIRFLOW PATTERN.
9. 4-WAY SUPPLY AIRFLOW PATTERN.
10. REFER TO CONTRACT SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS. VRF REFRIGERANT PIPING LAYOUT AND BV-SERIES BALL VALVES MAY VARY BASED ON MANUF. COORDINATE WITH VRF SYSTEM MANUF. TO PROVIDE NEW PIPING LAYOUT AND EQUIPMENT LOCATIONS WHERE DEVIATIONS EXIST BETWEEN CONTRACT DRAWINGS AND APPROVED VRF SYSTEMS. SUBMIT SCALED DRAWINGS OF ALL VRF SYSTEM LAYOUTS TO GOVERNMENT, SHOWING ALL VRF OUTDOOR AND INDOOR UNITS, SOLENOID VALVE KITS, REFRIGERANT PIPING, CONNECTED DUCTWORK, CONTROL PANELS, AND CABINETS TO HOUSE MANUF.'S LITERATURE. DO NOT INSTALL VRF SYSTEM BEFORE GOVERNMENT APPROVAL IS OBTAINED. INSTALL VRF SYSTEM PER MANUF.'S RECOMMENDATIONS. COORDINATE WITH ELECTRICAL CONTRACTOR BEFORE AND DURING VRF SYSTEM INSTALLATION.
11. PROVIDE WITH MANUF.'S SUPPLY BRANCH DUCT OPTION.
12. PROVIDE UNIT-MOUNTED FUSED DISCONNECT. PROVIDE MANUF.'S COATINGS FOR COASTAL ENVIRONMENT. COORDINATE WITH VRF SYSTEM MANUF. TO DETERMINE NOMINAL CAPACITY OF VRF OUTDOOR UNIT.

(1) BASED ON: COOLING @ 85°F DB OAT
HEATING @ 9°F DB OAT

SYM | DESCRIPTION | DATE | APPR

SEAL

A/E INFO

APPROVED
David Greenfield
David W Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON

SATISFACTORY TO DATE 06/18/2012

DES JLS | DRW JLS | CHK JM

<<PM/DM>> RS/NW

BRANCH MANAGER

CHIEF ENG/ARCH DWG

FIRE PROTECTION S/W

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION – NORFOLK, VIRGINIA
NORTHEAST IPT

NAVSUBASE NEW LONDON GROTON, CT

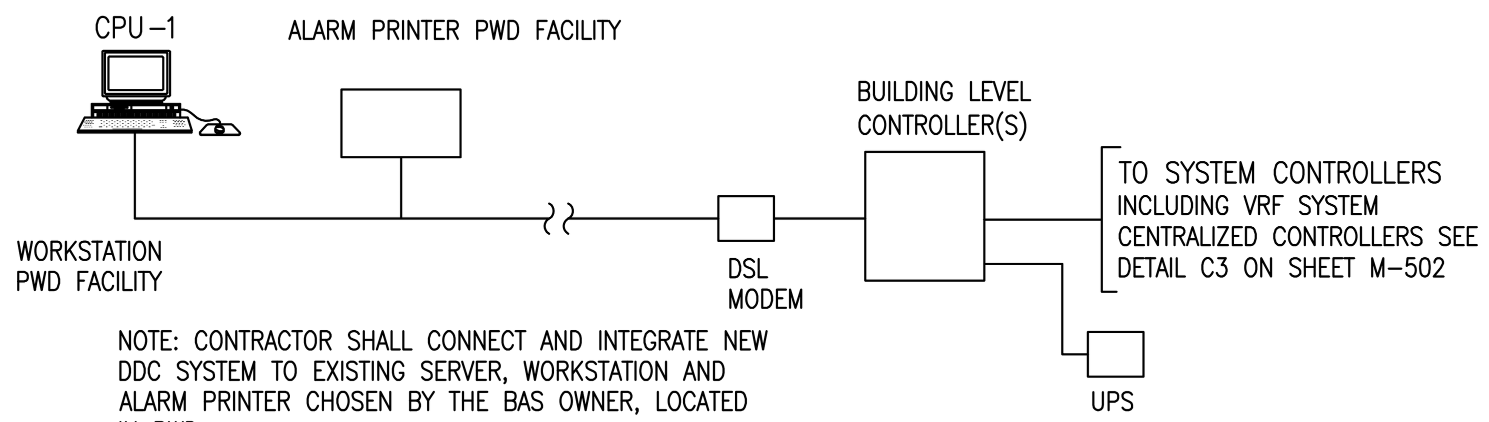
REPAIRS TO SUBMARINE A SCHOOL BQ 534

MECHANICAL SCHEDULES

SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624729
SHEET 156 OF 206

M-602

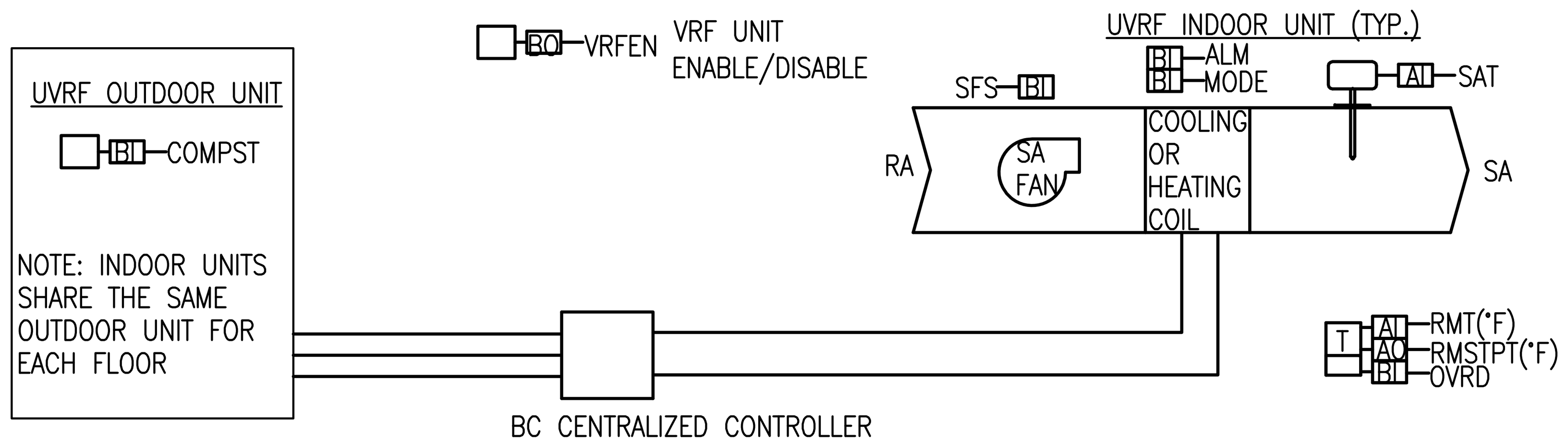
DRAWFORM REVISION: 10 MARCH 2009



NOTE: CONTRACTOR SHALL CONNECT AND INTEGRATE NEW DDC SYSTEM TO EXISTING SERVER, WORKSTATION AND ALARM PRINTER CHOSEN BY THE BAS OWNER, LOCATED IN PWD.

DDC CONNECTION TO EXISTING BAS

VARIABLE REFRIGERANT FLOW SYSTEM CONTROL DIAGRAM (TYP.)



NOTE: INDOOR UNITS SHARE THE SAME OUTDOOR UNIT FOR EACH FLOOR

SEQUENCE OF OPERATION

VARIABLE REFRIGERANT FLOW (VRF) SYSTEM (TYPICAL FOR ALL ROOMS)

THE DDC SHALL COMMUNICATE WITH THE MANUFACTURER'S CONTROLLER, AND THE MANUFACTURER'S CONTROLLER SHALL CONTROL THE VRF SYSTEM. SEE DETAIL C3 ON SHEET M-502.

VRF SYSTEM: THE SYSTEM SHALL RUN CONTINUOUSLY. SPACE TEMPERATURE SETPOINTS SHALL BE DETERMINED BY BUILDING OCCUPANCY MODE, AS DEFINED IN THE RUN CONDITIONS OF THE DOAS SEQUENCE OF OPERATION.

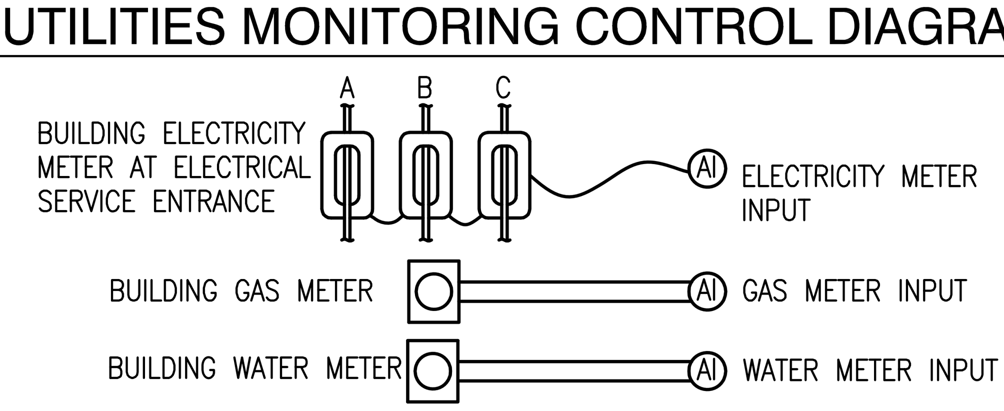
VRF INDOOR UNIT COOLING MODE: SPACE TEMPERATURE SETPOINTS – OCCUPIED 76°F (ADJUSTABLE), UNOCCUPIED 80°F (ADJUSTABLE). ON A RISE IN SPACE TEMPERATURE ABOVE SETPOINT, AS SENSED BY THE SPACE TEMPERATURE SENSOR, THE MANUFACTURER'S CONTROLLER SHALL MODULATE THE FAN SPEED AND REFRIGERANT FLOW TO MAINTAIN ITS COOLING SETPOINT.

VRF INDOOR UNIT HEATING MODE: SPACE TEMPERATURE SETPOINTS – OCCUPIED 70°F (ADJUSTABLE), UNOCCUPIED 65°F (ADJUSTABLE). ON A FALL IN SPACE TEMPERATURE BELOW SETPOINT, AS SENSED BY THE SPACE TEMPERATURE SENSOR, THE MANUFACTURER'S CONTROLLER SHALL MODULATE THE FAN SPEED AND REFRIGERANT FLOW TO MAINTAIN ITS HEATING SETPOINT.

VARIABLE REFRIGERANT FLOW SYSTEM (TYPICAL) – INPUT/OUTPUT SUMMARY

POINT	AI	AO	BI	BO	TREND	NOTES
(RMT) ROOM TEMPERATURE (A)	1				Y	ROOM TEMPERATURE SENSOR (WALL MOUNTED) DEGREES F
(RMSTPT) SPACE TEMP SETPOINT		1			Y	ADJUSTABLE THROUGH DDC AND LOCALLY DEGREES F
(OVRD) TENANT OVERRIDE			1		Y	INCLUDED WITH ROOM TEMPERATURE SENSOR
(SAT) SUPPLY AIR TEMPERATURE	1				Y	DUCT TEMPERATURE SENSOR DEGREES F
(MODE) HEATING OR COOLING MODE			1		Y	
(VRFEN) UNIT ENABLE/DISABLE				1	Y	ENABLE/DISABLE CONTROL RELAY
(SFS) SUPPLY AIR FAN STATUS (A)			1		Y	CURRENT SENSING RELAY
(COMPST) COMPRESSOR STATUS (A)			1		Y	CURRENT SENSING RELAY
(ALM) UNIT ALARM (A)			1		Y	
TOTAL	2	1	5	1		TOTAL POINTS

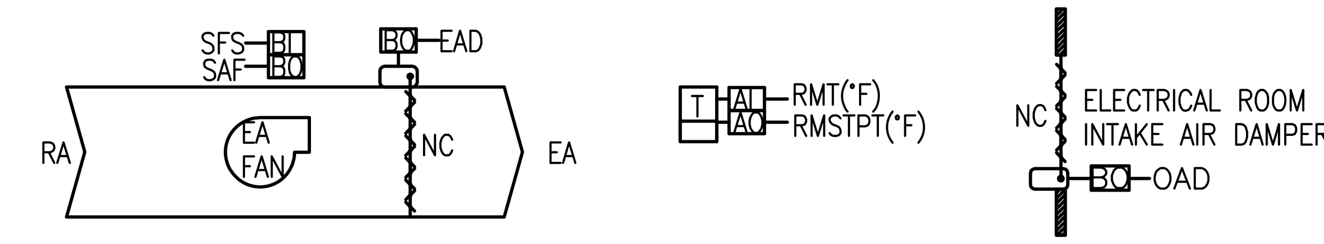
BUILDING UTILITIES MONITORING CONTROL DIAGRAM (TYP.)



BUILDING WATER, GAS AND ELECTRICAL MONITORING

POINT	AI	AO	BI	BO	TREND	NOTES
WATER METER	1				Y	
GAS METER	1				Y	
ELECTRICITY METER	1				Y	
	3					TOTAL POINTS

MECHANICAL ROOM EXHAUST FAN CONTROL DIAGRAM (TYP.)



SEQUENCE OF OPERATION

ELEVATOR MACHINE & MECHANICAL ROOM EXHAUST FANS (NEW & EXISTING)

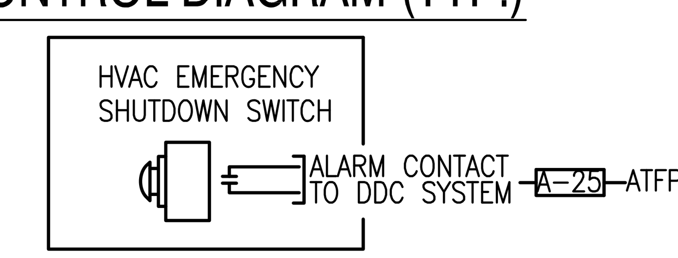
A DDC CONTROLLER, USING ELECTRIC ACTUATION, CONTROLS THE EXHAUST FAN AND INTAKE AND EXHAUST DAMPERS AS FOLLOWS:

EXHAUST FAN AND INTAKE AND EXHAUST DAMPER OPERATION: UPON A RISE IN ROOM SPACE TEMPERATURE ABOVE SETPOINT, 75°F (ADJ.), THE DDC SHALL OPEN THE INTAKE AND EXHAUST AIR DAMPERS, AND START THE EXHAUST FAN. UPON A FALL IN SPACE TEMPERATURE BELOW THE SETPOINT, THE REVERSE SHALL OCCUR. THE EXHAUST FAN STATUS SHALL BE MONITORED BY THE DDC THROUGH A CURRENT SWITCH.

ELEVATOR MACHINE & MECHANICAL ROOM EXHAUST FAN – INPUT/OUTPUT SUMMARY

POINT	AI	AO	BI	BO	TREND	NOTES
(SAF) ELEV. RM SUPPLY FAN START/STOP				1	Y	CONTROL RELAY
(SFS) ELEV. RM SUPPLY FAN STATUS (A)			1		Y	CURRENT SENSING RELAY
(RMT) ELEV. RM TEMPERATURE (A)	1				Y	ROOM TEMPERATURE SENSOR
(RMSTPT) ELEV. RM TEMPERATURE SETPOINT		1			Y	ADJUSTABLE THROUGH DDC AND LOCALLY
(EAD) SUPPLY FAN EXHAUST AIR DAMPER				1	Y	ELECTRONIC DAMPER ACTUATOR
(OAD) ELEV. RM OUTSIDE AIR DAMPER				1	Y	ELECTRONIC DAMPER ACTUATOR
	1	1	1	3		TOTAL POINTS

HVAC EMERGENCY SHUTDOWN SWITCH CONTROL DIAGRAM (TYP.)



SEQUENCE OF OPERATION

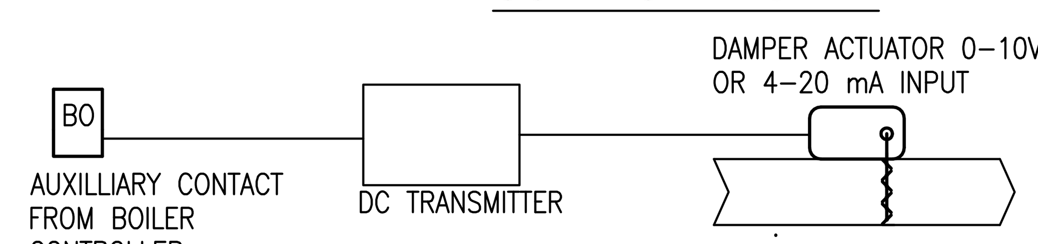
HVAC EMERGENCY SHUTDOWN (TYPICAL FOR ALL UNITS)

THE ONE HVAC EMERGENCY SHUTDOWN SWITCH SHALL BE WIRED TO THE MAIN DDC PANEL AND SHALL OPERATE AS FOLLOWS:

HVAC EMERGENCY SHUTDOWN SWITCH OPERATION: UPON ACTIVATION OF THE HVAC EMERGENCY SHUTDOWN SWITCH BY AN OCCUPANT, THE DEDICATED OUTSIDE AIR SYSTEM (DOAS) AND EXHAUST FANS SHALL SHUTDOWN. UPON DEACTIVATION OF THE SWITCH, THE DOAS AND EXHAUST FANS SHALL RETURN TO NORMAL OPERATION.

LOCATION: SHALL BE IN FIRST FLOOR LOBBY/QUARTERDECK THE ON THE WALL BY THE FIRE ALARM PULL BOX.

MECHANICAL ROOM COMBUSTION AIR LOUVER AIRFLOW CONTROL DIAGRAM



NOTE: CONTRACTOR SHALL PROVIDE AND INSTALL ALL ELECTRICAL EQUIPMENT REQ'D, INCLUDING TRANSFORMERS, TRANSMITTERS AND DAMPER ACTUATORS.

SEQUENCE OF OPERATION

MOTORIZED SUPPLY DAMPER IN MECHANICAL ROOM

MOTORIZED SUPPLY DAMPER OPERATION: MOTORIZED DAMPER SHALL BE CONTROLLED BY ITS RESPECTIVE EQUIPMENT CONTROLLER. UPON WATER START SIGNAL FROM HEATERS CONTROLLER THE MOTORIZED DAMPER SHALL OPEN. UPON STOP SIGNAL FROM HEATERS CONTROLLER THE MOTORIZED DAMPER SHALL CLOSE.

GENERAL SHEET NOTES

1. CONTROLS CONTRACTOR IS RESPONSIBLE FOR COORDINATING WITH THE BAS OWNER AND PROVIDING INTEGRATION OF THE NEW DDC SYSTEM TO THE EXISTING BACNET FRONT-END SERVER AND EXISTING APPLICATION SOFTWARE, WHICH SHALL BE CHOSEN BY THE BAS OWNER. DISPLAY ALL GRAPHIC FLOOR PLANS, EQUIPMENT GRAPHICS, DDC LADDER DIAGRAMS, AND SEQUENCE OF OPERATIONS GRAPHIC PAGES. PROVIDE ALL ALARMING, TREND SERVICES, SCHEDULES AND OTHER BACNET SERVICES AS DESCRIBED IN SPECIFICATION SECTION 23 09 23.13 20.
2. THE BAS OWNER AND DDC POC FOR THE PROJECT IS ? ? IN BUILDING A-?, DDC OFFICE, ROOM ?, ?. CONTACT THE BAS OWNER FOR ACCESS TO THE EXISTING BACNET SERVER AND INFORMATION REGARDING SOFTWARE, BACNET ADDRESSING, COMMUNICATIONS, TRAINING, SITE CONDITIONS, AND GENERAL TECHNICAL SUPPORT.
3. ALL OPERATOR WORKSTATION FUNCTIONS REQUIRING BACNET SERVICES, INCLUDING NAVIGATING THROUGH THE GRAPHIC DISPLAYS, TRENDING, ALARMING AND MONITORING OF THE NEW BACNET CONTROLS SYSTEM, MUST BE DEMONSTRATED FROM THE EXISTING BACNET FRONT-END SERVER USING ONLY THE EXISTING APPLICATION SOFTWARE, BOTH CHOSEN BY THE BAS OWNER, AND WITHOUT THE NEED TO LAUNCH OTHER APPLICATIONS OR LOGON TO OTHER VENDOR APPLICATIONS. INTEGRATE ALL NEW BACNET POINTS ON THE EXISTING SERVER SO THAT THERE IS A SEAMLESS LOGICAL FLOW FROM THE EXISTING FACILITIES TO THE NEW FACILITIES. TRENDED DDC POINTS SHALL INCLUDE ALL PHYSICAL AND SOFTWARE POINTS REQUIRED FOR REVIEWING AND ANALYZING THE BUILDINGS COMMISSIONED SYSTEMS.
4. CONTROLS CONTRACTOR IS RESPONSIBLE FOR COORDINATING WITH THE BAS OWNER FOR THE CORRECT DDC POINT NAMING CONVENTIONS.
5. MECHANICAL EQUIPMENT WHICH IS PROVIDED WITH MANUFACTURER'S INSTALLED CONTROLS SHALL BE STARTED UP BY THE MANUFACTURER'S REPRESENTATIVE IN CONJUNCTION WITH THE CONTROLS CONTRACTOR. CONTROLS CONTRACTOR IS RESPONSIBLE FOR COORDINATING WITH MANUFACTURER'S REPRESENTATIVE FOR THE COMMUNICATION OF THE DDC SYSTEM TO THE MANUFACTURER'S INSTALLED CONTROLS.

CONTROL LEGEND

T	SPACE TEMPERATURE SENSOR
CO2	SPACE CO2 SENSOR
P	WALL-MOUNTED POTENTIOMETER
	RELAY
	DUCT MOUNTED TEMPERATURE OR PRESSURE SENSOR
	CURRENT SENSING RELAY
	DAMPER ACTUATOR (ELECTRIC ACTUATOR)
	DAMPER
	FILTER
	FILTER DIFFERENTIAL PRESSURE SENSOR
AFMS	AIR FLOW MEASURING STATION

CONTROL ABBREVIATIONS

(A)	ALARMABLE DDC POINT
AI	ANALOG INPUT
AO	ANALOG OUTPUT
BI	BINARY INPUT
BO	BINARY OUTPUT
DISCH.	DISCHARGE
DSL	DIGITAL SUBSCRIBER LINE
DX	DIRECT EXPANSION COOLING COIL
EA	EXHAUST AIR
ELEC.	ELECTRICAL
RM	ROOM
LVG	LEAVING AIR
NC	NORMALLY CLOSED
NO	NORMALLY OPEN
OA	OUTSIDE AIR
RA	RETURN AIR
REC.	RECOVERY
SA	SUPPLY AIR
TEMP	TEMPERATURE
UPS	UNINTERRUPTABLE POWER SUPPLY
VFD	VARIABLE FREQUENCY DRIVE

SYM | DESCRIPTION | DATE | APPR
NAVFAC
APPROVED David W. Greenfield
FOR COMMANDER NAVFAC
ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON
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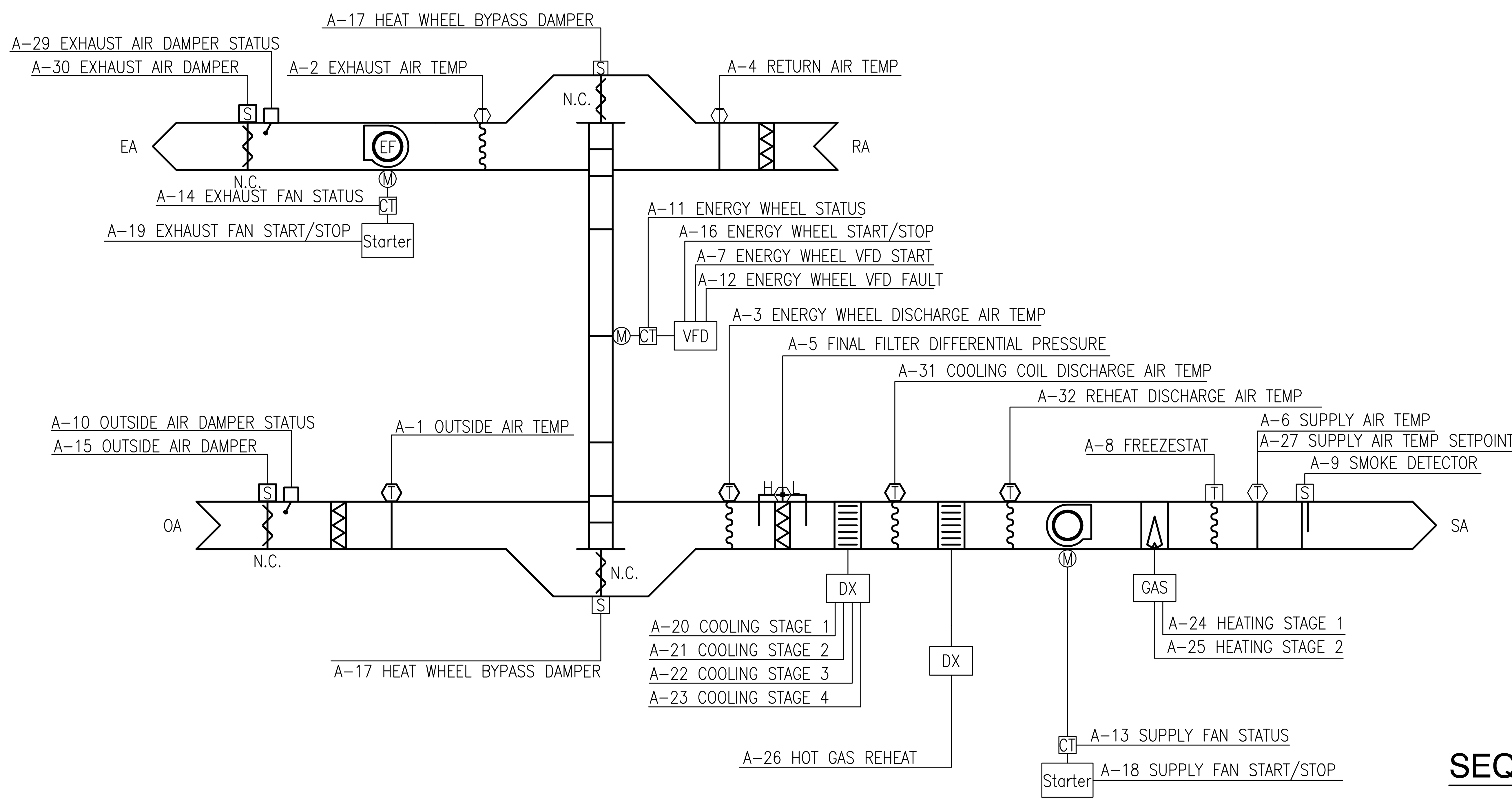
NAVAL STATION – NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON
GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
CONTROL DIAGRAMS
SCALE: AS NOTED
EPROJECT NO.: 1127117
CONSTR. CONTR. NO. N40085-12-R-1715
NAVFAC DRAWING NO. 12624730
SHEET 157 OF 206
M-701
DRAWFORM REVISION: 10 MARCH 2009

DOAS-1&2 CONTROL DIAGRAM



SEQUENCE OF OPERATION

DEDICATED OUTSIDE AIR UNIT

RUN CONDITIONS – CONTINUOUS: THE UNIT SHALL RUN CONTINUOUSLY.
EMERGENCY SHUTDOWN: THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING AN EMERGENCY SHUTDOWN SIGNAL.

FREEZE PROTECTION: THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING A FREEZESTAT STATUS.

SMOKE DETECTION: THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING A SMOKE DETECTOR STATUS.

OUTSIDE AIR & EXHAUST AIR DAMPERS: THE DAMPERS SHALL OPEN ANYTIME THE UNIT RUNS AND SHALL CLOSE ANYTIME THE UNIT STOPS. THE SUPPLY & EXHAUST FANS SHALL START ONLY AFTER THE DAMPER STATUS HAS PROVEN THE DAMPERS ARE OPEN. THE DAMPERS SHALL CLOSE 4SEC (ADJ.) AFTER THE FANS STOP.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- OUTSIDE AIR OR EXHAUST AIR DAMPER FAILURE: COMMANDED OPEN, BUT THE STATUS IS CLOSED.
- OUTSIDE AIR OR EXHAUST AIR DAMPER IN HAND: COMMANDED CLOSED, BUT THE STATUS IS OPEN.

ENERGY RECOVERY WHEEL – VARIABLE SPEED: THE CONTROLLER SHALL MODULATE THE ENERGY WHEEL FOR ENERGY RECOVERY AS FOLLOWS.

COOLING RECOVERY MODE: THE CONTROLLER SHALL MEASURE THE ENERGY WHEEL DISCHARGE AIR TEMPERATURE AND MODULATE THE ENERGY WHEEL SPEED TO MAINTAIN THE UNIT SUPPLY AIR TEMPERATURE SETPOINT OF 55°F. THE ENERGY WHEEL SHALL RUN FOR COOLING RECOVERY WHENEVER:

- UNIT RETURN AIR TEMPERATURE IS 5°F (ADJ.) OR MORE BELOW THE OUTSIDE AIR TEMPERATURE.
- AND THE UNIT IS IN A COOLING MODE.
- AND THE SUPPLY FAN IS ON.

HEATING RECOVERY MODE: THE CONTROLLER SHALL MEASURE THE ENERGY WHEEL DISCHARGE AIR TEMPERATURE AND MODULATE THE ENERGY WHEEL SPEED TO MAINTAIN THE UNIT SUPPLY AIR TEMPERATURE SETPOINT OF 55°F. THE ENERGY WHEEL SHALL RUN FOR HEAT RECOVERY WHENEVER:

- UNIT RETURN AIR TEMPERATURE IS 5°F (ADJ.) OR MORE ABOVE THE OUTSIDE AIR TEMPERATURE.
- AND THE UNIT IS IN A HEATING MODE.
- AND THE SUPPLY FAN IS ON.

PERIODIC SELF-CLEANING: THE ENERGY WHEEL SHALL RUN AT 5% SPEED (ADJ.) FOR 10SEC (ADJ.) EVERY 4HRS (ADJ.) THE UNIT RUNS.

FROST PROTECTION: THE ENERGY WHEEL SHALL RUN AT 5% SPEED (ADJ.) OR MANUFACTURER RECOMMENDED SPEED WHENEVER:

- OUTSIDE AIR TEMPERATURE DROPS BELOW 15°F (ADJ.)
- OR WHENEVER EXHAUST AIR TEMPERATURE DROPS BELOW 20°F (ADJ.) OR AS RECOMMENDED BY THE MANUFACTURER.

THE BYPASS DAMPERS WILL OPEN WHENEVER THE ENERGY WHEEL IS DISABLED.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- ENERGY WHEEL ROTATION FAILURE: COMMANDED ON, BUT THE STATUS IS OFF.
- ENERGY WHEEL IN HAND: COMMANDED OFF, BUT THE STATUS IS ON.

SEQUENCE OF OPERATION (CONTINUED)

- ENERGY WHEEL RUNTIME EXCEEDED: STATUS RUNTIME EXCEEDS A USER DEFINABLE LIMIT (ADJ.).
- ENERGY WHEEL VFD IN FAULT

SUPPLY FAN & EXHAUST FAN: THE FANS SHALL RUN ANYTIME THE UNIT IS COMMANDED TO RUN. TO PREVENT SHORT CYCLING, THE FANS SHALL HAVE A USER DEFINABLE (ADJ.) MINIMUM RUNTIME, UNLESS SHUTDOWN ON SAFETIES.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- SUPPLY FAN FAILURE: COMMANDED ON, BUT THE STATUS IS OFF.
- SUPPLY FAN IN HAND: COMMANDED OFF, BUT THE STATUS IS ON.
- SUPPLY FAN RUNTIME EXCEEDED: STATUS RUNTIME EXCEEDS A USER DEFINABLE LIMIT (ADJ.).
- EXHAUST FAN FAILURE: COMMANDED ON, BUT THE STATUS IS OFF.
- EXHAUST FAN IN HAND: COMMANDED OFF, BUT THE STATUS IS ON.
- EXHAUST FAN RUNTIME EXCEEDED: STATUS RUNTIME EXCEEDS A USER DEFINABLE LIMIT (ADJ.).

SUPPLY AIR TEMPERATURE SETPOINT – FIXED: THE CONTROLLER SHALL MONITOR THE SUPPLY AIR TEMPERATURE AND SHALL MAINTAIN A FIXED SUPPLY AIR TEMPERATURE SETPOINT OF 55°F (ADJ.).

COOLING STAGES: THE CONTROLLER SHALL MEASURE THE COIL DISCHARGE AIR TEMPERATURE AND STAGE THE COOLING TO MAINTAIN ITS COOLING SETPOINT OF 45°F (ADJ.). TO PREVENT SHORT CYCLING, THERE SHALL BE A USER DEFINABLE (ADJ.) DELAY BETWEEN STAGES, AND EACH STAGE SHALL HAVE A USER DEFINABLE (ADJ.) MINIMUM RUNTIME.

COOLING SHALL BE ENABLED WHENEVER:

- ENERGY WHEEL DISCHARGE AIR TEMPERATURE IS GREATER THAN 56°F FOR MORE THAN ONE CONSECUTIVE HOUR OR GREATER THAN 57°F (ADJ.).
- AND THE SUPPLY AIR TEMPERATURE IS ABOVE COOLING SETPOINT OF 55°F.
- AND THE FAN STATUS IS ON.

HOT GAS REHEAT: WHENEVER THE DX COOLING IS ENABLED HOT GAS REHEAT SHALL BE MODULATED TO MAINTAIN THE SUPPLY AIR TEMPERATURE SETPOINT OF 55°F.
HEATING MODE: GAS HEAT SHALL BE ENABLED WHENEVER:

- ENERGY WHEEL DISCHARGE AIR TEMPERATURE IS LESS THAN 54°F FOR MORE THAN ONE CONSECUTIVE HOUR OR LESS THAN 53°F (ADJ.).
- AND THE SUPPLY AIR TEMPERATURE IS BELOW SETPOINT OF 55°F.
- AND THE FAN STATUS IS ON.

HEATING STAGES: GAS HEAT SHAL MOUDALTE TO MAINTAIN SUPPLY AIR TEMPERATURE SET POINT OF 55°F.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- HIGH SUPPLY AIR TEMP: IF THE SUPPLY AIR TEMPERATURE IS GREATER THAN 58°F (ADJ.).
- LOW SUPPLY AIR TEMP: IF THE SUPPLY AIR TEMPERATURE IS LESS THAN 51°F (ADJ.).

FILTER DIFFERENTIAL PRESSURE MONITOR: THE CONTROLLER SHALL MONITOR THE DIFFERENTIAL PRESSURE ACROSS THE FILTERS.

ALARMS SHALL BE PROVIDED AS FOLLOWS: FILTER CHANGE REQUIRED: FILTER DIFFERENTIAL PRESSURE EXCEEDS A USER DEFINABLE LIMIT (ADJ.).

DEDICATED OUTSIDE AIR UNIT POINT LIST

UNIT LEVEL CONTROLLER

TAG	SYSTEM POINT DESCRIPTION (1)	TYPE	ALARM	TREND	NOTES
A-1	OUTSIDE AIR TEMP	AI		5 MIN	(3)
A-2	EXHAUST AIR TEMP	AI		5 MIN	(3)
A-3	ENERGY WHEEL DISCH AIR TEMP	AI		5 MIN	(3)
A-4	RETURN AIR TEMP	AI		5 MIN	(3)
A-5	FILTER DIFF PRESS	AI	ALARM		(3)
A-6	SUPPLY AIR TEMP	AI	50/60	5 MIN	(2) (3)
A-7	ENERGY WHEEL VFD	AO			(3)
A-8	FREEZESTAT	BI	ALARM	COV	(3)
A-9	SMOKE DETECTOR	BI	ALARM		(3)
A-10	OUTSIDE AIR DAMPER STATUS	BI	FAIL	COV	(3)
A-11	ENERGY WHEEL STATUS	BI	FAIL	COV	(3)
A-12	ENERGY WHEEL VFD FAULT	BI	FAIL		(3)
A-13	SUPPLY FAN STATUS	BI	FAIL	COV	(3)
A-14	EXHAUST FAN STATUS	BI	FAIL	COV	(3)
A-15	OUTSIDE AIR DAMPER	BO	FAIL	COV	(3)
A-16	ENERGY WHEEL START/STOP	BO	FAIL	COV	(3)
A-17	ENERGY WHEEL BYPASS DAMPER	AO	FAIL	COV	(3)
A-18	SUPPLY FAN START/STOP	BO	FAIL	COV	(3)
A-19	EXHAUST FAN START/STOP	BO	FAIL	COV	(3)
A-20	COOLING STAGE 1	BO	FAIL		(3)
A-21	COOLING STAGE 2	BO	FAIL		(3)
A-22	COOLING STAGE 3	BO	FAIL		(3)
A-23	COOLING STAGE 4	BO	FAIL		(3)
A-24	HEATING STAGE 1	BO	FAIL		(3)
A-25	HEATING STAGE 2	BO			(3)
A-26	HOT GAS REHEAT	AO	FAIL		(3)
A-27	SUPPLY AIR TEMP SETPOINT	AV		COV	(3)
A-28	EMERGENCY SHUTDOWN	BI	ALARM		(3)
A-29	EXHAUST AIR DAMPER STATUS	BI	FAIL		(3)
A-30	EXHAUST AIR DAMPER	BI	FAIL		(3)
A-31	COOLING COIL DISCHARGE AIR TEMP.	AI	40/55	5 MIN	(3)
A-32	HOT GAS REHEAT DISCHARGE TEMP.	AI	45/55	5 MIN	(3)

NOTES:
1. NOT ALL VIRTUAL POINTS ARE SHOWN. PROVIDE ALL POINTS REQUIRED FOR SOFTWARE OPERATION.
2. LOCKOUT ALARM WHEN UNIT IS DE-ENERGIZED. PROVIDE 15 MINUTE STARTUP DELAY PRIOR TO ENABLING ALARM.
3. SHOW OBJECT ON GRAPHIC.

A/E INFO

APPROVED David Greenfield
David W Greenfield
FOR COMMANDER NAVFAC

ACTIVITY PER DOCUMENT FROM CAPT. EMIL CASCIANO WITH SLC – SUBASE NEW LONDON
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DES JLS	DRW JLS	CHK JM

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FIRE PROTECTION SLW

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NORTHEAST IPT
NAVAL STATION – NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON
GROTON, CT

REPAIRS TO SUBMARINE A SCHOOL BQ 534

CONTROL DIAGRAMS

SCALE: AS NOTED
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CONSTR. CONTR. NO. N40085-12-R-1715
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SHEET 158 OF 206

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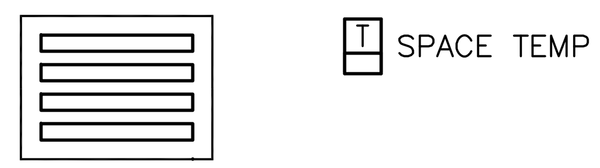
ELECTRIC UNIT HEATER CONTROL DIAGRAM (TYP.) (NON-DDC)

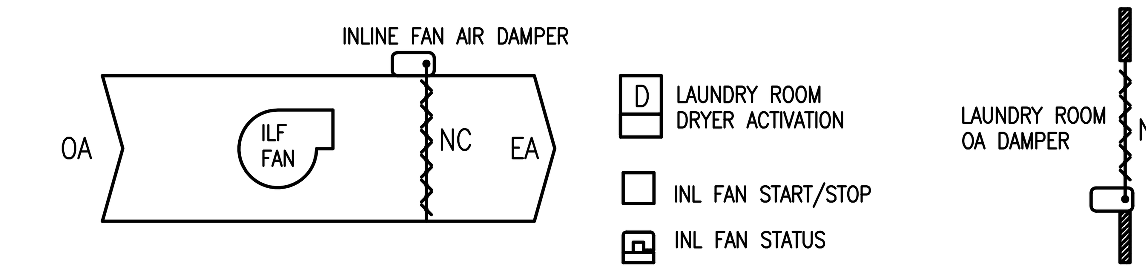


SEQUENCE OF OPERATION

ELECTRIC UNIT HEATER (UH-1&2)

HEATING OPERATION: THE ELECTRIC UNIT HEATER IN THE ELEVATOR MACHINE ROOM SHALL OPERATE AT A SETPOINT OF 50°F (ADJUSTABLE). THE HEATER WILL BE CONTROLLED FROM A MANUFACTURER PROVIDED WALL-MOUNTED THERMOSTAT.

LAUNDRY ROOM INLINE FAN CONTROL DIAGRAM



SEQUENCE OF OPERATION

LAUNDRY ROOM SUPPLY FAN

A DDC CONTROLLER, USING ELECTRIC ACTUATION, CONTROLS THE INLINE FAN AS FOLLOWS:

FAN AND OA DAMPER OPERATION: UPON A ACTIVATION OF A DRYER IN THE LAUNDRY ROOM THE DDC SHALL OPEN THE OUTSIDE AIR DAMPER AND EXHAUST DAMPER, START THE LAUNDRY ROOM EXHAUST FAN AND OPEN THE LAUNDRY ROOM OUTSIDE AIR DAMPER. UPON THE DEACTIVATION OF ALL DRYER, THE REVERSE SHALL OCCUR. THE EXHAUST FAN STATUS SHALL BE MONITORED BY THE DDC THROUGH A CURRENT SWITCH.

SAFETY MODE: THE LANDRY ROOM EXHAUST FAN AND OUTSIDE AIR DAMPER WILL ENTER A SAFETY SHUTDOWN MODE WHEN ANY OF THE FOLLOWING OCCUR:
AN ALARM INDICATION FROM THE BUILDINGS MAIN FIRE ALARM CONTROL PANEL OR THE EMERGENCY HVAC SHUTDOWN SWITCH IS ACTIVATED.

MECHANICAL RM EXHAUST FAN – INPUT/OUTPUT SUMMARY

POINT	AI	AO	BI	BO	NOTES
LANUDRY RM INLINE FAN START/STOP				1	
LAUNDRY RM INLINE FAN STATUS (A)			1		CURRENT SWITCH
LAUNDRY RM DRYER ACTIVATION	1				
SUPPLY OUTSIDE AIR DAMPER				1	
LAUNDRY RM INLINE FAN AIR DAMPER				1	
	1		1	3	TOTAL POINTS

LAUNDRY RM INLINE FAN CONTROLS

SYM DESCRIPTION DATE APPR

SEAL

A/E INFO

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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND
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NORTHEAST IPT NAVAL STATION – NORFOLK, VIRGINIA
NAVSUBASE NEW LONDON GROTON, CT
REPAIRS TO SUBMARINE A SCHOOL BQ 534
CONTROL DIAGRAMS

SCALE: AS NOTED
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M-703

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